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Contributions to the statistical quantification of uncertainties affecting the use of numerical models

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Abstract

This document summarizes several contributions in applied mathematics made during the period 2010-2023. Since my PhD [323], my work has mainly focused on stochastic modeling of uncertain information and its propagation within numerical models used for decision-helping, with several applications in resource management in a broad sense. This work has been carried out personally, or in collaboration with academic researchers and research engineers, and has also involved several PhD and post-doc students whom I have co-supervised. Its main results are illustrated by industrial and environmental examples, where the numerical models can be, typically, discretized simulation models (e.g., based on partial derivative equations), or models learned either from numerical experiments (*meta-models* or so-called *surrogates*) or from observational data (in a machine learning framework).

More specifically, in an industrial context, essentially dedicated to production, resource management requires taking into account the material risks associated with this activity. Thus, the service life (and more generally the indicators of ageing, fatigue, wear) of industrial components, under operational conditions, are key for decision-making (preventive or corrective maintenance), while endowed with uncertainty. The structural reliability of installations (such as civil engineering works) is alternatively more and more based on the use of complex and time-consuming computational codes, called *forward models* and possibly seen as *black boxes*, reproducing multi-physical phenomena [1]. For such models, uncertainty is generally viewed as stochastic, expressed on the code inputs, and require to choose probability models, e.g. by solving inverse problems and accounting for prior knowledge. Others sources of uncertainty are related to model and observational errors. Stochastic uncertainties allow to define probabilistic risk criteria on model outputs, such as probabilities of exceedance or quantiles. Such indicators need typically to be assessed using forward simulation, despite typically hight computational costs.

In an environmental context, the sustainable management of exploited resources (e.g. commercial fish stocks) requires the definition of an evolutionary model for these resources (e.g. a population model, an ecosystemic model), subject to controllable influences (e.g. extraction) and uncontrollable forcings (e.g. the environment). These models enable us to study the conditions under which these resources can return to and/or maintain equilibrium. Many of the same methodological problems remain, however.

Against this general backdrop, the contributions summarized here deal with:

- The influence of uncertainty on decisions (trying to improve them) based on numerical modeling;
- Bayesian modeling of uncertain knowledge (trying to be more guided and explain-

1. Such forward models and the associated algorithms have now become key components of *digital twins*.

able for a better use), in our case seen as an input of such models;
- Bayesian stochastic inversion of these numerical models, especially when such models are computationally costly and surrogate meta-modeling seems required;
- The fast estimation and the bounding of probabilistic decisional indicators based on the geometric properties of these numerical models, in relation with risk analyses.

1 Introduction

This dissertation deal with various contributions in applied mathematics –especially statistics– to the general field of uncertainty quantification (UQ) in decision support problems, conducted over the period 2010-2023. Accompanied with a curriculum vitae, it aims at demonstrating a personal capacity to formalize such problems and solve them through scientific researches conducted with colleagues and by supervising PhD students, post-doctoral researchers and teams of research engineers. These contributions are mainly motivated by industrial and environmental issues.

The UQ field has known many scientific formalisms, but since the 2000s, a mathematical framework has gradually emerged, under the influence of a dynamic statistical and engineering communities oriented towards the increasing use of numerical tools to support (and gradually replace) real experiments. It culminates today with the methodological scheme popularized under the name *Verification, Validation and Quantification of Uncertainty* (VVUQ ; [12, 240]) and the current paradigm of *digital twins* [106]. Their ambitions are to simulate an approximation of reality, sufficiently accurate to allow extrapolation and optimization under uncertainty, avoiding systematic recourse to new real, costly measurements. Basically, decision support with such tools is based on a formalism $Y = g(X)$ where X represents uncertain input parameters and g is a phenomenological model [2]. Integrating more and more scientific knowledge, the more complex phenomenological simulation models (or *computer models*) have now, in many disciplinary fields, acquired the status of *objective references* in the sense proposed by Karl Popper [3] [263]. Therefore the regulation of many activities, particularly those involving safety-critical systems, is based on using such *certified* computer models. Aircraft design [201], nuclear safety and reliability analyses [2] or ecosystem services [249] are just a few key examples.

In other situations where, on the contrary, data information can be massively collected, the strong development of statistical learning methods and tools also makes it possible to envisage “mimicking" the real phenomena, replacing g (or some particular sub-phenomenon encoded in g) by a so-called *surrogate* (or *meta-model*) $\hat{g}$ [191]. These two frameworks have gradually merged, adapting the principles of statistical learning to the study of phenomena which can alternatively be simulated by numerical models generating synthetic data. Indeed, the computational burden of such simulation tasks often increases with their complexity, and the intelligent exploration of numerical models, guided by surrogates, has also become an engineering necessity (e.g., see [116]). One example of this growing interplay is the massive research about *physics-inspired* machine learning models and algorithms [181].

In this context, the role of the statistician is to face multiple uncertainties limiting the informational power of data (observed or generated) on X and Y, threatening the

2. For instance, g denotes an operator that can rely on non-explicit solutions of equations (e.g., partial differential equations) and intern optimization procedures.

3. *By mutual decision of the protagonists*, they incarnate a scientific consensus.

promise of g and $\hat{g}$ to accurately predict the phenomenon in question. And in so doing, a sub-optimal or even misguided decision support when dealing with critical industrial or environmental resources [4]. These uncertainties are traditionally separated into two classes:

- The *aleatoric*, intrinsic uncertainties characterize the fact that knowledge about X cannot be fixed at a point value, no matter how much information (data) about X can be collected. The consensual modeling of these uncertainties is based on the probability theory [45];
- The *epistemic*, extrinsic uncertainties, on the contrary, characterize a lack of knowledge of the phenomenon of interest that can be reduced by providing additional information (about X, Y and g) [102]. While the tools of probability theory are still in the majority when it comes to to formalize this uncertainty, there is not the same scientific consensus on this choice [155, 280, 310].

The modeling of both uncertainties and the reduction of epistemic uncertainty affecting decision-making thus form a research program required for many mathematical engineering tasks, and this document summarizes some contributions to this work program. In particular, the reduction of epistemic uncertainty by statistical means is related to commonly encountered situations, which we list below, in connection with personal contributions:

1. *Accounting of modeling errors* in decision-making. Section 2 introduces this topic by presenting a contribution on the mathematical correction of a resource management decision related to the replacement of a deterministic model by a stochastic model, incorporating a so-called *process error*.
2. *Dealing with small samples*, which limits or even prohibits the real consistency of classical (frequentist) statistical estimators, and suggests to adopt informative Bayesian approaches if other sources of information are available (e.g., expert information). More precisely, this framework offers the possibility of integrating multiple sources of information, possibly subjective, and of taking into account the two types of uncertainty cited above [284, 158, 156, 90, 340]. Section 3 examines these informative Bayesian modeling problems, which take part into an active research epistemological program [43, 307]. To illustrate Bayesian benefits, the Bayesian optimization of experimental designs proposed in the contribution [10], allowing to complete destructive test data sets, which are costly in essence, offers an answer to the problem of reducing epistemic uncertainty. While most current research is focused on Bayesian computation methods, the search for formalprior modeling rules has in recent years focused strongly on going beyond the manipulation of tools with algebraic facilities (for example, conjugated models). The work summarized in this dissertation is in line with this ambition, focusing

4. According to common sense, specialized researchers and regulatory agencies (e.g., [291, 25]), resources are considered critical when an imbalance in their use can lead to high-risk situations (e.g., risk of major accident for industrial resources, risk of extinction for natural resources). Phenomenological modeling aims in particular at limiting these risks or mitigating their consequences by showing their probable origin.

on the intelligibility of these formal rules. Such intelligibility seems today to be indispensable to promote the acceptance of Bayesian informative modeling approaches, a tenacious issue [217, 306], if only because of their use within artificial intelligence tools.

3. *Dealing with incomplete data*, typically produced by agglomeration rules. In [252], my co-authors and I have considered in particular the generic case of discrete Markovian models to be estimated from such observations, which agglomerate trajectory data. We have produced adaptive stochastic algorithms using data augmentation mechanisms to extract the hidden information. Such problems also characterize latent state, or missing data, and stochastic inversion problems, which are particularly examined in Section 4.
4. *Determining the most influential variables of a problem*, in order to focus the work of reducing epistemic uncertainty on these target variables. Tied to the general problem of dimension reduction, the tools of sensitivity analysis (SA) are nowadays the most popular to approach this subject in the UQ context. In [197], my co-authors and I have proposed *goal-oriented sensitivity indices* based on variational techniques, whose interest for mathematical engineering has been demonstrated by a large number of subsequent works (e.g., [91, 120, 134]). Drawing on strong epistemological proximities between SA, robustness analysis and machine learning interpretability [270, 165], I am currently co-leading work along these lines in the latter framework [161, 162, 163].

Decision-making itself is often based on low-dimensional indicators. In particular, in a context of uncertainty treatment by probabilistic tools, the dimension of the variable of interest Y is often reduced to 1. Very generally, these indicators are probabilities as $P(Y < y_0)$ where y_0 represents a limit value, or (dually) the threshold of the quantile associated with a chosen value y_0. Such indicators are usually encountered in structural reliability [195], but also in different application fields of resource management (e.g., [246]). They are usually computed using forward simulation, and a key issue is to diminish the computational cost with a minimum set of hypotheses on g (e.g., avoiding intrusive techniques), to fit with concrete requirements on using complex models. Section 5 summarizes two contributions to this field (which originated in personal work, and were then developed in PhD co-supervisions):

- *Developing accelerated Monte Carlo methods* for estimating these indicators, taking into account the monotony and convexity properties of g. These geometrical properties of g generally correspond to known constraints (g can for example be related to a multi-objective optimization problem), explored beforehand, or constraints set in order to define a conservative version of the problem (from the point of view of a risk associated with the decision). Decision-making indicators can be bounded and consistently estimated, These contributions require tools from random space theory, martingale theory and computational geometry.
- *Determining penalizing correlations on X related to quantile minimization on Y.* While this contribution focuses on improving input modeling, it is guided by a concern to reach low values of Y, which can be compared to limit values, a key

assessment for decision-making for numerous safety studies involving critical systems. Involving elements of nonparametric analysis, methods developed in this work explore the space of penalizing correlations by imposing a parsimony constraint (sparsity) on large size (vine) copulas.

The remainder of this document does not aim at summarizing exhaustively all the contributions in statistical modeling and UQ, nor even all the research work carried out in the broader field of applied statistics (see the curriculum vitae for such a summary). It proposes a few works selected for their mathematical content, without being mentioned further:

1. elaborating methodologies for the selection of statistical models for X improving the use of physical models, applied to the study of industrial steels [186, 261, 10] (which gave rise to the patent [262]) ;
2. producing methodological summaries for a vast community of applied researchers [354, 317, 265] ;
3. leading the writing of a mathematical and engineering textbook on extreme values, including new research contributions [319] ;
4. co-leading theoretical and applied researches in robust post-hoc interpretability of machine learning models, as evoked before [161, 162, 164] ;
5. supervising applied researches in statistical learning that led (among others) to :
   - designing alarm systems based on a mixture of noisy and non-noise variables, in highly unbalanced situations (patent [85]) ;
   - automating and significantly improving the search for specific shapes in medical images and patterns within unstructured medical texts, using deep learning tools [303, 298] ;
   - automating translation tasks by hybridizing deep learning tools with rules-based system [148].
6. Improving the treatment of uncertainties and the statistical estimation of complex environmental resource models, generally requiring contributions in Bayesian modeling [55, 328, 326, 13, 96, 246].

2 Decision-making in the face of uncertainties

2.1 Context

To begin this dissertation, let us consider a class of resource management problems and highlight, through a mathematical contribution, the major influence of taking epistemic uncertainties into account on decision support (resource extraction). These uncertainties translate into the incorporation of a process error into a well-established deterministic model, of *state-space nature*, used in many fields for forecasting how a critical resource evolves. Therefore this work constitutes a natural prerequisite to the dissertation.

State space models are very generally derived from a numerical approximation of a dynamic equation. The simplest cases are ordinary differential equations (ODE), according to which a quantity of interest $X(t)$ changes over time t as follows:

$$\frac{dX(t)}{dt} = h(t, X(t)) \tag{1}$$

A "natural" state space model, providing a practical approximation of this equation, can be derived by choosing a time discretization scheme (ie., $\frac{dX(t)}{dt} \simeq X(t+1) - X(t)$). A case studied extensively, and widely used in resource management for its simplicity [221, 154, 88], is the so-called (logistic) Gray-Verhulst-Gordon-Schaefer function [290]

$$h(t, X(t)) = rX_t \left(1 - \frac{X_t}{K}\right) - \phi_t X_t, \quad t \geqslant 0, \tag{2}$$

that has known large success for modeling exploited natural resources (e.g., fisheries), and introduced the broader class of *surplus production models* (SPM) sharing common parameters: $r > 0$ is defined as a per capita growth rate of the resource, $K > 0$ is the carrying capacity (the point at which growth and decline rates of the resource are equal) and ϕ_t is an extraction rate ($\phi_t X_t$ being the harvested part of the resource at time t) [255]. While simplistic, this class of models can be easily extended to multimodality assessments (e.g., multi-species) and is still now very used for preliminary evaluations of harvesting strategies and assessment methods [228, 54, 56, 269]. Beyond the case of exploited natural resources, models of the form (1-2) has shown high relevance to simulate the number of tumor cells growing within an organism [108, 352]. More generally, numerous application fields, that require to forecast how a measurable or latent resource evolves in an environment characterized by a *saturation level* (K), regular *stress levels* (ϕ_t) and some *production mechanism* (r) make a large use of this model for forecasting. Among others, this was notably applied to the lifetime of Lithium-Ion batteries [356], harbour throughput [1], load forecasting of power systems [302], steel intensity [110], coal consumption [101] or CO_2 emissions [338]. See [38] for more applied references.

A key assessment of this class of models is, in the framework of resource management, the *maximum sustainable yield* (MSY), namely the maximal possible harvest (or

extraction) such that the resource can continue to *regenerate*, and the related amount of resource. It is generally defined in a deterministic way, by assuming $\frac{dX(t)}{dt} = 0$. Choosing h as (2) with constant extraction rate $\phi_t = \phi$ (defining the so-called *grey* version of the model [178, 302]), it leads to MSY $= rK/4$ [172], $X_{\text{MSY}} = K/2$ and $\phi_{\text{MSY}} = r/2$. Combining biological and economic concepts, the MSY is recommended as the optimal level of extraction by international organisations, as the Food and Agriculture Organization (FAO), since the 2002 World Summit on Sustainable Development [355] and a starting point for international environmental policies [159]. However, this approach, that does not account for the strong uncertainties that affect such basic models, independently of any data-based statistical inference on parameters, has been obviously, strongly criticized from years by researchers [199, 184]. Numerous empirical studies denounced this lack, concluded it often leads to overoptimistic decision-making then major resource collapses [253, 9] and extinction [8], and proposed other, more cautious reference points [282, 60, 171, 334]. The development of massive simulation methods from the 1990's helped to compute such points and define rules-of-thumbs. In parallel, theoretical works to define stochastic MSY were generally conducted in the frameworks of perturbation methods and diffusion theory, assuming continuous time and unpredictable additive perturbations. See [329] for a survey of such methods, applied to model

$$\frac{dX(t)}{dt} = h(t, X(t)) + \epsilon_t \tag{3}$$

where ϵ_t is a stochastic model error, or so-called *environmental process noise*. These approaches are based on understanding the notion of *regeneration*, or sustainability, as the *stationarity* assumption that X_{t-1}, X_t, X_{t+1} obeys a similar distribution f_X. They generally led to obtain approximate recalculations of bioeconomic reference points.

Another important interest in obtaining a stochastic MSY (and derived reference points), easy to calculate, comes from the use of SPM in the medical field, evoked hereinbefore. Indeed, the MSY could be interpreted as a measure (in terms of cell population number) of the minimal amount of curative treatment such that a tumorous cell population does not grow up (in a stochastic sense). Contrary to ecological situations where the sustainability requirement of a strongly exploited resource implies that the MSY should be considered as an extreme upper bound for any exploitation quota, the MSY could be thought as the minimal "level" of medicine to which the patient must be exposed. A too strong medical treatment (e.g., chimiotherapy) can severely attack the immune system of the patient, leading possibly to death. Therefore, a strategy of increasing slightly the amount of medicine at several time steps could be based on the study of extinction conditions and the deviation of the distribution of cell population from the law-at-equilibrium.

2.2 Problems and personal contributions

Additive stochastic process noises ϵ_t injected within SPM models (3) are much less used than multiplicative ones in the applied literature, as they appear more suitable to

generate relevant randomized trajectories. The optimal management levers arising from stationarity properties of discrete-time models as

$$X_{t+1} = \left(X_t + rX_t \left\{ 1 - \frac{X_t}{K} \right\} - \phi X_t \right) \epsilon_{t+1} \quad t \geqslant 0, \tag{4}$$

while of current use, have been little studied theoretically. I contributed to develop a full methodology for obtaining explicit conditions of stability, extinction and optimal management [329], whose main technical results are presented hereinafter. This methodology was later applied to other models by other authors, as [42]. Recently, my co-authors and I found it suitable to help improve the management of endangered species (e.g., small cetaceans captured accidentally – bycatches –[246]).

A preliminary result are general conditions for the existence of a unique, stationary distribution for X_t. The constructive theorem 2 improves the first result (from an operational perspective) by determining an explicit solution for this stationarity distribution, under the technical constraint (5) on the noise ϵ_t involving beta $\mathcal{B}_e$ distributions. This constraints appear mild in reality, as simulation-based comparisons highlight that for usual estimated values of the noise variance and r in applications, the distribution of ϵ_t no significantly differs from classic choices (see Figure 4).

THEOREM 1. **Existence of stationarity (B.** *et al.* **[329])**. *Denote $\gamma = r - \phi$. Assume that the initial state $X_0 \leqslant K(1+\gamma)/r$. Assuming that $\{\epsilon_t\}$ are continuous iid random variables bounded on $[0, 4(1+\gamma)^{-1}]$. Then the Markov chain defined by (4) is irreducible, aperiodic and admits a unique stationarity distribution.*

Most of these results were obtained from theoretical ingredients on stochastic discrete Markov models [218], by determining the properties of the deterministic control model associated to the Markovian kernel of the chain defined by (4), and findings drift conditions for proving Harris recurence (which means that the chain converges to a unique stationary distribution) for every starting point $X_0 \leqslant K(1+\gamma)/r$. Constraint (6) actually appears as a requirement for the nonextinction of the resource X_t, assuming the relevance of model (4), and implies a former condition of nonextinction provided by [271] in a simplified framework where at low levels the amount of resource increases with probability one.

THEOREM 2. **Explicit stationary distribution (B.** *et al.* **[329])**. *Assume that*

$$\epsilon_t = \frac{4\beta}{1+\alpha+\beta} U_t V_t \tag{5}$$

where $U_t \sim \mathcal{B}_e(\{\alpha+\beta+1\}/2, \{\alpha+\beta-1\}/2)$ and $V_t \sim \mathcal{B}_e(\{\alpha+\beta\}/2, \{\beta-\alpha\}/2)$ are independent beta random variables. Denote $Z_t = \frac{rX_t}{K(1+\gamma)}$ the reduced variable of interest. Given a choice of variance

$$\sigma^2 < \sigma_M^2 = 1 - \frac{1+\gamma+2\gamma^3}{(1+\gamma)^3}, \tag{6}$$

then there exist two unique explicit expressions for (α, β) *such that* $\mathbb{E}[\epsilon_t] = 1$, $\mathbb{V}[\epsilon_t] = \sigma^2$ *and the unique stationary distribution of* Z_t *is* $\mathcal{B}_e(\alpha, \beta)$.

A consequence of these results is that, under the previous conditions, refined approximations of MSY and related reference points can be derived, extending the deterministic results:

$$\begin{aligned} \mathrm{MSY}(\sigma) &= \frac{rK}{4}\left(1 - \frac{\sigma^2}{r(1-r/4)} + \frac{4\sigma^4}{r^2(4-r)^4}\left[r^4 - 4r^3 - 12r^2 + 48r - 16\right] + o(\sigma^5)\right), \\ X_{\mathrm{MSY}}(\sigma) &= \frac{K}{2}\left(1 - \frac{8\sigma^2}{r(4-r)^2} - \frac{8\sigma^4}{r(4-r)^5}\left[3r^3 - 18r^2 + 12r + 32\right] + o(\sigma^5)\right) \end{aligned}$$

and the optimal extraction rate is

$$\phi_{\mathrm{MSY}} = \frac{r}{2} - \frac{2(2-r)}{(4-r)^2}\sigma^2 + o(\sigma^3).$$

The differences with the deterministic framework, summarized above, are obvious and dramatically illustrate the major impact of epistemic uncertainties on decisional quantities. For some values of r, the functions $MSY(\sigma)/MSY(0)$ and $X_{MSY}(\sigma)/X_{MSY}(0)$ are displayed on Figures 1 and 2. Notice that for a small r and a value of σ large enough can reduce the deterministic $MSY(0) = rK/4$ more than 50%. A less expected result is the fact that $X_{\mathrm{MSY}}(\sigma)$ can be significantly lower than $K/2$, even if the decrease of $X_{\mathrm{MSY}}(\sigma)$ as a function of σ appears to be slower than the decrease of $\mathrm{MSY}(\sigma)$. A non-optimized extraction rate ϕ can induce a severe shortfall especially when σ is large (Figure 3).

In [329], retrospective studies on commercial resources exploited using deterministic approaches showed that not accounting for uncertainties can cost dozens of millions of dollars in useless harvesting effort. A tragic example of huge economic losses related to the absence of really accounting for uncertainties affecting environnemental resources is the challenges faced by Mozambique after the launch of the tuna bond in 2013. The real benefits of fisheries after some years reached at most 5% of the expected gains. Such overoptimism strongly contributed to the cessation of payment of this state in 2016 [26].

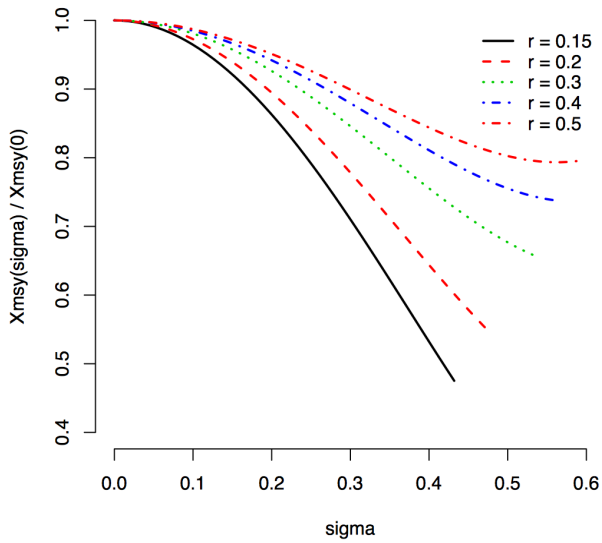


Figure 1 – Plots of the ratio $\mathrm{MSY}(\sigma)/\mathrm{MSY}(0)$ as a function of σ, for some representative values of r and $\sigma \in (0, \sigma_M)$ (cf. (6)). $\mathrm{MSY}(0)$ is the deterministic yield $rK/4$. Quantities are calculated using a higher order approximation than (7).

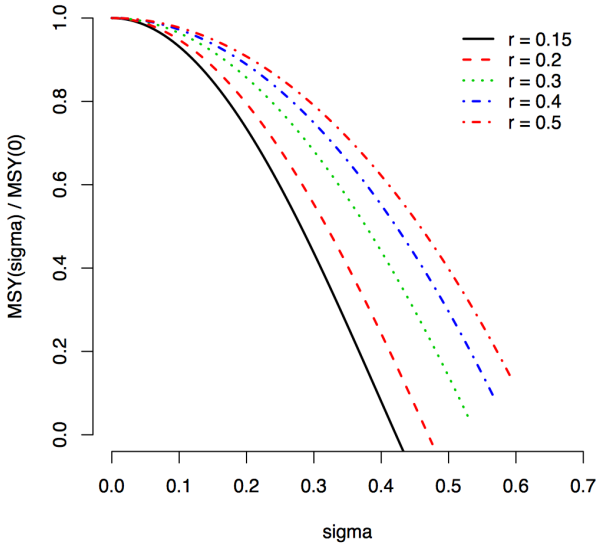


Figure 2 – Plots of the ratio $X_{\mathrm{MSY}}(\sigma)/X_{\mathrm{MSY}}(0)$ as a function of σ, for some representative values of r and $\sigma \in (0, \sigma_M)$ (cf. (6). $X_{\mathrm{MSY}}(0)$ corresponds to the deterministic maximally exploited biomass $K/2$. Plots are drawn using a higher order approximation than (7).

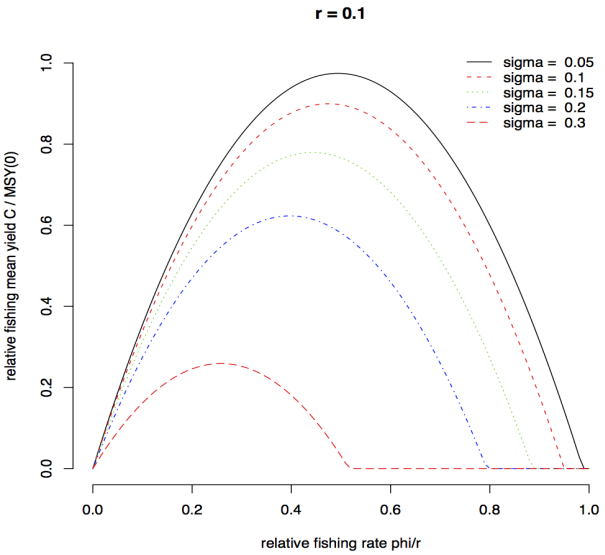


Figure 3 – Plots of the ratio of mean catches over the deterministic maximum sustainable yield $\phi\mathbb{E}[X_t]/MSY(0)$ as a function of the relative fishing effort ϕ/r, for $r = 0.1$ and some values of σ. There are no catches for large ratio ϕ/r and values of σ larger than (6). A non-optimized ϕ can induce a severe shortfall especially when σ is large.

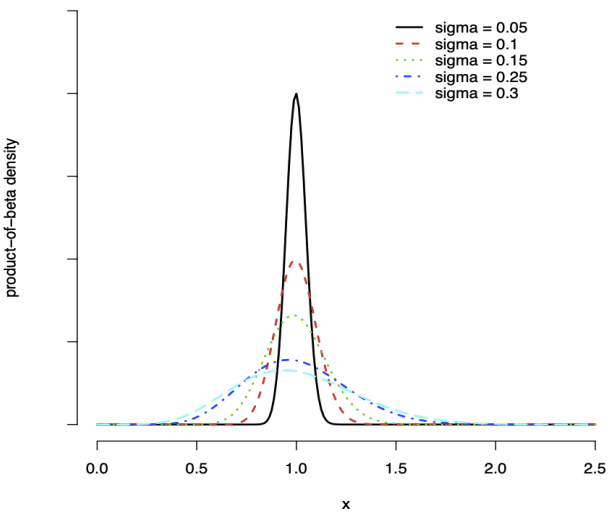


Figure 4 – Typical densities of noise ϵ_t generated by Equation (5).

3 Nonconjugate Bayesian modeling

This section takes up and extends a major part of the contributions published in [321].

3.1 Context

The Bayesian paradigm provides a theoretically adequate response to the problem of mixing aleatoric and epistemic uncertainties [155] and optimizing under uncertainty [254, 23, 125]. More formally, without recalling the fundamentals of Bayesian modeling, inference and decision theory, consider a randomized quantity X, involved in a decision-helping process, for which a parametric model with probability density function (pdf) $f(x|\theta)$ is assumed [5], where the parameter vector $\theta \in \Theta \subset \mathbb{R}^d$ with $d < \infty$. For technical or epistemological reasons, Θ is given a probabilistic space structure, and θ is assumed to follow a prior distribution with pdf $\pi(\theta)$. Denote $f(\mathbf{x_n}|\theta)$ a likelihood of available observations $\mathbf{x_n} = (x_1, \ldots, x_n)$, defined conditionally to θ. In such a framework [277], decision-helping on any quantity of interest $h(Y) = h(g(X))$ relies on a choice of a cost function and updating prior information on θ then X by conditioning to observations, through Bayes' theorem, which provides the *posterior distribution* on θ with pdf

$$\pi(\theta|\mathbf{x_n}) \quad \propto \quad \pi(\theta) f(\mathbf{x_n}|\theta) \tag{7}$$

and consequently the *posterior predictive* pdf on X

$$f(x|\mathbf{x_n}) \quad = \quad \int_\Theta f(x|\theta)\pi(\theta|\mathbf{x_n})\, d\theta. \tag{8}$$

Apart the non-trivial choice of the cost function (a research subfield in itself, see [173]), two key issues may limit the use of the methodology (now popularized by the term *Bayesian workflow*) [289, 128, 142]) in applications:

1. *Choosing $\pi(\theta)$ given the available prior information* (prior elicitation) ; this is certainly the most criticized point in practice [140, 307].
2. *Computing the posterior distribution* (which is key too for Bayesian model averaging, selection and Bayesian optimization).

The second issue is the permanent subject of numerous works, mostly based on numerical approximations and stochastic algorithms as Monte Carlo Markov Chains (MCMC), which focuses now on exploring high-dimensional models (see for instance the recent article [235]). Theoretical and practical improvement of Bayesian computation methods is an unequivocal necessity to combat the inherent brittelness of Bayesian learning, brilliantly highlighted by Owhadi *et al.* [248]. These latter authors, as well as seminal researchers as Gelman [141], among others, have also highlighted the need to resolve some deep-seated weaknesses related to the first issue, that still hamper the understanding and use of Bayesian approaches, despite a very important state of the art. The contributions

5. In many engineering situations, $f(x|\theta)$ can be chosen for historical reasons–as a *legacy model*– or on the basis of regulatory arguments (e.g., in extreme risk analysis).

summarized in this section concern this first issue and aim at contributing to the improvement of the Bayesian workflow.

More precisely, a lot of work has been done since the 1970s on the construction of so-called "objective", "neutral", "benchmark", "default", "flat" or *noninformative priors*[6] by formal rules [64]. These measures aim at regularizing the problem of inference on θ while avoiding the introduction of subjective information[7]. The Jeffreys' and Berger-Bernardo's prior measures, Zellner's Maximal Data Information Prior (MDIP) or the coverage matching priors, which all depend on the choice of $f(x|\theta)$ and various notions of "non-informativeness" (related to parametrization invariance, prior-posterior entropy maximization, etc.) are the most famous of the proposed measures [182]. The problems posed by these measures are well known: they are in general *improper* ($\int_\Theta \pi(\theta)\, d\theta = \infty$) which can, among other things, can render posterior distributions improper and unusable or lead to inconsistencies and paradoxes. Other authors have also proposed new formal rules depending only on Θ and a choice of loss function (scoring rule), in order to improve the possibility to produce true prior probability measures [194]. This theoretical problem responds to the long-standing search of the applied Bayesian community for weakly informative priors that may avoid crude (and somewhat dangerous) numerical approximations, in relation with the noticeable development of software computing tools [137, 139]. While it is still largely open to discussion on its underlying principles, this need was recently strongly advocated again in several communities (e.g., in ecology with [198] and in reliability with [335]), decided Mikkola *et al.* to produce an impressive review article on prior elicitation [219].

The search for non-informative measures precedes, in the Bayesian workflow, the search for informative ones, i.e., prior integrable measures, which may complete the information provided by the data or guide the collection of new experimental observations. Two sub-problems are then to be addressed:

(i) the elicitation of this information often considered as subjective (resulting from expertise, sorting of old data...) and its interpretation in statistical terms;

(ii) the choice of a (hyper)parametrized model $\pi(\theta) = \pi(\theta|\omega)$ where ω lives in a space of finite dimension.

Proposing formal, defensible rules to guide these steps appears as a logical requirement to improve the use of Bayesian statistics [306] and to fill in some of the "holes" in the theory evoked by Gelman [141]. Beyond, in the specific context of Bayesian sensitivity analysis [183, 135, 152, 364], such rules should ideally be used to produce more defensible generic methodologies to improve Bayesian studies and test the good behavior of posterior distributions against fluctuations in prior information, through the choice of a geometric variation class Γ_π of $\pi(\theta)$ around a selected prior $\pi_0(\theta)$. These classes are today lacking

6. Although this term is misleading and regularly questioned [168, 243, 117].

7. But similarly, such a point of view is nowadays questioned, subjective model choices being considered as unavoidable by several authors, especially A. Gelman [76, 138]

of interpretability. The $\varepsilon-$contamination approach [31, 353], for instance, considers that

$$\Gamma_\pi \quad = \quad \{\pi : \ \pi(\theta) = (1-\varepsilon)\pi_0(\theta) + \epsilon q(\theta), \ q \in \mathcal{D}\}$$

where $\mathcal{D}$ is a subset of the space $\mathcal{P}$ of all probability distributions. The classic usage is to choose $\mathcal{D}$ as the subset of symmetric distributions, and/or such that π be unimodal, etc. Other geometric deformations can be produced by considering exponential perturbations (*exponential twisting*) produced by minimizing relative entropy under moment constraints, or other customary discrepancies (e.g., Wasserstein distances) from optimal transport theory (as my co-authors and I did in [197, 161] in the somewhat different context of goal-oriented sensitivity and robustness analyses):

$$\mathcal{L} \quad = \quad \left\{\pi : \ \pi(\theta) = \arg\min_{\tilde{\pi}\in\mathcal{P}} \int_\Theta \pi_0(\theta) \log \frac{\pi_0(\theta)}{\tilde{\pi}(\theta)} \ d\theta, \ \mathbb{E}_\pi[h(\theta)] = c_i, \mathbf{1} \leqslant i \leqslant d \right\}. \quad (9)$$

Fixing h, ε or the constraints c_i remains, in practice, a non-trivial and highly subjective choice, which suffers today from the absence of clear formal rules. See [281] for a recent critical view on this theme about hierarchical models.

Problems **(i)** and **(ii)** have been cleared up and reviewed by many authors in relation with various application domains, especially [67, 170, 264, 133, 244], without addressing all of the barriers related to statistical interpretation and correction of subjective biases surrounding the incorporation of non-data information (and without providing a straightforward and complete methodology of prior modeling). The search for an algebraic form for $\pi(\theta)$ still remains focused on:

- *"cautious" approaches*[8] minimizing an amount of statistical information relative to a noninformative measure $\pi^N(\theta)$, under M informative constraints; so is the well-known maximum Shannon-Kullback entropy (Maxent) principle [170]

$$\pi^*(\theta) \quad = \quad \arg\max_{\pi\in\mathcal{P}} \int_\Theta \pi(\theta) \log \pi_(^N\theta) \ d\theta \quad \text{such that} \int_\Theta h_i(\theta)\pi(\theta) \ d\theta = d_i$$

  for $i = 1, \ldots, M$, which knows many variations (e.g., [100]) and is similar to the exponential twisting mechanism described in (9).
- *algebraically practical approaches*, such as the *conjugate prior distributions*, which can for those belonging to the exponential family be interpreted too as Maxent priors. In addition to their ease of use, they can also be interpreted as *posteriors of virtual samples and noninformative priors*, which can strongly contribute to justify their use (see § 3.2.2). The conjugated problems remain however rare.
- *"parsimonious" approaches* for high-dimensional / hierarchical problems, that consider the prior specification as a lever for controlling the balance between predictive power and parsimony. Maybe the most famous of such approaches are the PC priors [299] that penalize model complexity and, in regression frameworks, or spike-and-slab / sparse priors [222, 157, 48] (from which PC priors borrow key

8. In the sense that they attempt to minimize the subjectivity of a representation.

ideas). In particular, PC priors offer a way of building informative, "transparent" (in the sense suggested by Draper [99]) priors for hierarchical additive models with nontrivial latent structures, on the basis of several principles coming from Occam's razor to the ease of implementation. This great methodological advance is, however, reserved for certain classes of problems where the number of components is key and prior information is truly available on prior flexibility. Their interpretability is besides conditional to understanding entropic features.

Without there being a "canonical" approach, while formal rules have now become deeply studied for classes of hierarchical problems (see for instance [361]) in line with the development of PC priors, the landscape of prior modeling in view of conducting various analyses (e.g., prediction, selection) is thus dense. Among others, it is also enriched by important considerations on prior pooling (e.g., [7]), prior compatibility in model selection (e.g., [83]), or the need for agreement between prior information and data information, which can be processed either by "prior-data conflict" detection criteria [313, 111, 112] or by form constraints on the prior structure [245, 237, 5]. This major concern for the robustness of posterior inference to prior misspecification under finite information – an antagonistic goal to learning as highlighted by [248] – as well as the wish to minimize any subjective modeling approach, was also at the source of the development of *robust Bayesian analysis* in the 1990-2000 period [36, 29, 276]. However, the interest in this technique does not seem to have grown significantly in recent years, maybe because of some conceptual and computational issues raised by this approach (which are closely related to those encountered in possibility theory) [308], although its extensive use is part of the expectations about Bayesian statistics in the 21st century [30]. Recent results on the inherent *brittleness* of Bayesian robust analyses [248] probably did not help to increase interest in these approaches. Other wished constraints are regularly highlighted in the literature, which should feed any modern Bayesian informative prior modeling. For instance, [39] has recently made explicit a mode of convergence on strictly positive Radon measures for which a sequence of probability measures can admit an improper measure for limit, which provides a logical rule of "consistency" between the worlds of noninformative and informative measures (see Definition 1 beneath).

DEFINITION 1. **Q-vague convergence [39]** *A sequence* $\{\Pi_n(\theta)\}_{n\in\mathbb{N}}$ *of informative prior distributions converges* $q-$*vaguely towards a noninformative prior measure* $\Pi(\theta)$ *if it ensures the vague convergence on the quotient space* $\mathcal{R}/\simeq$ *where* $\simeq$ *is the equivalence relation defined by*

$$\Pi \simeq \Pi' \quad \Leftrightarrow \quad \exists \alpha > 0 \text{ such that } \Pi = \alpha\Pi'$$

Within this complex research environment, which would deserve to unify methodological and epistemological developments (in particular those carried out recently within the European Research Council (ERC) project *Making Scientific Inferences More Objective* led by J. Sprenger [9]), I am mostly interested in the following section in formal rules of

9. `http://msio.wordpress.com`

prior modeling for nonconjugate situations in UQ problems, a relevant issue for numerous applications.

3.2 Problems and personal contributions

3.2.1 Extending maximal data information priors

The search and selection non- or weakly informative prior measures by formal rules is, as previously stated, an important issue for many Bayesian tasks [299]. The MDIP approach originally proposed by Zellner [362, 363] aims at providing a common objective framework for the development of maximum entropy-type (Maxent) priors and non- or weakly informative priors based on an exact finite sample information-theoretic criterion functional. They can offer interesting alternatives to reference priors [182, 360], as the latter are based on asymptotic arguments. Let us recall its guiding principles.

The MDIP is chosen so as to maximize the average information $G(\boldsymbol{\Theta})$ in the data density relative to that in the prior: denoting $H(X|\theta)$ and $H(\boldsymbol{\Theta})$ the entropy of the sampling model and the (relative) entropy of the prior,

$$G(\boldsymbol{\Theta}) \quad = \quad \mathbb{E}_\theta \left[H(\boldsymbol{\Theta}) - H(X|\theta) \right].$$

Thus MDIP are broadly defined by

$$\pi^{MDI}(\theta) \quad = \quad \arg \max_{\pi(\theta) \geqslant 0} \int_\Theta \pi(\theta) Z(\theta) \; d\theta - \int_\Theta \pi(\theta) \log \frac{\pi(\theta)}{\pi^N(\theta)} \tag{10}$$

where $Z(\theta)$ is Shannon's information (or negative differential entropy) of the sampling distribution

$$Z(\theta) \quad = \quad \int_S f(x|\theta) \log f(x|\theta) \; dx$$

and π^N is not necessarily confounded with the (usually unwritten) dominating measure, but is more generally a noninformative measure. Zellner (1997) noticed that quantity $G(\boldsymbol{\Theta})$ gives "the total information provided by an experiment over and above the prior". Thus maximizing gain $G(\boldsymbol{\Theta})$ implies to minimize the information carried by $\pi(\theta)$ through the inference [304].

An issue which is frequently encountered using the Maxent priors or the MDIP is their impropriety. Building an encompassing family for both could help to increase the chances of getting proper priors, and this is the aim of the contribution [315], summarized here. The so-called *propriety* constraint $\int_\Theta \pi(\theta) \; d\theta = 1$ is known to be inoperative in the functional optimization of Maxent priors or in (10), as the associated Lagrange multiplier disappear in the proportionality constant of the maximization result. The main idea of [315] is proposing to replace this constraint by $\int_\Omega Z(\theta)\pi(\theta) \; d\theta = c < \infty$, which is induced

by $\int_\Theta \pi(\theta)\ d\theta < \infty$. This offers a way of producing an extension of both approaches. The maximization rule can be written as

$$\pi \quad = \quad \arg \max_{\pi(\theta)\geqslant 0} \int_\Theta \pi(\theta)(-\gamma_0) Z(\theta)\ d\theta - \int_\Theta \pi(\theta) \log \frac{\pi(\theta)}{\pi^J(\theta)}\ d\theta. \tag{11}$$

PROPOSITION 1. **Extended Maxent-MDIP (B. [315])**.
Considering that some prior information can be summarized, if available, in linear equations

$$\int_\Theta f_i(\theta)\pi(\theta)\ d\theta = c_i. \quad i = 1, \ldots, p, \tag{12}$$

the solution to (11) is

$$\pi(\theta) \quad \overset{a.s.}{\propto} \quad \pi^N(\theta) \exp\left(-\gamma_0 Z(\theta) - \sum_{i=1}^{p} \gamma_i f_i(\theta)\right). \tag{13}$$

where $(\gamma_0, \ldots, \gamma_p) \in \mathbb{R}^{p+1}$.

In addition to constraints (12), another constraint is required to assess γ_0, which can take another form (e.g., qualitative / quantitative). As it is defended in next section, relevant prior constraints can take the form of a prior predictive percentile (α, x_α) such that

$$\mathbb{E}_\pi\left[P(X < x_\alpha|\theta]\right. \quad = \quad \int_\Theta P(X < x_\alpha|\theta)\pi(\theta)\ d\theta \ = \ \alpha,$$

which, as a linear constraint in $\pi(\theta)$, could theoretically feed the exponential form (13), but would lead to an indicator function $f_i(\theta)$ which would strongly distort the prior and break the constraint of absolute continuity.

Expression (11) highlights that the encompassing prior tries to reach a compromise, which is modulated by γ_0, between the maximization of a prior entropy and the need for keeping true a relevant characteristic of the prior joint information (a finite average information in the sampling distribution). One gives more importance to the entropy maximization if $\gamma_0 \to 0$ and more importance to the inferential weight of the data if $\gamma_0 \to -1$.

The elicited prior knows application in both objective and subjective Bayesian frameworks. The latter case is exemplified in [315], in a reliability and risk assessment context, highlighting that a weakly informative (flat), proper and semi-explicit prior can be reach for Weibull models unlike the Maxent prior and MDIP. Considering the former case, the prior knows an interesting interpretation in *channel coding*. In this context the maximal value of the Bayes risk, or *channel transinformation* [210]

$$I(\boldsymbol{\Theta}, \mathbf{X^n}) \quad = \quad \int_\Theta \pi(\theta) \int_{S^n} f(\mathbf{x_n}|\theta) \log \frac{f(\mathbf{x_n}|\theta)}{f(\mathbf{x_n})}\ dx_1 \ldots dx_n\ d\theta$$

where $f(\mathbf{x_n}) = \int_\Theta \pi(\theta) f(\mathbf{x_n}|\theta)\, d\theta$ is the marginal likelihood, or *evidence*, is defined as the capacity of a channel in which information on $\mathbf{\Theta}$ is sent to n receivers who have to decode X. Noticing that $G(\mathbf{\Theta}) = \frac{1}{n}\left(I(\mathbf{\Theta}, \mathbf{X^n}) - H\left(\mathbf{X^n}\right)\right) + H(\Theta)$, the asymptotic behavior of the data entropy $H(\mathbf{X^n}) \sim (d \log n)/2 - cn$. In other terms, when the length of a message increases, the mean data information can be strongly modulated by the linear coefficient c. Choosing $c < 0$ will makes the data be little informative and have the effect of slacking the increase of the transinformation. Thus, the prior will have more chances to be informative. Conversely, a positive value for c will give more information to the data and decrease the chances of eliciting a proper prior. Thus, assuming an unknown but finite value for $E_\theta[Z] = c$ (typically expressed in bits) is forcing the channel's capacity to be mainly dependent on the information yielded by the data when $|nc/H(\mathbf{X^n})| \sim 0$. Since $I(\Theta, \mathbf{X^n}) \geqslant 0$ in informative contexts (i.e., when π is proper), all values of $c \in \mathbb{R}$ do not lead inevitably to proper elicitations, but this construction can avoid to obtain automatically improper priors, providing a range of possible value of c to keep π proper.

It follows that, provided a nonparametric estimator of $\hat{H}_0$ of $H(\mathbf{X^{n_0}})$ can be provided [206] with n_0 a small number of data, one can build informative priors with decreasing inferential weight when the total data size n increases, using $c \leqslant -\hat{H}_0/n_0$. This upper bound can be refined using the covariance matrix K of the $\mathbf{X^n}$ to select c as a function of an ideal channel capacity and a fixed number of data, through the following result (in bits):

$$I(\Theta, \mathbf{X^n}) \quad \leqslant \quad n\left(c + \frac{1}{2}\log\left\{2\pi \exp(1)|K|^{1/n}\right\}\right).$$

This work opens the way to the elicitation of new prior families and their comparative studies with "historical" approaches. Extensions appear also required. Indeed, being invariant only for the location-scale family and under linear transformations of the parameter, MDIP require additional side conditions to remain invariant under more general transformations, and the elicited prior should inherit of these difficulties. Beyond, this contribution could be probably pursued by considering formal rules that attempt to reconcile all prior inferential methods, as the one proposed in [346, 347].

3.2.2 Approximate posterior priors (APP)

Once a non-informative $\pi^N(\theta)$ prior measure for the problem is defined, it has been often suggested that an ideal representation of an informative prior distribution should be a *posterior prior* distribution:

$$\pi(\theta) \quad = \quad \pi^N(\theta|\tilde{\mathbf{x}}_\mathbf{m}) \propto f(\tilde{\mathbf{x}}_\mathbf{n}|\theta)\pi^N(\theta) \tag{14}$$

where $f(\tilde{\mathbf{x}}_\mathbf{m}|\theta)$ is the likelihood of imaginary data $\tilde{\mathbf{x}}_\mathbf{m} = (\tilde{x}_1, \ldots, \tilde{x}_m)$ of size m (which becomes a natural prior hyperparameter). This view is similar to the one proposed by Clarke [61], who used reference prior and a likelihood possibly defined by another sampling scheme than $X \sim f(x|\theta)$. As stated in [272], "the resulting likelihood function

is then interpreted as representing implicit data information in the target prior, relative to the reference prior".

The idea of missing, virtual or imaginary samples originally appeared in the statistical literature as a device, or trick, used to tackle issues related to the *impropriety* of regularizing prior measures [147, 305, 169]. Especially, the computation of Bayes factors, the most natural tools of model selection, has to be adapted when prior measures are not integrable (improper) ; preceding new approaches as the one proposed in [180], techniques like Intrinsic Bayes Factors [32] and Fractional Bayes Factors [242] are the most known of such methodologies, making use of such samples. They however needs to pick up real observations in an observed *training* dataset [33], in a spirit comparable to pre-trained classification in machine learning tasks.

Posterior priors build as (14) present some advantages in subjective Bayesian analysis, as explained in [330, 321].

1. They are uniquely defined from π^N and a likelihood choice, which offers a way of building more interpretable classes for sensitivity analysis. Such classes could be defined by modifying the size m, the neutral prior π^N or/and the likelihood. This could enable us to study in greater detail and counteract the brittleness observed in usual Bayesian robust analysis even with structured classes of priors [248].
2. Assuming that $X, \tilde{X} \sim f(x|\theta)$, the correlation structure between the components of θ is automatically assessed through the Bayes rule, coherently with the sampling model. In fits with the idea that an informative prior is not given the sense of a error model on $f(x|\theta)$, but should agree with the information carried by the sampling model [10]. In some sense, the fact that prior information is constrained by the algebraic structure of $f(x|\theta)$ carries out a certain *rationality in probabilistic encoding*.
3. This approach offers a reasonable solution to the problem of *prior aggregation* [84, 133, 62]: if several independent priors are made equal to posteriors, the aggregated prior can be build by successive Bayes rules and is similar both to the posterior of the consensus virtual sample (being an aggregation of all virtual data) and to the logarithmic pooling of all informative priors:

$$\tilde{\pi}(\theta) \quad \propto \quad \left(\prod_{i=1}^{M} f(\tilde{\mathbf{x}}_{\mathbf{m_i}}|\theta) \right) \pi^N(\theta). \tag{15}$$

which avoids the well-known *paradox of externality* [143] related to linear pooling (but not a marginalization paradox). Situations where prior pieces of information (each providing a sample $\tilde{\mathbf{x}}_{\mathbf{m_i}}$) can be assessed but are dependent were examined by [7], between others, using a *supra-Bayes* pooling approach [136] [11]. Note

10. This principle is at the basis of the huge work on prior-data conflict, evoked hereinbefore.

11. Other possible approaches could intuitively be based on composite likelihoods or copula-based techniques. A large literature is more generally dedicated to this kind of problem in the field of composite modeling and aggregation of predictors, adressing the problem of allocating weights to prior distributions or predictors on the basis of equilibrium rules.

that extensions of rule (15), including weights relativizing the importance of these pieces of information, were defined as *power prior distributions* by [160] in regression problems, by *fractional posteriors* by Bhattacharya [37] and as a *generalized Bayesian approach* by Grumwald [150]. They are based on reusing methodological arguments pointed out by O'Hagan [242] to define fractional Bayes factors in Bayesian selection. Priors defined by (14) and (15), where the baseline prior $\pi^N(\theta)$ is not an integrable measure (improper prior), can be extended by considering prior forms like

$$\tilde{\pi}(\theta) \quad \propto \quad \left(\prod_{i=1}^{M} \{L(\tilde{\mathbf{x}}_{\mathbf{m_i}}|\theta)\}^{\omega_i} \right) \pi^N(\theta) \tag{16}$$

where the "flattening" weight ω_i is defined by

$$\omega_i \quad = \quad \frac{\tilde{m}_i}{m_i},$$

m_i being the *minimal training sample* size [33], such that each marginal posterior $\tilde{\pi}_i(\theta) \propto L(\tilde{\mathbf{x}}_{\mathbf{m_i}}|\theta)\pi(\theta)$ be proper, and $\tilde{m}_i$ having the sense of the equivalent iid sample size of each piece of prior information. This hyperparameter allows to modulate the choice of m_i, which can be constrained to a minimal value by the algebraic properties of the corresponding marginal likelihood. In the case where $M = 1$, the choice $\omega_i = \omega = 1/m_i = 1/m$ should intuitively lead to a prior yielding as much information as one single imaginary data, assuming data $\tilde{\mathbf{x}}_{\mathbf{m}}$ are iid.

4. Modeling choice (14) naturally accords with the principle of conjugate priors for the exponential family. In such cases the meaning of prior hyperparameters is clear, among other interesting properties highlighted in particular by Diaconis and Ylvisaker. They can be expressed as sizes and mean of virtual data samples, that summarizes the "strength" and the quantitative value of prior information, which can be directly compared to those characterizing real data (see for instance [264], § 5.3.3, and [278]). More formally, assume that $f(x|\theta)$ belongs to the natural exponential family (NEF). It can be written as (up to a given parametrization)

$$f(x|\theta) \quad = \quad h(x) \exp\left\{\theta^T x - \psi(\theta)\right\}$$

with $\mathbb{E}_\theta[X] = \nabla\psi(\theta)$ and $\text{cov}(X_i, X_j) = \frac{\partial^2\psi}{\partial\theta_i\partial\theta_j}(\theta)$. Assume now the conjugate prior form

$$\pi(\theta|x_0, m) \quad \propto \quad \exp\{\theta \cdot m x_0 - m\psi(\theta)\} \tag{17}$$

Then the prior predictive mean is

$$\mathbb{E}[X] \quad = \quad \mathbb{E}[\mathbb{E}[X|\theta]] \quad = \quad \mathbb{E}\left[\nabla\psi(\theta)\right] \quad = \quad x_0$$

and from [94] the posterior predictive mean, given an observed sample $\mathbf{x_n} = (x_1, \ldots, x_n)$, is

$$\mathbb{E}[X|\mathbf{x_n}] \quad = \quad \frac{m x_0 + n\bar{x}_n}{m + n}. \tag{18}$$

Then x_0 as the sense of the mean $\bar{\tilde{x}}_m$ of the imaginary sample of size m. The Diaconis-Ylvisaker theorem demonstrates that (18) implies (17) if the underlying dominating measure is Lebesgue-continuous. An unusual example is given by the (little known) conjugate prior found by [57] for the Gumbel distribution $P(X < x|\theta) = \exp\left\{-\exp\left(-\frac{x-\mu}{\sigma}\right)\right\}$ (where $\sigma > 0$):

$$\pi(\mu, \sigma) \quad \propto \quad \sigma^{-m} \exp\left(m\frac{(\mu - \bar{\tilde{x}}_m)}{\sigma} - \sum_{i=1}^{m} \exp\left\{-\frac{\tilde{x}_i - \mu}{\sigma}\right\}\right) \tag{19}$$

which is proper provided $m \geqslant 3$, and where hyperparameters $(m, \bar{\tilde{x}}_m, \tilde{x}_1, \ldots, \tilde{x}_m)$ correspond respectively to the size of a prior virtual sample, its mean and the virtual data themselves.

5. Finally, we can suspect (and investigations in this sense take part in a research program I recently proposed in [321]) that the $q-$vague convergence constraint towards appropriate noninformative priors (see Definition 1) be automatically respected by an APP rule. This was already proved for Jeffreys Conjugate Priors [39] (see besides Proposition 2 further in the text).

Since the conjugate models, the all for which the algebraic structure can be completely explained from rule (14), appear rarely, modelers should in practice build informative priors as follows (at least). Choose a convenient form (e.g. Gaussian) and define the *Prior Equivalent Sample Size* (PESS) m assessing the inferential subjective weight of $\pi(\theta)$. This line of research, based on works made by [234, 224, 205], has been studied by numerous authors (see a thorough review in [272]), although there is no consensus yet on a statistical estimator. Roughly speaking, the first approach is to define the PESS as an intrinsic feature of $\pi(\theta)$, independent of any posterior quantity of interest, by minimizing a discrepancy between $\pi(\theta)$ and $\pi^N(\theta|\tilde{x}_1, \ldots, \tilde{x}_m)$. Morita and his co-authors are the proponents of this approach [224, 225]. Alternatively, [236] and [272] propose to define the PESS as a possibly negative quantity, summarizing both the strength of prior information and its agreement with the objective information provided by observations $x_1, \ldots, x_n$. The asymptotic (i.e., frequentist) limitations of their approaches have been recently be relaxed by [177].

Choosing the form (algebraic structure) of $\pi(\theta)$ remains today challenging, as well as the calibration itself, which depends on the statistical interpretation of prior information. Indeed, the choice of the algebraic structure must remain interpretable to improve the defense of Bayesian choice in applications. In different works [314, 330, 318, 319, 321], I have proposed to produce approximations of the idealized choice (14) to define this structure, for some nonconjugate situations. They are called *Approximate Posterior Priors* (APP) in the following. Basically, the methodological program that I have undertaken to study can be summarized as follows, each step being discussed below:

---

1. Provide a statistical translation χ of available prior information.

2. Choose a neutral prior $\pi^N(\theta)$ for $X|\theta \sim f(x|\theta)$ producing a proper posterior given $m \geqslant m_0$ observations.
3. Given $m \geqslant m_0$ and an iid imaginary sample $\tilde{\mathbf{x}}_\mathbf{m} = (\tilde{x}_1, \ldots, \tilde{x}_m)$ such that $\tilde{\mathbf{x}}_\mathbf{m}|\theta \sim f(x|\theta)$, define a proper posterior $\pi^N(\theta|s(m, \tilde{\mathbf{x}}_\mathbf{m}))$.
4. Select an algebraic prior structure $\pi(\theta|\omega(m,\eta))$ that approximates $\pi^N(\theta|s(m, \tilde{\mathbf{x}}_\mathbf{m}))$.
5. Produce a calibration rule for η conditional to the PESS m:

$$\omega(m, \eta^*(m,\chi)) \quad = \quad \arg\min_{\omega(m,\eta)} \mathcal{D}\left(\pi(\theta|\omega(m,\eta)) \; ; \; \chi\right)$$

6. Produce a specific calibration rule for m.

As stated before, solutions for **Step 1** are provided in an abundant literature (e.g., [244]). In my works, I mainly considered with Garwaithe, Kadane and others [133] that the statistical translation of human predictors ("experts") knowledge should be mainly conducted in terms of *prior predictive percentiles* on an univariate X (these experts expressing mainly their opinion on observable, anchor variables) [12]. Deciding that a prior couple (x_α, α) is such that

$$P\left(X \leqslant x_\alpha\right) \quad = \quad \int_\Theta P\left(X \leqslant x_\alpha|\theta\right) \pi(\theta) \; d\theta \; = \; \alpha \tag{20}$$

is based, following the decisional principles of Bayesian analysis [277] [13], on interpreting α as the ratio $\alpha = c_1/c_1 + c_2$ where $\mathbb{E}[c_1|x_\alpha - X|]$ and $\mathbb{E}[c_2|x_\alpha - X|]$ are the expected first-order costs due to underestimate (*resp.* overestimate) the unknown value of X with x_α. The expression of α can be modified to incorporate the costs of erroneous under/overestimations, following the Bayesian classification principles.

Another usual interpretation of χ can rely on using legacy data rather than expert opinions. This frequently occurs, for instance, in extreme value analysis [319]. In such settings, these (extreme) legacy data may be interpreted as *censoring values* for $X|\theta \sim f(x|\theta)$, which can lead to conduct an objective posterior inference based on an extended likelihood and a prior $\pi^N(\theta)$ accouting for this particular structure (e.g., [288]).

Considering **Step 2**, many works are dedicated too to choose $\pi^N(\theta)$ (e.g., [64]). Obviously, $m_0 \geqslant d = \dim \theta$ but this does not necessary lead the final parametric structure of $\pi(\theta|\omega(m,\eta))$ to be constrained by $m \geqslant m_0$. For instance by assuming m on the half-line

12. In [321], I suggest to be less categorical about *qualitative* prior choices, for instance on the fact that covariates are influent or not in regression analysis

13. Note that the underlying interpretation of the predictor providing a prior couple (probably after some elicitation steps [244]) as a *risk-neutral minimizer* of an expected cost/loss function, as well as the choice (even approximate) of this cost/loss function, independently of the experimental context, appear as somewhat arbitrary, although they often lead to good practical solutions.

of positive reals, the prior information may be given the informational weight of a fraction of information brought by a real data, provided $\pi(\theta|\omega(m,\eta^*(m,\chi))$ remain proper.

Step 4 is articulated around several ideas.

- Assume that $\pi^N(\theta|\tilde{\mathbf{x}}_\mathbf{m}) = \pi^N(\theta|s(\tilde{\mathbf{x}}_\mathbf{m}))$ where $s(\tilde{\mathbf{x}}_\mathbf{m})$ is a sufficient set of statistics; this occurs in NEF-conjugate situations, where the algebraic structure of $\pi(\theta|\omega(m,\eta))$ can be easily found. In a natural way, these unknown statistics play the role of hyperparameters $\omega(m,\eta)$ to calibrate. A illustrative example involving a Gaussian model with conjugate prior was recently discussed in [6].
- In other situations and in multidimensional cases, it can be profitable to search for an *ordered structure* of $\theta \sim \pi^N(\theta|\tilde{\mathbf{x}}_\mathbf{m})$ involving parametric, non sufficient statistics:

$$\pi^N(\theta|\tilde{\mathbf{x}}_\mathbf{m}) \quad = \quad \pi_1(\theta_1|s_1(\tilde{\mathbf{x}}_\mathbf{m},\boldsymbol{\theta}_{>1}))\pi(\theta_2|s_2(\tilde{\mathbf{x}}_\mathbf{m},\boldsymbol{\theta}_{>2}))\ldots\pi(\theta_d|s_d(\tilde{\mathbf{x}}_\mathbf{m},\theta_d)) \quad (21)$$

where $\boldsymbol{\theta}_{>i} = (\theta_{i+1},\ldots,\theta_d)$. Approximating $\pi^N(\theta|\tilde{\mathbf{x}}_\mathbf{m})$ may rely on approximating the statistics $s_i(\tilde{\mathbf{x}}_\mathbf{m},\boldsymbol{\theta}_{>i})$ by (hyper)parametric functions involving some hyperparameters homogeneous to the observations, that have prior predictive sense (see Example 1). Another close approach could be based on a *Gibbs sampling* decomposition of $\pi^N(\theta|\tilde{\mathbf{x}}_\mathbf{m})$, ie.:

$$\{\theta_i|\boldsymbol{\theta}_{-i},\tilde{\mathbf{x}}_\mathbf{m} \quad \sim \quad \pi_i(\theta_i|s_i(\tilde{\mathbf{x}}_\mathbf{m},\boldsymbol{\theta}_{-i}))\}_{1\leqslant i\leqslant d}$$

where $\boldsymbol{\theta}_{-i} = (\ldots,\theta_{i-1},\theta_{i+1},\ldots)$. This would require to verify that these modified conditional distributions (replacing the $s_i(\tilde{\mathbf{x}}_\mathbf{m},\boldsymbol{\theta}_{-i})$ by parametric functions) are *compatible* in the sense proposed by [58] and [231], i.e. the underlying prior $\pi(\theta)$ is proper and unique. Else, $\pi(\theta)$ could be described as a limit distribution of a "Potentially Incompatible Gibbs Sampler" [58].

Note that $\pi^N(\theta|\tilde{\mathbf{x}}_\mathbf{m})$ is not necessarily completely explicit (see Proposition 4 for an example). This does not forbid prior and posterior sampling (e.g., by approximation and numerical methods [351, 235]). Conceptually, the approach which consists in defining $\pi(\theta)$ as the limit distribution of a simulation algorithm is close to generative tools currently used in machine learning (e.g., [151]). However, obtaining explicit forms for the prior $\pi(\theta|m,\omega(m))$ can be useful to conduct several important tasks with limited computational effort, as calibration tasks, the sensitivity analyses or the review of possible prior-data conflicts mentioned above, or improving *prior compatibility* to avoid biased model comparison in the senses considered by [83, 283, 52, 65, 257].

EXAMPLE 1. **(APP for the Weibull model ; B. [314])** *Considering a two-dimensional (nonconjugate) Weibull model pdf*

$$f(x|\mu,\beta) \quad = \quad \mu x^{\beta-1}\exp\left(-\mu x^\beta\right)\mathbf{1}_{\{x\geqslant 0\}}$$

and its corresponding Berger-Bernardo reference prior $\pi^N(\mu, \beta)$, an example of the previous approach was proposed in [314] and reused in [109]:

$$\begin{aligned} \mu|\beta, \tilde{\mathbf{x}_\mathbf{m}} &\sim \mathcal{G}\left(m, s_1(\tilde{\mathbf{x}}_\mathbf{m}, \beta)\right), \\ \pi^N(\beta|\tilde{\mathbf{x}_\mathbf{m}}) &\propto \frac{\beta^{m-1}}{s_1^m(\tilde{\mathbf{x}}_\mathbf{m}, \beta)} \exp\left(m \frac{\beta}{s_2(\tilde{\mathbf{x}}_\mathbf{m}, \beta)}\right) \end{aligned}$$

where $s_1(\tilde{\mathbf{x}}_\mathbf{m}, \beta) = \sum_{i=1}^m \tilde{x}^\beta$, $s_2(\tilde{\mathbf{x}}_\mathbf{m}, \beta) = (\sum_{i=1}^m \log \tilde{x}_i)^{-1}$ and $\mathcal{G}(a, b)$ denotes the gamma distribution with mean a/b and shape parameter a. Replacing $s_1(\tilde{\mathbf{x}}_\mathbf{m}, \beta)$ by $\omega_1(m, \eta) = ((1-\alpha)^{-1/m} - 1)x_\alpha^\beta$ allows to respect exactly the prior predictive specification (20) independently of m and $s_2(\tilde{\mathbf{x}}_\mathbf{m}, \beta)$. Then $\pi^N(\beta|\tilde{\mathbf{x}_\mathbf{m}})$ can be approximated by a gamma distribution.

PROPOSITION 2. **Q-vague convergence for a Weibull prior (B. [321])**.
Consider the Weibull prior defined in Example 1. Then it converges $q-$vaguely when $1/m \to \infty$ towards the Berger-Bernardo reference prior π^N.

Choosing formally such functions, based on the amount of total prior information available and the quality of approximation with $\pi^N(\theta|\tilde{\mathbf{x}}_\mathbf{m})$, remains in general an unsolved problem today. Methodologies for selecting and qualifying the relevance of an algebraic form $\pi(\theta|\omega(m, \eta))$ are to be defined, and it is likely that they should ideally rely on expansions of $\pi^N(\theta|\tilde{\mathbf{x}}_\mathbf{m})$ (e.g., Taylor, Edgeworth, kernel-based).

The philosophy of **Step 5** is to estimate $\pi(\theta|\omega(m, \eta))$, for a fixed m, such that the prior respect *as best as possible* the statistical specification χ of prior information. Obviously the *matching criterion* $\mathcal{D}$ should depend on the nature of χ. For instance, [87] considers classes of central moments in a machine learning context. When χ is established as a set of K prior predictive percentiles (x_{α_k}, α_k) linked by Equation (20), with $0 = \alpha_0 < \alpha_1 < \ldots < \alpha_K < \alpha_{K+1} = 1$, a choice made in the contributions beneath is to minimize a discretized Kullback-Leibler (KL) criterion originally advocated by Cooke [67] for estimating the accuracy of expert information:

$$\mathcal{D}\left(\pi(\theta|\omega(m, \eta)) \; ; \; \chi\right) = \sum_{k=0}^{K} (\alpha_{k+1} - \alpha_k) \log \frac{\alpha_{k+1} - \alpha_k}{\alpha_{k+1}(\omega(m, \eta)) - \alpha_k(\omega(m, \eta))} \quad (22)$$

where $\alpha_k(\omega(m, \eta))$ is the effective prior predictive quantile order associated to x_{α_k}. Since the Wasserstein distances are able to compare distributions that are not absolutely continuous with respect to each other, this second choice could be examined, as well as the conditions required to get a unique, optimal solution. Summaries of calibration relevance are based on explaining the difference between the elicited percentiles and the effective percentiles, due to the limited flexibility of the whole Bayesian model. More generally, tools for prior predictive checking [128] have now become common for carrying out Bayesian workflows [289, 87].

Then, the PESS m, as a key indicator of the "strength" granted to subjective assessments and certainly one the most meaningful hyperparameters, could deserve a specific calibration (**Step 6**), according to the situation. It is often recommended that the statistician be able to modulate the subjective information such that the behavior of the posterior be more dependent on observed data information, and m appears as a lever for this task, mostly. For instance, a diffuse hierarchical prior on m/n (where n is the size of truly observed data) was proposed by [52] in a regression context involving Zellner's $g-$prior. See [129] for more details on the interpretability of this prior. In [329], I proposed to calibrate m as the minimal size in a bounded domain such that a prior-data conflict criterion be reached. A default choice is fixing $m = m_0$ while, to be coherent with the ideas expressed in the SHELF methodology of prior elicitation based on histogram-type methods [149], one may argue that providing a given number of prior percentiles should be funded on a minimal number m of imaginary data.

As it can be seen, this methodology remains strongly idealized, and most of these points still lack justified formal rules, in particular concerning approximation and validation, which constitutes a research program for the future. Reaching high dimensions and hierarchical structures will certainly present non-trivial difficulties. My contributions have focused until now on case-by-case approaches which provide good results in applications, allowing to illustrate the different points of this idealized methodology and to highlight some difficulties. I present two of them below. Beyond, I am aware that the choice of a certain rigidity conferred to the prior is against the current of widespread approaches in the Bayesian community, which aims precisely at making priors more flexible, and that the proposed choice may seem very reductive. However, it seeks to propose a trade-off between flexibility and explicability, with the objective of encouraging an increased use of Bayesian statistics in applications, and avoiding too much versatility in posterior results, which can be feared from practice and on the basis of strong theoretical arguments [248].

3.2.2.1 Gamma processes. Stochastic Gamma Processes (SGP) are a particular class of stochastic models widely used in reliability and maintenance [4, 343, 342]. Such processes can be efficiently used to describe continuous natural phenomena as erosion, corrosion and concrete creep [50, 66, 123]. Consider a set of M identical components submitted to intensive effort. It is assumed that the deterioration of a individual component $k \in \{1, \dots, M\}$ is characterized by a positive measurement (e.g., crack size) $X_{k,t}$, observed at inspection times $t \in \{t_{1,k}, \dots, t_{n_k,k}\}$ (to alleviate notations let us remove the second index k for the inspection times). $X_{k,t}$ is therefore positive and monotonously increasing through time. $X_{k,t}$ is assumed to be a SGP with following features:

- for all k, $X_{k,0} = 0$ and $(X_{k,t})_{t \geqslant 0}$ has independent increments $Z_{k,i} = X_{k,t_i} - X_{k,t_{i-1}}$;
- for all k and $0 \leqslant s \leqslant t$, the increment $X_{k,t} - X_{k,s}$, follows a gamma probability distribution function with density function

$$f_{\alpha(t-s),\beta}(x) = \frac{1}{\Gamma(\alpha_i(t-s))} \cdot \frac{x^{\alpha(t-s)-1} e^{-\frac{x}{\beta}}}{\beta^{\alpha(t-s)}} \mathbb{1}_{\{x \geqslant 0\}}. \tag{23}$$

In [330], a rationale is provided to choose $\pi^N(\theta)$ as the Jeffreys prior for a standard Gamma distribution. From [360], denoting $\Psi_1(.)$ the trigamma function:

$$\pi^N(\theta) \;=\; \pi^N(\alpha,\beta) \;\propto\; \frac{1}{\beta}\sqrt{\alpha\Psi_1(\alpha)-1}$$

PROPOSITION 3. **APP for Gamma processes (B.** *et al.* **[330])**.
Recall that $\mathcal{G}(a,b)$ denotes the gamma distribution with mean a/b and shape parameter a, and denote $\mathcal{IG}(a,b)$ the corresponding inverse gamma distribution. Consider an iid imaginary sample of SGP increments $\tilde{\mathbf{z}}_\mathbf{m}=(\tilde{z}_1,\ldots,\tilde{z}_m)$ observed at times $\tilde{\mathbf{t}}_\mathbf{m}=(\tilde{t}_1,\ldots,\tilde{t}_m)$. Consider the following statistics:

$$\tilde{t}_{e,1}=\frac{1}{m}\sum_{i=1}^{m}\tilde{t}_i,\qquad \tilde{z}_e=\tfrac{1}{m}\sum_{i=1}^{m}\tilde{z}_i,\qquad \tilde{t}_{e,2}=\frac{1}{m}\sum_{i=1}^{m}\tilde{t}_i\log\frac{\sum_{j=1}^{m}\tilde{z}_j/\tilde{z}_i}{\sum_{j=1}^{m}\tilde{t}_j/\tilde{t}_i}.$$

Then the inverse gamma-gamma mixture

$$\beta|\alpha \;\sim\; \mathcal{IG}\left(\alpha m\tilde{t}_{e,1}, m\tilde{z}_e\right),\qquad \alpha \;\sim\; \mathcal{G}\left(m/2, m\tilde{t}_{e,2}\right)$$

is a first-order approximation of the posterior prior $\pi(\theta)=\pi^N(\theta|\tilde{\mathbf{z}}_\mathbf{m},\tilde{\mathbf{t}}_\mathbf{m})$. Consequently, given a truly $n-$observed sample $\mathbf{d}$ (mixing possibly right-censored, left-censored and correct data), $\pi(\beta|\alpha)$ is conditionally conjugate:

$$\beta|\alpha,\mathbf{d} \;\sim\; \mathcal{IG}\left(\alpha\left\{m\tilde{t}_{e,1}+n\delta_1(\boldsymbol{d})\right\},\left\{m\tilde{z}_e+n\delta_2(\boldsymbol{d})\right\}\right),$$

where $\{\delta_1(\boldsymbol{d}),\delta_2(\boldsymbol{d})\}$ are explicit statistics of $\mathbf{d}$ homogeneous to $\{t,z\}$, respectively.

The APP above is convenient since the meaning of hyperparameters $\tilde{t}_{e,1}$ and $\tilde{z}_e$ is straightforward for the practitioner: $\tilde{t}_{e,1}$ is an average time of observation and $\tilde{z}_e$ is an average crack increase, estimated from an imaginary sample of size m. Furthermore, from Proposition 1 in [330], the most probable value of the mean crack increasing during a time interval Δ_i, an anchor variable for experts, is

$$\hat{r}(\Delta_i) \;=\; \frac{\tilde{z}_e\Delta_i}{\tilde{t}_{e,1}}$$

which is independent of $\tilde{t}_{e,2}$. This later hyperparameter, less easily connected to the speed of cracking, can thus be seen as a "tuning" parameter, helping to get a flexible prior in the limits allowed by assuming that the prior information remains coherent with the sampling model.

3.2.2.2 Extreme value distributions. The models of block maxima or minima, of great importance in insurance, finance and modeling natural hazards [107, 322], are usually chosen amongst the Fréchet (Type 3), Reverse Weibull (Type 3) and Gumbel (Type

1) distributions according to the Fisher-Tipett-Gnedenko asymptotic theorem. Most often, they are generally studied through the encompassing generalized extreme value distribution (GEV). The estimation of its shape parameter ξ, that strongly governs the tail behavior and the calculation of return levels, is essential as it allows to differentiate these particular behaviors. An informative Bayesian framework is particularly adapted to this aim [63] (chapter 9). However, in order to avoid situations where a credibility domain of ξ can contain negative and positive values, resulting in very different, incompatible behaviors (e.g., for physical reasons when adressing the problems raised by natural hazards), I have proposed in [319] to consider separately the Fréchet, Reverse Weibull and Gumbel distributions in order to select a most relevant *basin of attraction*[14]. While the case of the Gumbel distribution has been mentioned above, the Fréchet $\mathcal{F}$ and Reverse Weibull $\mathcal{RW}$ distributions

$$\begin{aligned}
\mathcal{F}: \quad P(X<x|\theta_{\mathcal{F}}) &= \exp\left\{-\left(\frac{x-\mu_{\mathcal{F}}}{\sigma_{\mathcal{F}}}\right)^{-1/\xi_{\mathcal{F}}}\right\} \\
&\quad \text{with } \sigma_{\mathcal{F}}>0,\ \xi_{\mathcal{F}}>0,\ \mu_{\mathcal{F}}\in\mathbb{R} \text{ and } x\geqslant\mu_{\mathcal{F}}, \\
\mathcal{RW}: \quad P(X<x|\theta_{\mathcal{RW}}) &= \exp\left\{-\left(\frac{\mu_{\mathcal{RW}}-x}{\sigma_{\mathcal{RW}}}\right)^{1/\xi_{\mathcal{RW}}}\right\} \\
&\quad \text{with } \sigma_{\mathcal{RW}}>0,\ \xi_{\mathcal{RW}}>0,\ \mu_{\mathcal{RW}}\in\mathbb{R} \text{ and } x\leqslant\mu_{\mathcal{RW}},
\end{aligned}$$

dot not have conjugate priors.

The interpretability of an APP relies on a notion of *balance between the information brought by the m imaginary data and n truly observed data*. The following proposition provides such a result for the Fréchet distribution. A symmetric one can be easily produced for the Reverse Weibull distribution, since $X\sim\mathcal{F}(\theta_{\mathcal{F}})\Leftrightarrow Y=\mu_{\mathcal{RW}}+(X-\mu_{\mathcal{F}})^{-1}\sim\mathcal{RW}(\theta_{\mathcal{RW}})$ with $\sigma_{\mathcal{RW}}=\sigma_{\mathcal{F}}$ and $\xi_{\mathcal{RW}}=\xi_{\mathcal{F}}$.

PROPOSITION 4. **APP for the Fréchet distribution (B. *et al.* [319]).** *Reparametrize the distribution with $\theta=(\mu,\nu,\xi)$ with $\nu=\sigma^{1/\xi}>0$. Let $(x_{e,0},x_{e,1},x_{e,2})\in\mathbb{R}^3$ such that $x_{e_0}<x_{e_1}<x_{e_2}$. Define the prior*

$$\pi_{\mathcal{F}}: \quad \left\{\begin{array}{lcll} \nu|\mu,\xi & \sim & \mathcal{G}\left(m,s_1(\mu,\xi)\right) & \text{with } s_1(\mu,\xi)=m(x_{e_1}-\mu)^{-1/\xi} \\ \xi|\mu & \sim & \mathcal{IG}\left(m,s_2(\mu)\right) & \text{with } s_2(\mu)=m\log\left(\frac{x_{e_2}-\mu}{x_{e_1}-\mu}\right) \\ \pi(\mu) & \propto & \dfrac{\mathbf{1}_{\{x_{e,0}\leqslant\mu\leqslant x_{e_1}\}}}{(x_{e_2}-\mu)^m s_2^m(\mu)} & \end{array}\right. ,$$

Then $\pi_{\mathcal{F}}(\nu,\mu,\xi)$ is proper for any $m>0$ and is conjugated for ν given (μ,ξ). Furthermore, given a iid Fréchet sample $\mathbf{x_n}=(x_1,\ldots,x_n)$, the posterior is proper and can

14. In a spirit similar to [250] who considered the *Peaks Over Threshold* framework, the theoretical alternative to the block maxima framework.

be described by conditional distributions which are symmetrically balanced between the imaginary and virtual data ; denoting $\tilde{\bar{x}}(\mu) = \mu + \prod_{i=1}^{n}(x_i-\mu)^{1/n}$*, it comes :*

$$\begin{aligned}
\nu|\mu,\xi,\mathbf{x_n} &\sim \mathcal{G}\left(m+n, s_1(\mu,\xi) + \sum_{i=1}^{n}(x_i-\mu)^{-1/\xi}\right),\\
\pi(\xi|\mu,\mathbf{x_n}) &\propto \frac{\xi^{-n-m-1}\exp\left\{-\frac{1}{\xi}\left(m\log(x_{e_2}-\mu)+n\log(\tilde{\bar{x}}(\mu)-\mu)\right)\right\}}{\left(\sum_{i=1}^{m}(x_{e_1}-\mu)^{-1/\xi}+\sum_{i=1}^{n}(x_i-\mu)^{-1/\xi}\right)^{m+n}},\\
\pi(\mu|\nu,\xi,\mathbf{x_n}) &\propto (\tilde{\bar{x}}(\mu)-\mu)^{-n}(x_{e_2}-\mu)^{-m}\left\{\exp\left(-\nu\left[\sum_{i=1}^{m}(x_{e_1}-\mu)^{-1/\xi}+\sum_{k=1}^{m}(x_k-\mu)^{-1/\xi}\right]\right)\right.\\
&\qquad \left. -\exp\left(\frac{1}{\xi}\left[m\log(x_{e_2}-\mu)+n\log(\tilde{\bar{x}}(\mu)-\mu)\right]\right)\right\}
\end{aligned}$$

As can be noticed, this type of approach leads to new prior forms, hyperparametrized by two values belonging to the range of observations, which have been shown to be easily calibrated by rule (22) [319] to fit with available prior information (a small number of prior predictive percentiles). However, they remain imperfect, because a fully-balanced posterior would imply that the hyperparameters $(x_{e,1}, x_{e,2})$ depend on (μ,ξ) and μ, respectively. Indeed, formally speaking they have the sense of a shifted inverse arithmetic sum and shifted geometric mean of the imaginary data:

$$x_{e_1}(\mu,\xi) = \mu + \left(\sum_{i=1}^{m}(\tilde{x}_i-\mu)^{-1/\xi}\right)^{-\xi} \tag{24}$$

$$x_{e_2}(\mu) = \mu + \prod_{i=1}^{m}(\tilde{x}_i-\mu)^{1/m}. \tag{25}$$

In this case $\pi(\nu,\mu,\xi) = \pi^N(\nu,\mu,\xi|\tilde{\mathbf{x}}_\mathbf{m})$ where $\pi^N(\mu,\nu,\xi)\propto(\nu\xi)^{-1}\mathbf{1}_{\mu\geqslant x_{0,1}}$ is the Berger-Bernardo reference prior provided by [3] (truncated on μ). Unfortunately, using frequentist arguments to find pivotal quantities and to increase artificially the prior dimensionality, in the spirit of [301, 300] – for instance based on the fact that

$$(x_{e_1}(\mu,\xi)-\mu)^{\xi}\,|\nu,\xi,\mu \sim \mathcal{G}(m,1/\nu)$$

would lead to try establishing a hierarchical prior form requiring higher levels of prior information on the unknown parameters than percentiles on observed quantities. Such an approach seems too idealistic and not usable in practice, except in the situation where *expected posterior priors* can be build (see § 3.2.2.3). Providing a better APP while keeping the above decomposition valid could rather be based on replacing $x_{e,1}$ and x_{e_2} by simple (hyper)parametrized functions of μ, approximating (24-25). Prior predictive features, as the fact that

$$P_\pi(X\leqslant x_{e,1}) = \mathbb{E}_\pi[P(X<x_{e,1}|\theta] = 1/(1+1/m)^m$$

(which is the median when the PESS $m = 1$), should besides be used to justify fixing a certain degree of rigidity inside the prior structure. Principles coming from PC priors could be an interesting alternative to study.

3.2.2.3 Towards expected posterior priors. Avoiding prior predictive specification can make sense for some applications where, typically, a prior point estimate θ_e or more generally a sampler $g(\cdot)$ of X is available (for instance based on a historical calculation). This section introduces the following idea (and an original result) as an additional way to be explored in the context of prior modeling. The *expected posterior prior* (EPP) under g

$$\pi(\theta|m, g) \quad = \quad \int_{\chi^m} \pi^N(\theta|\tilde{x}_1, \ldots, \tilde{x}_m) \prod_{i=1}^{m} g(\tilde{x}_i)\ d\tilde{x}_i \tag{26}$$

defines a formal approach to elicit priors that pool the knowledge of the predictor and the knowledge of the sampling model. EPP were originally defined by Pérez and Berger [260] and generalized by Fouskakis *et al.* [121, 122], under the name *power expected priors*, in the specific context of Bayesian model selection involving an improper benchmark prior $\pi^N(\theta)$. Following this idea, the aggregation rule should be extended, under Fubini-Lebesgue conditions, in:

$$\tilde{\pi}(\theta) \quad \propto \quad \left(\prod_{i=1}^{M} \int_{\chi_i^m} \{f(\tilde{\mathbf{x}}_{\mathbf{m_i}}|\theta)\}^{\omega_i} \prod_{j=1}^{m_j} g_i(\tilde{x}_j)\ d\tilde{x}_i \right) \pi^N(\theta). \tag{27}$$

In accordance with the sampling model, a default choice for $g(x)$ is

$$g(x) \quad = \quad f(x|\theta_e) \tag{28}$$

Accordingly, $\pi(\theta|m, g) = \pi(\theta|m, \theta_e)$. Assumption (28) reflects how a "perfect" predictor could say about X (in the sense "perfectly in accordance with the sampling model"). For human predictors the assessment of θ_e reconciles the marginal approach to prior elicitation with the numerous elicitation techniques that try to connect prior knowledge with point estimates (see for instance [319] for a review in the specific context of extreme value models). Predictors defined as point estimates of θ based on historical data, usual in the engineering literature, are natural candidates for providing θ_e, especially when θ is a set of location parameters. This appears obvious when studying a simple exponential model.

EXAMPLE 2. *Exponential model.*
Assume to dispose of an estimate x_e of the median lifetime of a device with failure rate λ. Then the prior point estimate of λ is $\lambda_e = x_e^{-1} \log 2$. Using the Jeffreys prior $\pi^N(\lambda) \propto \lambda^{-1}$, then from (26)

$$\pi(\lambda|m, \lambda_e) \quad = \quad m^{m+1} \lambda_e \frac{\lambda^{m-1}}{(\lambda_e + m\lambda)^{m+1}}.$$

This nonconjugate, shifted Pareto distribution provides larger uncertainty about λ than the usual conjugate Gamma prior.

REMARK 1. *Note that (28) differs from the original EPP approach proposed by [260], where $g(x)$ is chosen as the marginal distribution*

$$g_0(x) \quad = \quad \int_\Theta f(x|\theta)\pi_0(\theta)\ d\theta.$$

In this context of Bayesian model selection, $g_0(x)$ is selected amongst the competing models. Such an approach is based on former elicitation of the corresponding competing priors, and cannot be used to the elicitation task itself. Empirical approaches seem besides not realistic, as they would require a large sample of observations of $g(x)$.

Clearly, apart cases involving conjugacy, Definition (26) threatens to lead to untractable prior forms, especially in multidimensional settings. A variational approximation $\pi(\theta|\omega)$ with practical form, parameterized by ω living in a finite-dimensional space Ω, could be searched such that

$$\omega^* \quad = \quad \arg\min_{\omega\in\Omega} \mathrm{KL}\left(\pi(.|m,\theta_e)||\pi(.|\omega)\right) \tag{29}$$

where KL means the Kullback-Leibler divergence (or more generally a Bregman divergence), that offers convexity in its argument $\pi(\theta|\omega)$, ensures identifiability and unicity of ω^* provided $\Omega : \omega \mapsto \pi(.|\omega)$ is injective and twice differentiability in ω. Selecting an algebraic structure $\pi(\theta|\omega)$ (e.g., a mixture) as well as estimating ω^* can stand on mixing samples from $\pi(\theta)$ and stochastic optimization procedures, as usual in numerous machine learning tasks. Strategies for decreasing the high computational time required for such procedures have to be found. We can for example save heavy calculations by deciding to choose π in a class of densities which are both adapted to this type of resolution and yet flexible enough to be flexible enough to be easily simulated. For example, if the prior is bounded, [19] propose to use sequences of exponential families to approximate $\pi(.|\omega)$ in the KL sense. When examining multivariate settings allowing conditioning, such strategies could be based on the following result (proof provided in § 3.2.4).

PROPOSITION 5. *Let $(\theta_1,\theta_2) \in \Theta_1 \times \Theta_2 \subset \Theta$ and $(\omega_1,\omega_2) \in \Omega_1 \times \Omega_2 \subset \Omega$. Assume that $\pi(\theta|\omega) = \pi(\theta_1|\theta_2,\omega_1)\pi(\theta_2|\omega_2)$. Then $\omega^* = (\omega_1^*,\omega_2^*)$ where*

$$\left\{ \begin{array}{rcl} \omega_2^* & = & \arg\min_{\omega_2} \mathcal{D}_2(\omega_2), \\ \omega_1^* & = & \arg\min_{\omega_1} \mathcal{D}_1(\omega_1) \end{array} \right.$$

with

$$\begin{array}{rcl} \mathcal{D}_1(\omega_1) & = & KL\left\{\pi(\theta|\theta_e) \ ||\ \pi(\theta_1|\theta_2,\omega_1)\right\}, \\ \mathcal{D}_2(\omega_2) & = & KL\left\{\pi(\theta_2|\theta_e)||\ \pi(\theta_2|\omega_2)\right\}, \end{array}$$

3.2.3 A last research perspective

By taking a step back, it is essential to place the use of Bayesian statistics and its promise to produce posterior credibility zones in a modern context where statistical models become of very high dimension (typically, neural networks), a context where the notion of confidence introduced by *conformal prediction* is becoming increasingly prevalent in many applications requiring probabilistic forecasting [294, 119].

The success of this approach lies in the low number of assumptions required to produce well-calibrated predictors (in particular, data exchangeability), and the absence of distributional assumptions (e.g., gaussianity). Several applied statisticians have recently pointed out (although it is well known from [82, 93, 232, 81, 124]) the "failure" of Bayesian approaches to produce correct [15] prediction zones when the prior has an informational mismatch with the data (see for instance [292]), although posterior predictive distributions can be considered as conformity measures [118]. A review of recent attempts to "frequentizing" Bayesian approaches and technical advances are made by these authors.

This showdown, that seems to which seems to be a rehash of the historical debate on the meaning of the notion of trust or credibility, inspires to me the following comments:

(a) the way of interpreting prior information (for example via marginal representation) does not presume that a similar interpretation is required a posteriori: for instance, it may be very well founded to use Pareto-like distributions in extreme situations, and to use Bayesian parameter estimators (rather than a Bayesian estimator of the entire distribution). The choice of a posterior cost function is crucial for determining the value of a forecast.

(b) the crucial need to guard against (and therefore detect in advance) prior-data conflicts [329, 111, 112]. Viewing a prior as a posterior of virtual data requires us to assume that these virtual data constitute a "missing" sub-sample, and that the data actually observed typically suffer from selection bias. As the notion of conformal confidence is based on the positioning of future data (or a future label) relative to past observations, a fair comparison between a classical conformal forecast and its Bayesian counterpart should take into account the fact that a relevant prior should be a legacy data provider. I suspect that the proposed APP/EPP approaches can naturally aim to facilitate this comparison.

3.2.4 Chapter appendix

Proof of Proposition 5. We consider the case where $\pi(\theta|\theta_e)$ is defined by (26). With $\pi^N(\theta) = \pi^N(\theta_1|\theta_2)\pi^N(\theta_2)$, and denoting $\tilde{\mathbf{x}}_\mathbf{m} = \tilde{x}_1, \ldots, \tilde{x}_m$, one has

$$\pi_a^N(\theta|\tilde{x}_1, \ldots, \tilde{x}_m) \quad = \quad \pi_a^N(\theta_1|\tilde{\mathbf{x}}_\mathbf{m})\pi_a^N(\theta_2|\tilde{\mathbf{x}}_\mathbf{m})$$

15. In a frequentist sense.

Denote $H = -\int_\Theta \pi(\theta|\theta_e)\log\pi(\theta|\theta_e)\,d\theta$ the entropy of $\pi(.|\theta_e)$. Denote $\nu_e(\tilde{\mathbf{x}}_\mathbf{m}) = \prod_{i=1}^{q} f(\tilde{x}_i|\theta_e)$. Then

$$\begin{aligned}
\mathcal{D}(\omega) &= -H - \int_{\Theta_2}\int_{\Theta_1}\int_{\chi^q} \pi^N(\theta_1|\theta_2,\tilde{\mathbf{x}}_\mathbf{m})\pi^N(\theta_2|\tilde{\mathbf{x}}_\mathbf{m})\log\{\pi(\theta_1|\theta_2,\omega_1)\pi(\theta_2|\omega_2)\}\ \nu_e(\tilde{\mathbf{x}}_\mathbf{m})d\tilde{\mathbf{x}}_\mathbf{m}\ d\theta_1 d\theta_2,\\
&= -H - \int_{\Theta_2}\int_{\chi^q}\left\{\int_{\Theta_1}\pi^N(\theta_1|\theta_2,\tilde{\mathbf{x}}_\mathbf{m})\log\pi(\theta_1|\theta_2,\omega_1)\ d\theta_1\right\}\pi^N(\theta_2|\tilde{\mathbf{x}}_\mathbf{m})\nu_e(\tilde{\mathbf{x}}_\mathbf{m})\ d\tilde{\mathbf{x}}_\mathbf{m}d\theta_2\\
&\quad - \int_{\Theta_2}\int_{\chi^q}\pi^N(\theta_2|\tilde{\mathbf{x}}_\mathbf{m})\log\pi(\theta_2|\omega_2)\ \nu_e(\tilde{\mathbf{x}}_\mathbf{m})\ d\tilde{\mathbf{x}}_\mathbf{m}d\theta_2,\\
&= \int_{\Theta_2}\int_{\chi^q}\mathcal{D}_1(\omega_1|\tilde{\mathbf{x}}_\mathbf{m},\theta_2)\ \nu_e(\tilde{\mathbf{x}}_\mathbf{m})\pi^N(\theta_2|\tilde{\mathbf{x}}_\mathbf{m})\ d\tilde{\mathbf{x}}_\mathbf{m}d\theta_2\ +\ \mathcal{D}_2(\omega_2) + \quad \Delta
\end{aligned}$$

$$\text{where} \quad \mathcal{D}_1(\omega_1|\tilde{\mathbf{x}}_\mathbf{m},\theta_2) = \mathrm{KL}\left\{\pi^N(\theta_1|\theta_2,\tilde{\mathbf{x}}_\mathbf{m})\ ||\ \pi(\theta_1|\theta_2)\right\}$$

and

$$\begin{aligned}
\Delta &= -H - \int_{\Theta_1}\int_{\Theta_2}\int_{\chi^q}\pi^N(\theta_1|\theta_2,\tilde{\mathbf{x}}_\mathbf{m})\pi^N(\theta_2|\tilde{\mathbf{x}}_\mathbf{m})\log\left\{\pi^N(\theta_1|\theta_2,\tilde{\mathbf{x}}_\mathbf{m})\right\}\ \nu_e(\tilde{\mathbf{x}}_\mathbf{m})d\tilde{\mathbf{x}}_\mathbf{m}\ d\theta_1 d\theta_2\\
&\quad - \int_{\Theta_2}\int_{\chi^q}\pi^N(\theta_2|\tilde{\mathbf{x}}_\mathbf{m})\log\left\{\int_{\chi^q}\pi^N(\theta_2|\tilde{\mathbf{x}}_\mathbf{m})\nu_e(\tilde{\mathbf{x}}_\mathbf{m})\ d\tilde{\mathbf{x}}_\mathbf{m}\right\}\nu_e(\tilde{\mathbf{x}}_\mathbf{m})d\tilde{\mathbf{x}}_\mathbf{m}\ d\theta_2.
\end{aligned}$$

Since Δ does not depend on ω, minimizing $\mathcal{D}(\omega)$ is tantamount to minimize

$$\int_{\Theta_2}\int_{\chi^q}\mathcal{D}_1(\omega_1|\tilde{\mathbf{x}}_\mathbf{m},\theta_2)\ \nu_e(\tilde{\mathbf{x}}_\mathbf{m})\pi^N(\theta_2|\tilde{\mathbf{x}}_\mathbf{m})\ d\tilde{\mathbf{x}}_\mathbf{m}d\theta_2\ +\ \mathcal{D}_2(\omega_2).$$

These two terms are positive and independent. The minimizers are thus

$$\begin{aligned}
\omega_2^* &= \arg\min_{\omega_2\in\Omega_2}\mathcal{D}_2(\omega_2),\\
\omega_1^* &= \arg\min_{\omega_1\in\Omega_1} -\int_{\Theta_2}\int_{\Theta_1}\left\{\int_{\chi^q}\pi^N(\theta_1|\theta_2,\tilde{\mathbf{x}}_\mathbf{m})\pi^N(\theta_2|\tilde{\mathbf{x}}_\mathbf{m})\ \nu_e(\tilde{\mathbf{x}}_\mathbf{m})d\tilde{\mathbf{x}}_\mathbf{m}\right\}\log\pi(\theta_1|\theta_2)\ d\theta_1 d\theta_2
\end{aligned}$$

which is equivalent to the statement of the proposition.

4 Stochastic inversion of numerical models

4.1 Context

One of the key issues related to the increasing use of numerical models requiring uncertainty quantification (UQ) is the need to solve inverse problems. Inverse problems appear in many scientific fields, as process engineering, image processing or financial mathematics [113]. In the broadest sense, such problems consist of reconstructing a signal X from indirect observations Y^*. Many situations consider deterministic linear operators g linking the signal to observations. Despite this linearity, solving these problems remains usually rather difficult and require regularization methods [49]. These difficulties increase when g is nonlinear, Y^* suffers from noise or/and model error ε, and X is stochastic. Such situations frequently occur in the UQ field [144], when X is an unknown or weakly known random input vector in $\mathbb{R}^p$, $g : \mathbb{R}^p \times \mathbb{R}^r \to \mathbb{R}^q$ is a numerical (e.g., so-called *computer*) phenomenological model, and Y^* typically comes from sensor or experimental measurements. A common, simplified formalization of such problems is assuming

$$Y^* \quad = \quad g(X, z) + \varepsilon \tag{30}$$

where $\varepsilon \sim \mathcal{N}_q(0, R)$ is a stochastic error in $\mathbb{R}^q$, with known covariance matrix R, and $z \in \mathbb{R}^r$ is a set of observed (environmental) parameters. Considering X as random and estimating the features of its pdf f_X helps to address the practical concern of accounting for uncertainties surrounding the use of g to perform engineering tasks [68].

Given f_ε, the methods for reconstructing (estimating) f_X based on an observed sample $(\mathbf{y_n^*}, \mathbf{z_n}) = (y_1^*, \ldots, y_n^*; z_1, \ldots, z_n)$ generally take advantage of the missing data structure of the problem, and use the expectation-maximization (EM) principle in a frequentist setting [51, 18], using a likelihood build from data $\mathbf{y_n^*}$ only. See [259] for a recent survey of such algorithms and [89, 273, 193] for details on their theoretical convergence properties. While nonparametric frameworks have been extensively studied in the literature [53, 41, 349, 79], several authors in statistical engineering (e.g. [18, 17]) choose to simplify the problem by assuming a multivariate Gaussian input (possibly at the price of a reparametrization)

$$X \quad \sim \quad \mathcal{N}_p(\mu, \Gamma) \tag{31}$$

and aim to estimate the mean and covariance $(\mu, \Gamma) \in \mathbb{R}^p \times S_p^{++}(\mathbb{R})$, where $S_p^+(\mathbb{R})$ is the space of symmetric, positive-definite matrices. The rationale for this practical choice is that the shape of the distribution of X (or a reparametrization of X) must obey certain plausibility constraints (e.g., the distribution of a physically-meaning variable may be not multimodal [18, 126]), and to avoid possible difficulties related to a lack of model identifiability, in relation with the size n of $\mathbf{y_n^*}$. In order to be verified, the latter can require simplifications of g (e.g., linearization in [51]).

Let us mention three motivating case studies on which I have worked intensively during the last years:

1. The stochastic inversion of the friction parameters X of a free surface hydraulic flow model, contextualized in Figure 5. These friction parameters are unobservable, and the data $(\mathbf{y_n^*}, \mathbf{z_n})$ correspond to measurements of water head and flow (plus topographical features), respectively [126, 127, 324].
2. The stochastic inversion of parameters X describing the thickening behavior of a sick arm fitted with a connected health object (Figure 6), for which we dispose of a simulator and $(\mathbf{y_n^*}, \mathbf{z_n})$ measurements related to arm circumference and contextual [35].
3. The stochastic inversion of the parameters X of an animal growth model from DEB theory (Figure 7), for which the $(\mathbf{y_n^*}, \mathbf{z_n})$ data correspond animal sizes measured through capture-recapture experiments, and contextual measurements [95, 325].

4.2 Problems and personal contributions

4.2.1 Bayesian stochastic inversion with kriging-based meta-modeling

Several difficulties have been progressively overcome in recent years in the particular case of stochastic inversion of a deterministic numerical model. In response to the twofold need to account for various kinds of uncertainties neglected by frequentist procedures, and take advantage of the possible expert knowledge available on many engineering topics, the development of a Bayesian framework for stochastic inversion using Markov Chains Monte Carlo Methods (MCMC) has been conducted recently by different authors [80, 126, 233, 17]. Gibbs samplers, the usual Bayesian counterpart of EM-type algorithms in such missing data problems [336], were highlighted as the most practical tools for problems of moderate dimension.

Especially, my PhD student S. Fu, her co-supervisors and I participated in this research dynamic by advocating in [126] and [127] the use of a general Bayesian prior for the parameter vector (μ, Γ) of the problem (30-31):

$$\mu|\Gamma \sim \mathcal{N}_p\left(\mu_0, \Gamma/\alpha\right), \tag{32}$$

$$\Gamma \sim \mathcal{IW}_p(\Upsilon, \nu). \tag{33}$$

where $\mathcal{IW}$ denotes the inverse Wishart distribution on $S_p^{++}(I\!R)$, and where the prior hyperparameters $(\mu_0, \alpha, \Upsilon, \nu)$ are fixed. On the one hand, the meaning of the latter is straightforward and allows to establish formal rules for prior elicitation [127], and in the other hand this choice obviously offers facilities due to conjugation properties (conditional form stability) in a data augmentation scheme, resulting in a general Gibbs sampler approximating the target posterior distribution $\pi(\mu, \Gamma|y_1^*, \ldots, y_n^*, z_1, \ldots, z)$. Assuming a diagonal R, given initializing values $(\mu^{[0]}, \Gamma^{[0]})$ for the parameters, and initialized values $X^{[0]}$ for the missing variables $X = (X_1, \ldots, X_n)$, it generates the following Markov chain for $k = 0, 1, 2, \ldots$ as follows:

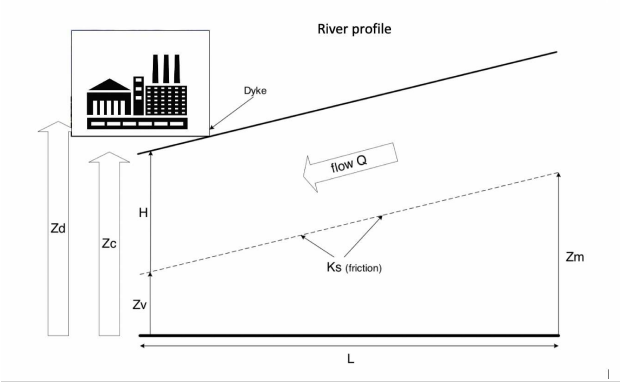


Figure 5 – Scheme of dike dimensioning for the protection of an industrial structure. We observe the downstream water heights ($H = \mathbf{y}_n^*$) and the upstream flows ($Q \subseteq \mathbf{z}_n$). We try to estimate in several locations (ideally, mesh centroids) the distribution of the friction parameters $X = K_s$, for known topological parameters defining an instance of a numerical model of free surface flow.

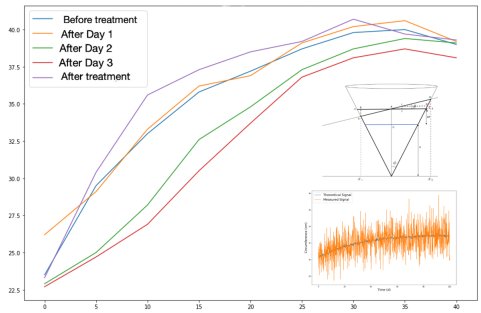


Figure 6 – Measurements of the evolution of arm circumferences affected by lymphedema. Such observations can be simulated by taking into account the measurement noise (bottom right sub-figure) from a non-linear model whose input parameters X incorporate geometric characteristics of a moving arm (schematically represented as a cone, on the other sub-figure). Figures taken from [35].

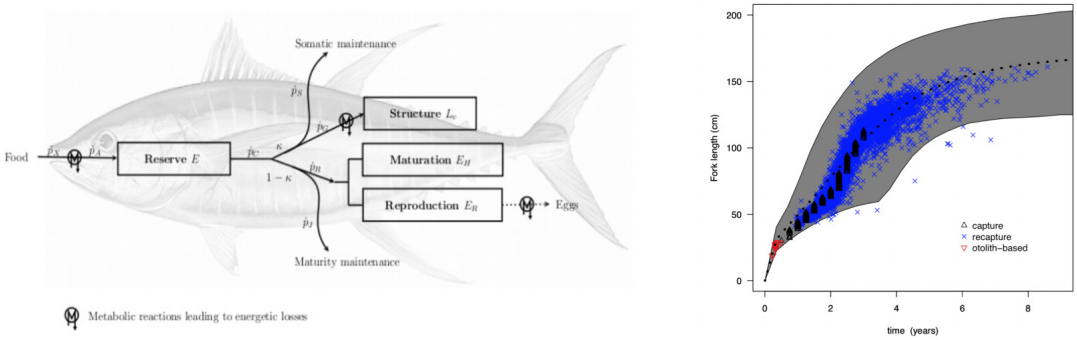


Figure 7 – Right: Conceptual representation of a Dynamic Energy Budget model for yellowfin tuna, summarizing energy fluxes defined using hidden state equations. Left: Bayesian credible area for the fork length-at-age, produced by Approximate Bayesian Computation, highlighting capture-recapture observations. Figures taken from [95] and [325], respectively.

$$
\begin{aligned}
1.\ \Gamma^{[k+1]} &\sim \mathcal{IW}_p\left(\Upsilon + \sum_{i=1}^{n}(\mu^{[k]} - X_i^{[k]})(\mu^{[k]} - X_i^{[k]})^T + \alpha(\mu^{[k]} - \mu_0)\times \right. \\
&\qquad \left. (\mu^{[k]} - \mu_0)^T,\ \nu + n + 1\right), \qquad (34)\\
2.\ \mu^{[k+1]} &\sim \mathcal{N}_p\Big(\frac{\alpha}{n+\alpha}\mu_0 + \frac{n}{n+\alpha}\overline{\mathbf{X}_n^{[k]}},\ \frac{\Gamma^{[k+1]}}{n+\alpha}\Big) \quad \text{with } \overline{X_n^{[k]}} = \frac{1}{n}\sum_{i=1}^{n} X_i^{[k]},\\
3.\ X^{[k+1]} &\sim \tilde{\pi}\left(X^{[k+1]}\middle|\ g, \mathbf{y}^*, \Sigma, \mu^{[k+1]}, \Gamma^{[k+1]}\right),
\end{aligned}
$$

this latter conditional density being proportional to

$$
\begin{aligned}
\exp\Big\{ -\frac{1}{2}\sum_{i=1}^{n}\Big[&\left(X_i^{[k+1]} - \mu^{[k+1]}\right)^T \left(\Gamma^{[k+1]}\right)^{-1}\left(X_i^{[k+1]} - \mu^{[k+1]}\right) \\
&+ \left(y_i^* - g(X_i^{[k+1]}, z_i)\right)^T \Sigma^{-1}\left(y_i^* - g(X_i^{[k+1]}, z_i)\right)\Big]\Big\} \qquad (35)
\end{aligned}
$$

Since (35) does not belong to a closed form family of density functions, a specific Metropolis-Hasting (MH) step based on a multivariate random walk was designed in [126] to simulate $X^{[k+1]}$ from its full conditional distribution.

However, this type of approach is reasonable when the cost of calling the g model is negligible; indeed, the convergence of this class of hybrid algorithms requires in practice several thousands (or tens of thousands) of iterations. For a class of problems involving expensive numerical models g for which this direct approach is prohibitive, we have provided a solution by replacing g by a stochastic interpolator based on a design of numerical experiments (DOE) D_N. More precisely, we used a Gaussian process (kriging) meta-model of g, now an usual tool of model emulation [216], assessed using maximin Latin Hypercube Sampling [16]: it is assumed that

$$
\{g(X,z),\ldots,\} \ \sim\ \mathcal{N}\left(\hat{g}_{D_N}(X,z), \hat{K}_{D_N}(X,\ldots)\right)
$$

where $\hat{g}_{D_N}(.)$ and $\hat{K}_{D_N}(.,.)$ are respectively the mean and covariance functions of the Gaussian process. This rather usual choice implies the following adaptation of Equation (35) to account for the *emulator error* [126], the magnitude of it being estimated by well-known indicators as the coefficient of predictability Q_2 [344] and the Malahanobis distance MD [20]:

$$
\begin{aligned}
3.\ \hat{\pi}\left(X^{[k+1]}|\ldots\right) \ \propto\ & |\mathbf{\Sigma}(X^{[k+1]})|^{-\frac{1}{2}} \cdot \exp\left\{ -\frac{1}{2}\sum_{i=1}^{n}(X_i^{[k+1]} - m^{[k+1]})\Big[\Gamma^{[k+1]}\Big]^{-1} \right. \\
&\left. (X_i^{[k+1]} - m^{[k+1]}) - \frac{1}{2}(\mathbf{Y} - \hat{\mathbf{g}}(X^{[k+1]}))\Big[\mathbf{\Sigma}(X^{[k+1]})\Big]^{-1}(\mathbf{Y} - \hat{\mathbf{g}}(X^{[k+1]}))\right\}
\end{aligned}
$$

where $\hat{\mathbf{g}}(X) = (\hat{g}(X_1, z_1) \cdots \hat{g}(X_n, z_n))$ (vector of kriging predictions) with $\hat{g}(X_i, z) = (\hat{g}^1_{D_N}(X_i, z) \cdots \hat{g}^p_{D_N}(X_i, z))$, and where $\mathbf{\Sigma}(X) = \mathbf{R} + \hat{\mathbf{K}}(X)$ (covariance matrix of the observations) with $\mathbf{R}$ (covariance matrix of the measurement errors) the block-diagonal matrix made up with n diagonal blocks R of size p and with the entry in the i-th row and the j-th column of $\hat{\mathbf{K}}(X)$ satisfying $\left[\hat{\mathbf{K}}(X)\right]_{\substack{i=p(k_i-1)+l_i \\ j=p(k_j-1)+l_j}} = \hat{K}^{l_i}_{D_N}(X_{k_i}, X_{k_j})$ if $l_i = l_j$, 0 otherwise, for all $1 \leqslant k_i, k_j \leqslant n$ and $1 \leqslant l_i, l_j \leqslant p$. The addition of $\hat{\mathbf{K}}(X)$ to $\mathbf{R}$ enables to take account of the uncertainty caused by replacing g by the meta-model.

4.2.2 Optimizing adaptive meta-modeling designs

In this model approximation framework, the question arises of the general convergence of the stochastic inversion algorithm: is the approximate posterior distribution obtained by sampling reasonably close to the targeted theoretical posterior distribution? Two difficulties may prevent this:

1. the meta-model is of good quality and the approximation error is related to the Monte Carlo sampling, which implies to continue this sampling, while improving the instrumental laws used in the MH step ;
2. the meta-model is of insufficient quality, and a different choice must be made, or the DOE must be improved to build this interpolating meta-model.

In parallel to the recent work conducted by [77, 47, 185] on the structural choice of meta-models for model emulation, we have extended previous works carried out during my PhD thesis to propose a methodology of iterative improvement of the stochastic inversion algorithm. This methodology is based on the Bayesian DAC criterion (data-agreement criterion), which originally measures a discrepancy (in the parametric space) between a prior measure and data information [313]. It has thus been adapted to measure the efficiency of a DOE, allowing to define a framework of sequential enrichment, a key problem in the domain of computer experiments.

The enrichment of the DOE itself is the subject of the article [127]. With my co-authors I proposed an adaptation of the well-known Expected Improvement (EI) criterion originated from [176], which showed to provide good results in small to moderate dimensions, alternatively to a Stepwise Uncertainty Reduction (SUR) strategy [345, 21] which was found more challenging from a computational perspective. Ideally, the posterior distribution of the parameters $\theta = (\mu, \Gamma)$ after adding a new point $\tilde{x}_{(N+1)}$ to the current DOE D_N in the product space $\Omega = I\!R^p \times I\!R^r$ should be as close as possible to the posterior distribution knowing the original function g, i.e. a relevant discrepancy measure between the two relative distributions must be minimized. Choosing the Kullback-Leibler (KL) divergence

$$\mathrm{KL}\Big(\pi(\theta|\mathbf{Y}, \mathbf{z}, g) \,||\, \pi(\theta|\mathbf{Y}, \mathbf{z}, \hat{\mathbf{g}}_{D_N} \cup \{g(\tilde{x})\})\Big),$$

as the discrepancy criterion, where $\hat{\mathbf{g}}_{D_N}$ crudely denotes both the current design and meta-model, a new design point $\tilde{x}_{(N+1)}$ should be ideally chosen as

$$\begin{aligned}
\tilde{x}_{(N+1)} &= \arg\min_{\tilde{x}\in\Omega} \mathrm{KL}\Big(\pi(\theta|\mathbf{Y},\mathbf{z},g)\,||\,\pi(\theta|\mathbf{Y},\mathbf{z},\hat{\mathbf{g}}_{D_N}\cup\{g(\tilde{x})\})\Big), \\
&= \arg\min_{\tilde{x}\in\Omega} \Big\{\mathrm{KL}\Big(\pi(\theta|\mathbf{Y},\mathbf{z},g)\,||\,\pi(\theta|\mathbf{Y},\mathbf{z},\hat{\mathbf{g}}_{D_N}\cup\{g(\tilde{x})\})\Big) \\
&\qquad\qquad - \mathrm{KL}\Big(\pi(\theta|\mathbf{Y},\mathbf{z},g)\,||\,\pi(\theta|\mathbf{Y},\mathbf{z},\hat{\mathbf{g}}_{D_N}\Big)\Big\}, \\
&= \arg\max_{\tilde{x}\in\Omega} \int_{\theta\in\Omega} \pi(\theta|\mathbf{Y},\mathbf{z},g)\,\log\frac{\pi(\theta|\mathbf{Y},\mathbf{z},\hat{\mathbf{g}}_{D_N}\cup\{g(\tilde{x})\})}{\pi(\theta|\mathbf{Y},\mathbf{z},\hat{\mathbf{g}}_{D_N})}\,d\theta. \qquad (36)
\end{aligned}$$

The consistency of the approach was supported by theoretical analyses showing that if the surrogate converges at a certain rate in the prior-weighted L_2 norm, then the approximate posterior distribution converges to the true posterior at least two times faster in the KL sense [358, 359]. Note however that these major authors in the field do not consider directly the KL divergence to define a specific enrichment criterion. Replacing: (a) the intractable target density $\pi(\theta|\mathbf{Y},\mathbf{z},g)$ by its best available approximation $\pi(\theta|\mathbf{Y},\mathbf{z},\hat{\mathbf{g}}_{D_N}\cup\{g(\tilde{x})\})$; (b) each time-consuming evaluation of $g(\tilde{x})$ by $\hat{g}_{D_N}(\tilde{x})$, with Gaussian distribution $\pi(\hat{g}_{D_N}(\tilde{x}))$ under the kriging assumption, a suboptimal criterion was proposed for replacing (36), relying on the key principle of Efficient Global Optimization (EGO) algorithm proposed by [176]. We now select $\tilde{x}_{(N+1)}$ as modifying maximally the current posterior $\pi(\theta|\mathbf{Y},\mathbf{z},\hat{g}_{D_N})$ as much as possible, on average, in an information-theoretic sense:

$$\tilde{x}_{(N+1)} = \arg\max_{\tilde{x}\in\Omega} \mathrm{ED}(\tilde{x}) \qquad (37)$$

with ED the *expected divergence* criterion

$$\mathrm{ED}(\tilde{x}) = \mathbb{E}_{\pi(\hat{\mathbf{g}}_{D_N}(\tilde{x}))}\mathrm{KL}\Big(\pi(\theta|\mathbf{Y},\mathbf{z},\hat{\mathbf{g}}_{D_N}\cup\{\hat{g}_{D_N}(\tilde{X})\})\,||\,\pi(\theta|\mathbf{Y},\mathbf{z},\hat{\mathbf{g}}_{D_N})\Big). \qquad (38)$$

While tractable in theory, in practice this criterion must be adapted again to avoid the huge computational burden related to the fact that testing a new design point $\tilde{x}$ requires to run a Gibbs sampler. In [127] we first proposed to consider rather the *expected conditional divergence* criterion

$$\mathrm{ECD}(\tilde{x}) = \mathbb{E}_{\pi(h_N)}\mathbb{E}_{\mathbf{X}_1(\tilde{x})}\mathbb{E}_{\mathbf{X}_2}\left[\mathrm{KL}\Big(\pi(\theta|\mathbf{X}_1(\tilde{x}))\,||\,\pi(\theta|\mathbf{X}_2)\Big)\right], \qquad (39)$$

where, ideally, the missing data samples $\mathbf{X}_1(\tilde{x})$ and $\mathbf{X}_2$ follow $\pi(\mathbf{X}|\mathbf{Y},\mathbf{z},\hat{\mathbf{g}}_{D_N}\cup\{\hat{g}_{D_N}(\tilde{x})\})$ and $\pi(\mathbf{X}|\mathbf{Y},\mathbf{z},\hat{\mathbf{g}}_{D_N})$, respectively, but are practically sampled at each step $k+1$ of the Gibbs algorithm:

$$\begin{aligned}
\mathbf{X}_2^{[k+1]} &\sim \pi\left(\cdot|\mathbf{Y},\mathbf{z},\theta^{(k+1)},\hat{\mathbf{g}}_{D_N}\right), \\
\mathbf{X}_1^{[k+1]}(\tilde{x}) &\sim \pi\left(\cdot|\mathbf{Y},\mathbf{z},\theta^{(k+1)},\hat{\mathbf{g}}_{D_N}\cup\{\hat{g}_{D_N}(\tilde{x})\}\right).
\end{aligned}$$

There is no simple ordering relation between $\mathrm{ECD}(\tilde{x})$ and $\mathrm{ED}(\tilde{x})$, despite the good behavior of the adaptive design observed on several examples. But it can be noticed that from a simple Jensen inequality,

$$\mathrm{ED}(\tilde{x}) \quad \leqslant \quad \mathbb{E}_{\pi(h_N)}\mathbb{E}_{\mathbf{X}_1(\tilde{x})}\left[\mathrm{KL}\Big(\pi(\theta|\mathbf{X}_1(\tilde{x}))\,||\,\pi(\theta|\mathbf{Y},\mathbf{z},\hat{\mathbf{g}}_{D_N})\Big)\right]. \tag{40}$$

This upper bound could help to get theoretical guarantees on the maximization of (38). Note that to improve tractability, approximations of (38), (39) or (40) could rely on Bernstein-von Mises' theorem, assuming that involved posterior distributions are close to Gaussian distributions. This would fit with usual approaches emphasized in the community of *physical experiments* [10], and lead to a more explicit SUR-type criterion. But one can suspect that such an approximation would "smooth" the differences in the sequence of posterior distributions, while it is precisely the contrast between these densities that brings the necessary information to propose a new design point. Such studies remain to be conducted.

Beyond improvements on tractability, this methodology (or similar ones based on, for instance, replacing KL by a q-Wasserstein distance) partially implemented in the Python software OpenTURNS [16], although having shown to be efficient for examples of small to moderate dimensions, faces several important and connected problems, all related to the exploration of high dimensionality: choosing a good meta-model and using scalable computational techniques. The recent period corresponding to the studies [126, 127] has been very fertile in many parallel developments aiming at solving these problems.

From a computational point of view, scalable MCMC stands on the fine use of instrumental distributions (e.g., adaptive MCMC [11], stochastic gradient Langevin dynamics (SGLD)-based approaches [103], natural gradient methods [215]), the approximation of likelihood posterior (e.g., polynomial chaos expansion [233], dimension reduction for both the input parameters and model state spaces [72], low rank tensor approximations [238], radial basis functions (RBF) [44], variational approximation, embedding within a Reproducing Kernel Hilbert Space (RKHS)), approximations of Markov kernels features (e.g., connections with stochastic gradient descent [235]) or transport methods between prior and posterior distributions [251].

On the other side, the choice of a meta-model is a necessary compromise between the known characteristics of the phenomenon represented by g (e.g., its regularity properties), the cost of numerical trials, the possibility or not of conducting an intrusive exploration of g (to which is linked the possible availability of reduced models of g, of degraded accuracy but reduced cost – the so-called multi-fidelity problem). The improvement of the choice of the meta-model, in any case in the most interesting areas of the input space for the inversion task, must therefore be articulated with the choice of an inferential method. Such a research is today driven, among others, by advances in meta-model comparison [348, 71, 187], hybridation (e.g., the combination of polynomial chaos expansions and Gaussian processes (*PC-Kriging*) [293] ; see [132] for a combination of kriging, RBF and

16. `http://openturns.github.io/`

quadratic functions) and averaging [180, 185]. More and more, this research axis is in connection with some of the most recent works in statistical machine learning (e.g., deep Gaussian processes [266, 190, 202]).

4.2.3 Covariance prior constraints for meaningful inversion problems

Other problems may in practice limit the resolution of an inverse problem. Statistical validity is based on sufficient exploration of the parameter space by the frequentist or Bayesian stochastic algorithms mentioned above. They have to browse a large number of values of (μ, Γ), and it can occur, for instance, that a Bayesian posterior distribution obtained by sampling under finite computational cost lacks of accuracy and still places importance on implausible values [145]. From a UQ point of view, for many applications it is key to diminish the possibility that a lack of accuracy in the estimation of the distribution of X result in underestimating the input uncertainty [179, 97].

Beyond theoretical statistical validity, in view of using the model for prediction tasks, I have formalized in [324] the notion of a *meaningful solution*, by imposing that the pair (μ_i, Γ_i) is such that the instance of the model (30) defined by

$$Y_i = g(X_i) \quad \text{with} \quad X_i \sim \mathcal{N}(\mu_i, \Gamma_i) \tag{41}$$

generates plausible values of Y_i, namely values that remain close to real observations $\mathbf{y}_\mathbf{n}^*$ and reflect most of their variability. This implies that the underlying phenomenon explaining this variability is globally well reproduced. If this were not the case, it would mean that the error $\varepsilon_i \sim f_\varepsilon$ has to strongly distort the values of Y_i to obtain values of $Y_i^* = Y_i + \varepsilon_i$ close to $\mathbf{y}_\mathbf{n}^*$. Hence the model g should not be used for simulation purpose. Plausibility should therefore be a consequence of the validation of g, that ideally *helps to ensure that the computational model has sufficiently rigor for the context of use* [229]. However, model validation is usually conducted using statistical testing strategies, following a calibration task that estimates particular values of X by minimizing a metric with respect to selected values of Y^* (e.g., limit cases) [239, 240]. This leads to validate g only for a selected range of situations, without getting a full guarantee that these situations automatically include the X_i generated by (41).

This problem is of particular concern when ε_i is a mixture of observational noise ε_1, approximation error ε_2 due to inherent inaccuracies related to discretization [241] and model error ε_3. In general, ε_1 is known and ε_2 can be bounded [258]. The model error ε_3 is often estimated by residual analyses from regression approaches that do not use the generative model $X \sim \mathcal{N}(\mu, \Gamma)$, but rather use selected values of X [311]. This prior estimation diminishes the risk of identifiability issues encountered in stochastic inversion [77]. Especially, such approaches based on designs of numerical experiments are used when (g, ε_3) includes a stochastic lower-cost approximation –a *surrogate* (or *meta-model*)– of a computationally expensive model (e.g., (g, ε_3) is chosen as a Gaussian-process based kriging meta-model [15]). Such a concern, in particular, arose in my previous co-supervising work [126].

Restricting ourselves to the unbiased case where the expectation of ε equals zero, an intuitive way to limit the risks of such solutions is to constrain the search for the values of Γ in subspaces of $S_p^{++}(\mathbb{R})$, by imposing that the variability of Y_i^* remains mainly explained by the variability of Y_i, i.e. that of X_i (g being deterministic). This can be interpreted through the prism of *sensitivity analysis* (SA ; [166]): this means that an existing solution is such that a sensitivity indicator of Y^* related to X takes values higher than this same sensitivity indicator related to ε. This UQ concern can be seen as a supplementary condition to the classical conditions of Hadamard's well-posed inverse problems: not only a solution should exist, be unique and data-continuously dependent, but it has to be meaningful too. Then constraining Γ is similar in spirit to the use of prior information for attempting to transform an ill-posed inversion problem into a well-posed problem using regularization techniques [49].

In [324], I proposed with my co-authors several mathematical interpretations of what could be a meaningful stochastic inversion problem, considering the problem (30-41). The following two first definitions arise from natural connections with SA.

DEFINITION 2. *Let (S_X, S_ε) be the generalized grouped Sobol indices proposed by [130, 131], quantifying the uncertainty on Y^* explained by X and ε, respectively. Provided X and ε are independent, the stochastic inversion problem (30-41) can be said meaningful in Sobol' sense if*

$$S_X \quad > \quad S_\varepsilon. \tag{42}$$

DEFINITION 3. *The stochastic inversion problem (30-41) is said to be meaningful in the entropic sense if*

$$\mathbb{E}_\varepsilon \left[\mathcal{E}(Y^*)|\varepsilon\right] \quad > \quad \mathbb{E}_X \left[\mathcal{E}(Y^*)|X\right]. \tag{43}$$

Then, in the simple linear case $Y = aX$, (42) is equivalent to the covariance constraint $\mathrm{T}_r(a\Gamma a^T) > \mathrm{T}_r(\Sigma)$ and (43) is equivalent to to the alternative covariance constraint $\left|a\Gamma a^T\right| > |\Sigma|$. These inequalities reflect a similar constraint on the covariance matrix Γ when $a\Gamma a^T$ is invertible and close to the identity. In such cases, indeed,

$$|I_q + \varepsilon A| \quad = \quad 1 + \varepsilon \mathrm{T}_r(A) + O(\varepsilon^2).$$

This kind of result offers the possibility of deriving proper Jeffreys priors for Γ, assuming $|\Gamma| < \infty$, which is often true in applied situations.

Beyond these connections with SA, information theory offers a way of formalizing more broadly the notion of meaningful problem and provide covariance constraints, under the classical assumptions of existence of Fisher information [69]. Denote by $I_{g(X)}(\theta) \in S_q^{++}(\mathbb{R})$ and $I_{Y^*}(\theta) \in S_q^{++}(\mathbb{R})$ the Fisher information matrices provided respectively by $g(X)$ and Y^* about θ. Since the impact of ε is to degrade information, it can be expected that $I_{g(X)}(\theta)$ be greater than $I_{Y^*}(\theta)$, according to some order relation $\succcurlyeq$ defined for

symmetric matrices. We argued that we can restrict the usual Loewner order [208, 296] to the usual total order $>$ stated in (44):

$$I_{g(X)}(\theta) \quad > \quad I_{Y*}(\theta) \tag{44}$$

where $A > B$, for two squared matrices A and B, means that $A - B$ is a positive definite matrix. Stating that most of information on θ in Y^* is transmitted from $g(X)$ implies that the difference between $I_{g(X)}(\theta)$ and $I_{Y*}(\theta)$, which is a measure of the information loss because of the noise ε, should not be greater than a fraction $(1 - 1/c)I_{g(X)}(\theta)$ where $c > 1$. We therefore proposed the following definition.

DEFINITION 4. *For $c > 1$, the stochastic inversion problem is meaningful in Fisher's sense if*

$$I_{g(X)}(\theta) > I_{Y*}(\theta) > \frac{1}{c} I_{g(X)}(\theta). \tag{45}$$

Studying the case of Gaussian linear models $g : x \mapsto Hx$ allows to provide formal arguments to select a relevant value for c. More precisely, considering the simplified assumption $\Gamma = \tau^2 I_p$, we obtain constraints over τ^2, which can be interpreted as minimum variance bounds, from necessary and sufficient conditions to (45). Under other assumptions inducing the commutativity of HH^T and Σ, such equivalent bounds are of the form

$$\sum_{i=1}^{q} \left(\frac{\tau^2 \lambda_i^{HH^T}}{\tau^2 \lambda_i^{HH^T} + \lambda_i^{\Sigma}} \right)^2 > \frac{q}{c}.$$

where $\left\{\lambda_i^A\right\}_{1 \leqslant i \leqslant n}$ denote the eigenvalues of any invertible matrix A in $R^{n,n}$. Simpler sufficient conditions can be obtained by assuming $\Sigma = \sigma^2 I_q$

$$(\sqrt{c} - 1)\tau^2 > \frac{\max\limits_{1 \leqslant i \leqslant q} (\lambda_i^{\Sigma})}{\min\limits_{1 \leqslant i \leqslant q} (\lambda_i^{HH^T})}$$

or, without this hypothesis, by $\left(\sqrt{c} - \mathrm{cond(HH^T)}\right) \tau^2 \geqslant \max_{1 \leqslant \mathrm{i} \leqslant \mathrm{q}} \left(\lambda_i^{\Sigma}\right) / \min_{1 \leqslant \mathrm{i} \leqslant \mathrm{q}} \left(\lambda_i^{HH^T}\right)$ where $\mathrm{cond}(HH^T) = \|HH^T\| \|(HH^T)^{-1}\|$ is the usual *condition number* of HH^T with respect to the matrix norm induced by the Euclidian norm on $\mathbb{R}^q$. A more general result is related to next theorem:

THEOREM 3. **(derived from [324])**. *Denote $\Psi = \Sigma^{-1/2} HH^T \Sigma^{-1/2}$. Then*

$$I_{Y*}(\sigma^2) \quad = \quad \frac{1}{2} \sum_{j=1}^{q} \left(\frac{\lambda_i^{\Psi}}{1 + \tau^2 \lambda_i^{\Psi}} \right)^2.$$

This result helps to show first that the left inequality of (45) is always true: $I_{g(X)}(\theta) > I_{Y*}(\theta)$ (as expected, since the noise degrades information). Furthermore, a sufficient

(*resp.* necessary) condition for the right inequality of (45) to be true is $(\sqrt{c}-1)\tau^2 > (\min_{1\leqslant i\leqslant q}\{\lambda_i^\Psi\})^{-1}$ (*resp.* $(\sqrt{c}-1)\tau^2 > (\max_{1\leqslant i\leqslant q}\{\lambda_i^\Psi\})^{-1}$). Besides, it must be noticed that the necessary condition can be rewritten as $c > (1+1/R(\tau,\Psi))^2$ where

$$R(\tau,\Psi) = \tau^2 \max_{1\leqslant i\leqslant q}\{\lambda_i^\Psi\}$$

can be interpreted as the *signal over noise ratio* (SNR). It seems reasonable to assume that $R(\tau,\Psi) \geqslant 1$. Hence, necessarily, $c \geqslant 4$, which allows to recover the conditions required for Sobol and entropy-based notions of meaningfulness.

Much more results when $q \geqslant 1$, often under the form of necessary conditions for (45), are provided in [324]. Especially, as a preliminary result we provide a new characterization of the Fisher matrix for

$$Y_\alpha \quad = HX + \alpha\varepsilon \quad \text{with } \alpha \in \{0,1\}, \tag{46}$$

that encompasses the cases $Y_0 = Y$ and $Y_1 = Y^*$. Denoting

$$\Psi_\alpha(\Gamma) \quad = \quad \Sigma^{-\alpha/2} H\Gamma H^T \Sigma^{-\alpha/2}$$

(with the convention $\Sigma^0 = I_q$), which can be seen as a multivariate generalization of the SNR, the following decomposition can be obtained.

THEOREM 4. **(B.** *et al.* **[324])**. *Denote $I_{Y_\alpha}(\theta)$ the Fisher information matrix for (46), assuming $\theta = (\mu,\Gamma) \in \mathbb{R}^p \times S_p^{++}(\mathbb{R})$. Let $\partial^2/\partial\Gamma^2$ indicate the double differentiation with respect to the free components of Γ. Then, assuming that $\alpha I_q + \Psi_\alpha(\Gamma)$ is inversible,*

$$I_{Y_\alpha}(\theta) \quad = \quad \begin{pmatrix} I_{Y_\alpha}(\mu) & \mathbf{0} \\ \mathbf{0} & I_{Y_\alpha}(\Gamma) \end{pmatrix}$$

where

$$\begin{aligned} I_{Y_\alpha}(\mu) &= H^T\Sigma^{-\alpha/2}\left(\alpha I_q + \Psi_\alpha(\Gamma)\right)^{-1}\Sigma^{-\alpha/2}H, \\ I_{Y_\alpha}(\Gamma) &= -\frac{1}{2}\sum_{i=1}^{q}\frac{\partial^2}{\partial\Gamma^2}\log\left(\alpha + \lambda_i^{\Psi_\alpha(\Gamma)}\right). \end{aligned} \tag{47}$$

Beyond linear cases, a variational approximation of $g(x)$ by a linear model allows to propose a strategy for constraining the covariance Γ: this heuristic is based on producing an approximation constrained to be meaningful in Fisher's sense (in addition to being relevant in terms of interpolation). It provides a new lower bound for Γ. It can be shown that the linear model $\tilde{Y} = HX + y_0 + U$, incorporating an error term $U \sim \mathcal{N}(0,S)$, that minimizes $\mathrm{KL}(\tilde{Y},Y)$, is such that $y_0 = \mathbb{E}[Y]$ and

$$H^T + S \quad = \quad \mathbb{E}\left[(Y-y_0)(Y-y_0)^T\right]. \tag{48}$$

This reflects the well-known following result: the minimization of the KL divergence between the distribution f_Y of Y (*resp.* f_Y^* of Y^*) and the parametric distribution f_H of $\tilde{Y}$ (*resp.* f_H^* of $\tilde{Y}^*$) is similar to estimate the parameter structure of f_H (*resp.* f_H^*) by the maximum likelihood (ML) estimators of an infinite sample $(y_1, \ldots, y_n, \ldots)$ from f_Y (*resp.* f_Y^*) [220]. The quantities $\mathbb{E}[Y]$ and $\mathbb{E}\left[(Y - y_0)(Y - y_0)^T\right]$ can be consistently estimated from the available set of noisy observations $\mathbf{y_n^*}$. The variational linear model is meaningful in Fisher's sense if Ψ is replaced by

$$\Psi_{\Gamma,Z} = (\Sigma + S)^{-1/2} H_{\Gamma,Z} H_{\Gamma,Z}^T (\Sigma + S)^{-1/2}.$$

where $H_{\Gamma,Z}$ is of the form $O_Z \tilde{\Sigma}_Z \Gamma^{-1/2}$ with O_Z in the orthogonal group $\mathcal{O}(q, I\!R)$ and $\tilde{\Sigma}_Z \in I\!R^{q,p}$. Assuming again the simplification $\Gamma = \tau^2 I_p$, the last equation simplifies in

$$\Psi \quad = \quad (\Sigma + S)^{-1/2} \left(\mathrm{Cov}(Y^*) - S - \Sigma\right) (\Sigma + S)^{-1/2}.$$

which provides a tractable constraint on τ^2, provided S can be estimated. Since it expresses the error between g and a linear model, this latter task requires the use of a DOE for which the computation of g is required in a relevant subspace (this one being naturally provided using an informative prior in a Bayesian framework). The estimation of S can then be consistently made by the covariance matrix of the residuals in a multiple regression.

Several numerical experiments conducted in [324] with the default choice $c = 4$, adapting the algorithms built in [126, 127], have highlighted a real numerical advantage to constrain inferiorly the covariance of the variable X to be inverted: the predictive posterior distributions obtained on X present a better closeness with the simulation distributions than those which do not take account this constraint.

Beyond the numerous desired improvements (obtaining new and more tractable theoretical guarantees, management of numerical design of experiments in a parallel way to the use of real data, approximation of models by Gaussian mixtures of linear models), this work suggests new ideas to define what might be a *meaningful* inversion problem, by establishing connections with regularity, preconditioning, sensitivity analysis, and should allow us to question the procedures which rely on simulation in the treatment of uncertainties.

5 Quantification of decisional indicators

5.1 Context

A situation frequently encountered in numerous UQ studies, including structural reliability [213], is the estimation of a so-called limit state probability [195], or failure probability, defined by

$$p \;=\; P(g(X) \leqslant 0) \;=\; \int_{\mathbb{U}} \mathbf{1}_{\{g(x)\leqslant 0\}} f(x)\; dx,$$

where g is a deterministic function and X is a d-multidimensional input random variable with probability distribution function f, taking its values in a $d-$dimensional space $\mathbb{U}$. Most often, g is implemented as a black-box computer model. It requires that p can only be estimated in a non-intrusive way, by means of a well-tailored design of numerical experiments. In such contexts, Monte Carlo (MC) variance reduction techniques are currently used to diminish the size of the design and limit the computational burden. See [223] for a recent survey of such techniques. These approaches aims to improve the usual MC method, that estimates p by $\hat{p}_n = n^{-1} \sum_{k=1}^{n} \mathbf{1}_{\{g(X_k)\leqslant 0\}}$ where n is large and the X_k are independently sampled from f. Avoiding regularity hypotheses on g, this unbiased estimator presents good convergence properties and an estimation error independent on d. Unfortunately, this strategy often appears inappropriate in practice when p reaches low values, since g can be time-consuming and the computational budget may be limited: a good estimation of a probability $p \sim 10^{-q}$ typically requires at least 10^{q+2} calls to g [196]. Furthermore, $\hat{p}_n$ has the theoretical defect not to be robust, in the sense given in [146]: its relative error, namely its coefficient of variation, does not tend to a finite limit when $p \to 0^+$, given any finite number n of trials.

Many non-intrusive strategies have been proposed to accelerate the MC approach. Traditional methods from the engineering community in structural reliability (FORM/-SORM) treat the estimation of p as an optimization problem. The computational work is usually fast but the estimators suffer from weakly or non-controllable errors. Statistical approaches are judged in terms of reduction rate with respect to the MC estimator variance $\mathbb{V}[\hat{p}_n] = p(1-p)/n$. Methods like quasi-MC, sequential MC or importance sampling [192, 22, 309] are based on selecting a *design of experiments* (DOE), namely a set of points in $\mathbb{U}$ on which g is tested, such that $\mathbb{U}$ be explored in areas close to the limit state surface $\Gamma = \{X \in \mathbb{U} \;;\; g(X) = 0\}$. Most advanced methods as *subset sampling* often get rid of the time-consuming difficulties by emulating the behavior of g, for instance using kriging techniques [46, 23] which presuppose smoothness conditions on g [312]. The constant quest to obtain mild hypotheses and resolve computational difficulties has made this an ever-dynamic field of research, as shown by recent works in line sampling by [357], high-dimensional importance sampling by [105, 59] or using concentration properties of Lipschitz functions by [34].

5.2 Problems and personal contributions

5.2.1 Accelerated Monte Carlo techniques under geometric constraints

Minimizing the strength of regularity hypotheses placed on g underlies the development of specialized acceleration methods. For instance, computer codes can suffer from edge effects which restrict smoothness conditions [230]. On the other hand, the reality of the phenomenon can imply various form constraints on $g(X)$. Especially, the assumption that g is monotonic with respect to X is a standard problem in regression analysis [92, 104]. In the area of numerical experiments, monotonicity properties of computer models have been considered theoretically and practically, e.g. proving the MC acceleration of Latin Hypercube Sampling for the estimation of expectancies [212], carrying out screening methods for sensitivity analyses [204], constraining response surfaces as kriging meta-models (surrogates) [189, 188, 75, 211], kernel models [153] or neural networks [78], predicting the behavior of network queuing systems [268], computing flood probabilities [86] or estimating the safety of a nuclear reactor pressure vessel [230]. Specific engineering works in structural reliability highlighted the possibility of bounding and estimating p significantly faster than using a MC approach. Under the name of *monotonic reliability methods* (MRM), a class of algorithms contouring the limit state surface and enclosing p between deterministic bounds which dynamically narrow were proposed by [86]. A similar idea was explored in [267], and a parallelization of such algorithms was implemented by [203], but these works lacked of theoretical foundations.

At the time of this research, previous methods exploiting model monotonicity properties were only explored from an empirical point of view, and the proposed estimators of p remained crude. In [316], I build a theoretical argument to formally understand how the exploitation of monotonic properties can drastically accelerate the calculations of p and produce robust estimators, under a set of mild hypotheses and based on martingale theory. These results emerged at a time when an enlarged community of statisticians and numericians became particularly interested in the fact that shape constraints on g, especially monotonicity, could be used to build highly efficient approximation or computation algorithms [275, 75, 200, 207]. Some years before, maybe inspired by many works on isotonic regression and multi-objective optimization (e.g., [333]), machine learning researchers forecasted this growing interest (e.g., [297, 40]). This provided inspiration for physics-inspired approaches to statistical learning [181]. The detection of monotonicity properties within observed or simulated data, a requirement for such studies, was itself explored in [24] with several colleagues and students, through Bayesian approaches based on co-kriging meta-modeling. This theme of research is now taking on new importance in causality analysis [227] and more generally in the field of model interpretability.

I also introduced the main concepts that allowed us, with my PhD. student V. Moutoussamy and his co-advisor Thierry Klein (IMT), to understand how to assess the deeper problem of estimating the limit state surface $\Gamma = \{X \in \mathbb{U} \ ; \ g(X) = 0\}$, using the tools of random set theory. The results we obtained, published in [332], have strong links with classification tasks in machine learning.

This research, and current results and avenues, are summarized in the following sections. Most of the tools have been implemented within the R package `mistral`.

5.2.1.1 Mathematical framework Let X be a random vector uniformly distributed on $\mathbb{U} = [0,1]^d$, and g be a measurable, deterministic function from $\mathbb{U}$ to $\mathbb{R}$. This function g discriminates between two sets of input situations, leading either to undesirable or safe events. These classes are denoted $\mathbb{U}^- = \{X \in \mathbb{U} : g(X) \leqslant 0\}$ and $\mathbb{U}^+ = \{X \in \mathbb{U} : g(X) > 0\}$, with $\mathbb{U} = \mathbb{U}^- \cup \mathbb{U}^+$. Hence the probability p that an undesirable event may occur is

$$p = \mathbb{P}(g(X) \leqslant 0) = \mathbb{P}(X \in \mathbb{U}^-).$$

Characterizing the monotonicity of g requires the use of Pareto dominance between two elements of $\mathbb{R}^d$:

Definition 5. *Let $u = (u_1, \ldots, u_d)$ and $v = (v_1, \ldots, v_d) \in \mathbb{R}^d$. The vector u is said to be (resp. strictly) dominated by v, denoted by $u \preceq v$ (resp. $u \prec v$), if for all $i = 1, \ldots, d$, $u_i \leqslant v_i$ (resp. $u_i < v_i$).*

Next assumption is mild: any monotonic function can be reparametrized to be increasing with respect to all its inputs.

Assumption 1. *Let $g : \mathbb{U} \subset \mathbb{R}^d \to \mathbb{R}$. Assume that g is globally increasing, i.e. for all $u, v \in \mathbb{U}$ such that $u \preceq v$, then $g(u) \leqslant g(v)$.*

This property results in a simple exploration of $\mathbb{U}^-$ and $\mathbb{U}^+$ and provides deterministic bounds on p. Let $x \in \mathbb{U}^-$ (*resp.* $\mathbb{U}^+$) and $y \in \mathbb{U}$. Since g is increasing, if $y \preceq x$ (*resp.* $y \succeq x$), then $y \in \mathbb{U}^-$ (*resp.* $\mathbb{U}^+$). More generally, consider a set A of elements on $\mathbb{U}$ and define

$$\mathbb{U}^-(A) = \bigcup_{x \in A \cap \mathbb{U}^-} \{u \in [0,1]^d : \ u \preceq x\},$$
$$\mathbb{U}^+(A) = \bigcup_{x \in A \cap \mathbb{U}^+} \{u \in [0,1]^d : \ u \succeq x\}.$$

Then, provided these sets are nonempty (so that p exists in $]0,1[$), the monotonicity property implies the following result, since $\mathbb{U}^-(A) \subset \mathbb{U}^- \subset [0,1]^d \backslash \mathbb{U}^+(A)$.

Proposition 6. *Denote μ the Lebesgue measure on $\mathbb{U}$. For any countable set A,*

$$\mu(\mathbb{U}^-(A)) \leqslant p \leqslant 1 - \mu(\mathbb{U}^+(A)).$$

Consequently, the limit state surface Γ, which is formally defined by

$$\Gamma = \left(\overline{\mathbb{U}}^- \backslash \mathring{\mathbb{U}}^-\right) \cap \left(\overline{\mathbb{U}}^+ \backslash \mathring{\mathbb{U}}^+\right),$$

where $\overline{E}$ and $\mathring{E}$ are respectively the closure and the interior of a set E, is surrounded by the dominated subsets $\mathbb{U}^-(A)$ and $\mathbb{U}^+(A)$. Hence the *non-dominated* subset $\mathbb{U}(A) = \mathbb{U}\backslash(\mathbb{U}^-(A) \cup \mathbb{U}^+(A))$ contains Γ and is the only subset of $\mathbb{U}$ for which a deeper exploration could be useful for refining this bounding of p (and also, for the applied setting, improving the upper bound $1 - \mu(\mathbb{U}^+(A))$ seen as a conservative assessment of p), and improving the knowledge of Γ.

Indeed, improving this knowledge of Γ can be viewed as refining the classification threshold between unsafe and safe inputs. The next (mild) assumption is required to define the problem of estimating Γ as a binary classification problem with perfectly separable classes, and furthermore will be needed to study the convergence of the dominated sets to $\mathbb{U}^-$ and $\mathbb{U}^+$.

ASSUMPTION 2. *Assume that* $\mu(\Gamma) = 0$.

This assumption appears somewhat intuitive, in the sense that due to the monotonicity of g, Γ is a simply connected surface. In particular, this is ensured if g is continuously differentiable on a non-empty measurable subset of Γ [316]. These results open the way for strategies for bounding p and surrounding Γ. In the remainder of this chapter, designs for simulated numerical experiments $\{X_i, g(X_i)\}_{i=1,\dots,n}$ are considered, and A is selected as $\{X_1, \dots, X_n\}$. Indeed, deterministic strategies have exponential, prohibited cost and can be only used to initialize such strategies [316]. Consequently, the sets $\mathbb{U}^-(A)$ and $\mathbb{U}^+(A)$ become random and the inequality in Proposition 6 becomes an almost sure inequality. A two-dimensional illustration of such sets is provided in Figure 8.

To simplify notation, denote for all $n \geqslant 1$,

$$\mathbb{U}_n^- = \mathbb{U}^-\{X_1, \dots, X_n\},$$
$$\mathbb{U}_n^+ = \mathbb{U}^+\{X_1, \dots, X_n\},$$

and the non-dominated set $\mathbb{U}_n = [0,1]^d\backslash(\mathbb{U}_n^- \cup \mathbb{U}_n^+)$. The Lebesgue measures (or hypervolumes) of sets $(\mathbb{U}_n^-, \mathbb{U}_n^+)$ are then denoted p_n^- and $1 - p_n^+$, such that, from Proposition 6, it comes with probability one that

$$p_n^- \leqslant p \leqslant p_n^+. \tag{49}$$

REMARK 2. *These two bounds* (p_n^-, p_n^+) *can be computed using the sweepline algorithm described in [316], originally devoted to solve Klee's measure problem [341], with an exponential cost. A faster approximation method is provided in Section 4 in [332].*

Note that the simplifying, starting assumption that $\mathbb{U} = [0,1]^d$ can be removed under conditions. In real situations, $x_1, \dots, x_d$ can be defined as *transformed inputs*, as usual in structural safety problems [213]. In such cases one can write $X = T(\mathbf{y})$ where $\mathbf{y} = (y_1, \dots, y_d)$ is a vector of *physical* inputs and T is their multivariate distributional

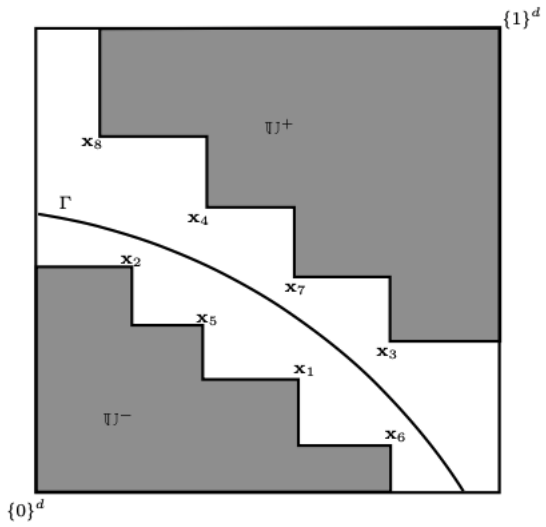


Figure 8 – Illustration for $d = 2$ of the designs $\mathbb{U}^-(x_1, x_2, x_5, x_6)$ and $\mathbb{U}^+(x_3, x_4, x_7, x_8)$, where the x_i are sampled over $\mathbb{U} = [0, 1]^2$. The probability p corresponds to the volume under the limit state surface Γ.

transform. Therefore $g = \tilde{g} \circ T^{-1}$ where $\tilde{g}$ is a mononotic function and T has to preserve this monotonicity. When the y_i are independent, T is reduced to the vector of marginal cdfs $(F_1, \ldots, F_d)$ and is obviously increasing, so Assumption 2 is not restrictive. In [316], technical requirements on T are provided to extend the relevance of this assumption, using the generalized Rosenblatt's transform [287].

LEMMA 1. *Assume that for $i = 2, \ldots, d$, there exists a mapping h_i and a set of (possibly random) parameters θ_i independent of $Y_1, \ldots, Y_i$ such that:*

(i) $Y_i = h_i(Y_1, \ldots, Y_{i-1}, \theta_i)$,

(ii) *h_i is a globally increasing function of $Y_1, \ldots, Y_{i-1}$;*

then $T^{-1}(X)$ is an increasing function of X.

5.2.1.2 Main results The theoretical and applied features of the convergence of random sets $\mathbb{U}_n^-$ and $\mathbb{U}_n^+$, as well as the definition of a fast and robust estimator of the probability p, were examined. The following results are based on two sampling strategies: (a) a simple, static Monte Carlo approach; (b) a nested Monte Carlo approach defined by sampling each new point X_{n+1} in the current non-dominated subset $\mathbb{U}_n$, conditionally to the $\sigma-$algebra $\mathcal{F}_n$ generated by the $n-$sequence of dominated and non-dominated spaces.

5.2.1.3 Convergence of random sets Such strategies implies that the sequences of bounds (p_n^-, p_n^+) are supermartingale and submartingale, respectively. The almost sure convergence of these bounds to p stands on Robbins-Siegmund theorem and the convergence of $\mathbb{U}_n^-$ and $\mathbb{U}_n^+$ to $\mathbb{U}_n$ according to the Hausdorff distance, defined as follows.

DEFINITION 6. *Let $\|\cdot\|_q$ be the $\mathcal{L}^q$ norm on $\mathbb{R}^d$, that is for $0 < q < +\infty$, $\|x\|_q = (\sum_{i=1}^d |x_i|^q)^{1/q}$, and for $q = +\infty$, $\|x\|_\infty = \max_{i=1,\dots,d} x_i$. Let (A, B) be two non-empty subsets of the normed vector space $([0,1]^d, \|\cdot\|_q)$. The Hausdorff distance $d_{H,q}$ is defined by*

$$d_{H,q}(A,B) = \max(\sup_{y\in A} \inf_{X\in B} \|x-y\|_q;\ \sup_{x\in B} \inf_{y\in A} \|x-y\|_q).$$

It is always finite since $(A, B) \subset \left([0,1]^d\right)^2$ are bounded, and its usual form is defined for $q = 2$. To alleviate notations, it is denoted $d_H(.,.) = d_{H,2}(.,.)$ and $\|.\| = \|.\|_2$ in the following.

In [332], general conditions on stochastic samplings are determined such that, applying the Borel-Cantelli lemma and under Assumption 2,

$$\left(d_H(\mathbb{U}_n^-, \mathbb{U}^-),\ d_H(\mathbb{U}_n^+, \mathbb{U}^+)\right) \xrightarrow[n\to+\infty]{a.s.} (0,0).$$

These conditions are respected for any sequence of iid random variables with bounded from below densities on $[0,1]^d$, especially in the uniform sampling situations introduced hereinbefore.

More refined results on the convergence speed can be obtained assuming moreover that the limit state surface Γ is $r-$convex [350]. This means that for each boundary point $X \in \Gamma$, there exists $y \in \mathbb{U}^-$ such that X belongs to an open ball $\mathbf{B}(y, r)$ with radius r. A more intuitive interpretation is that each point of Γ can be reached by a rolling ball (*Blaschke's rolling condition*). Deeply related to the (α, γ)-regularity introduced in [14] (Definition 7) to prove minimax rates in classification problems, and used by [70, 279] to state convergence rates when estimating sets, this condition appears mild, excluding pathological sets $\mathbb{U}^-$, such as those having infinitely many sharper and sharper peaks, or with a jagged border Γ.

DEFINITION 7. *A set K is said to be (α, γ)-regular if there exist $\alpha, \gamma > 0$ such that for all $0 < \varepsilon \leqslant \gamma$ and for all $X \in K$, one has*

$$\mu(\mathbf{B}(X, \varepsilon) \cap K) \geqslant \alpha\mu(\mathbf{B}(X, \varepsilon)).$$

In such regular cases, general convergence results on estimators $(\tilde{\mathbb{U}}_n^-, F_n)$ of U^-, build from the experiments, were obtained in [332], a major result of V. Moutoussamy's PhD. thesis:

PROPOSITION 7. **Convergence of random sets** (B. *et al.* [332]). *Let $(X_k)_{k\geqslant 1}$ a sequence of iid random variables uniformly distributed on $[0,1]^d$ and $(\tilde{X}_k)_{k\geqslant 1}$ be a sequence of iid random variables uniformly distributed on $\mathbb{U}^-$. Denote $\tilde{\mathbb{U}}_n^- = \mathbb{U}^-(\tilde{X}_1,\ldots,\tilde{X}_n)$. Let $(F_n)_{n\geqslant 1}$ be a sequence of measurable subsets of $[0,1]^d$ such that for all $n \geqslant 1, \mathbb{U}_n^- \subset F_n \subset [0,1]^d \backslash \mathbb{U}_n^+$. Then,*

(i) $d_H(F_n, \mathbb{U}^-) \xrightarrow[n\to+\infty]{a.s.} 0$ *and* $\mu(F_n) \xrightarrow[n\to+\infty]{a.s.} p$.

(ii) *If $\mathbb{U}^-$ is regular, then almost surely*

$$d_H(\tilde{\mathbb{U}}_n^-, \mathbb{U}^-) = O\left((\log n/n)^{1/d}\right).$$

(iii) *Furthermore, if $\mathbb{U}^+$ is also regular, and if g is continuous, then almost surely*

$$d_H(F_n, \mathbb{U}^-) = O\left((\log n/n)^{1/d}\right). \tag{50}$$

More precise results, establishing in-law convergence of random set $\mathbb{U}_n^-$ towards $\mathbb{U}^-$, were obtained in dimension 1 or in the extreme situation where $\Gamma = \{1\}^d$ (permanent failure: $p = 1$). The interpretation of such results provide idealistic minimal bounds, in terms of precision (or upper bounds in terms of convergence speed), for the estimation of the failure set $\mathbb{U}^-$ in more general settings. By symmetry, a possible practical use of these results could be to define procedures for testing the nullity of p, which is an open problem in situations where a preliminary analysis cannot check if $p > 0$ (e.g., using brute-force Monte Carlo sampling).

Moreover, in problems when $\mathbb{U}^-$ is convex (see Remark 3), the convergence results summarized hereinbefore helped to define a new class of monotonic binary classifiers, based on the aggregation of linear Support Vector Machines (SVM), to estimate consistently the limit state surface Γ. Indeed, if g is monotonic then Γ is $p-$monotonic, in the sense specified in the Corollary of Definition 8. The theoretical aspects of these tools are described in [332]; they showed to converge significantly faster, in a more robust way, than traditional monotonic neural networks .

DEFINITION 8. *Let $\alpha \in]0,1[$ and S be a set in $[0,1]^d$. The set S is said α-monotonic if for all $u, v \in S$, u is not strictly dominated by v and if $\mu(\mathbb{U}^-(S)) = \alpha$.*

Corrolary. A consequence of the monotonicity of g is that the set $\{x \in [0,1]^d,\ g(x) = \alpha\}$ is $F(\alpha)$-monotonic, where F is the cumulative distribution function of $g(X)$. In particular, the limit surface Γ is p-monotonic.

REMARK 3. *The convexity of $\mathbb{U}^-$ can be interpreted as a conservative requirement in risk analysis (but not in robust conception): it means that any convex mixture of two unsafe situations (belonging to $\mathbb{U}^-$) results in a third unsafe situation.*

5.2.1.4 Fast, robust and consistent estimation of extreme probability A framework for estimating statistically p can be defined in the situation (b), where each new design point X_{n+1} is sampled uniformly in the current non-dominated subset $\mathbb{U}_n$, conditionally to the $\sigma-$algebra $\mathcal{F}_n$. I personally introduced and studied this problem in [316]. The proposed methodology stands on the fact that, at step $n+1$, the occurrence of the event $\mathbf{1}_{\{g(X_{n+1})\leqslant 0\}}$ follows a Bernoulli distribution $\mathcal{B}(\gamma_{n+1})$ conditional to $\mathcal{F}_n$, with

$$\gamma_n \;=\; P\left(g(x)\leqslant 0 | x\in\mathbb{U}_n\right) \;=\; \frac{p-p_n^-}{p_n^+-p_n^-}.$$

According, a likelihood of dependent (*simulated*) "observations" yields the whole information carried by the set of n successive design experiments:

$$L_n(p) \;=\; \prod_{k=1}^{n}\left(\frac{p-p_{k-1}^-}{p_{k-1}^+-p_{k-1}^-}\right)^{\mathbf{1}_{\{g(x_k)\leqslant 0\}}}\left(\frac{p_{k-1}^+-p}{p_{k-1}^+-p_{k-1}^-}\right)^{1-\mathbf{1}_{\{g(x_k)\leqslant 0\}}}.$$

The uniquely defined corresponding maximum likelihood estimator (MLE) $\hat{p}_n\in]p_{n-1}^-,p_{n-1}^+[$, expressed as

$$\hat{p}_n \;=\; \frac{\sum_{k=1}^{n}\tilde{\omega}_k\left(\hat{p}_n\right)p_k}{\sum_{k=1}^{n}\tilde{\omega}_k\left(\hat{p}_n\right)}, \tag{51}$$

$$\text{with}\quad \tilde{\omega}_k(x) \;=\; \left[\left(x-p_{k-1}^-\right)\left(p_{k-1}^+-x\right)\right]^{-1}$$

$$\text{and}\quad p_k \;=\; p_{k-1}^- + \left(p_{k-1}^+-p_{k-1}^-\right)\mathbf{1}_{\{g(x_k)\leqslant 0\}}, \tag{52}$$

can be theoretically studied using the martingality of the score process $n\mapsto\{\ell_n'(p)\}_n$ where $\ell_n'(p)=\log L_n(p)$. Denoting $V_n^{MC}(p)=p(1-p)/n$ the usual MC variance, the Fisher information related to (51) is

$$J_n^{-1}(p) \;=\; \left(\sum_{k=1}^{n}\mathbb{E}\left[\tilde{\omega}_k(p)\right]\right)^{-1} \;\leqslant\; V_n^{MC}(p)\frac{n}{\sum_{k=1}^{n}(1-c_{k-1})^{-1}} \;<\; V_n^{MC}(p),$$

$$\leqslant\; V_n^{MC}(p)\left(\frac{p\rho_0^2}{1-p}\right)$$

where $c_0=0$, $\rho_0=(p_0^+-p_0^-)/p_0^-$ and $\forall\ k>1$,

$$c_k \;=\; \mathbb{E}\left[\frac{p_k^-}{p}+\frac{1-p_k^+}{1-p}-\frac{p_k^-(1-p_k^+)}{p(1-p)}\right].$$

Next theorem follows, with a robustness corollary, that allows to estimate and compute approximately the asymptotic distribution of estimator $\hat{p}_n$. Among the required assumptions, a law of large numbers should ensure that the sum of weights $\tilde{\omega}_k(p)$ cannot diverge

faster than $\mathcal{O}(J_n(p))$ from its mean behavior when $n \to \infty$. Although difficult to check in practice, this behavior seems rather intuitive because the sampling trajectories mainly vary at the first steps of the algorithm, when the non-dominated set $\mathbb{U}_n$ is still wide.

THEOREM 5. **Asymptotic normality (B. [316])**. *Denote* $\hat{J}_n(p) = \sum_{k=1}^{n} \tilde{\omega}_k(p)$. *Under mild assumptions, then*

$$J_n^{1/2}(p)\,(\hat{p}_n - p) \xrightarrow{\mathcal{L}} \mathcal{N}(0,1),$$
$$\frac{\hat{J}_n^{\,5/2}(p)}{|\hat{J}_n'(p)|}\left(\hat{J}_n^{-1}(\hat{p}_n) - J_n^{-1}(p)\right) \xrightarrow{\mathcal{L}} \mathcal{N}(0,1).$$

Furthermore, asymptotically $CV[\hat{p}_n] \simeq \frac{\rho_0}{\sqrt{n}}$.

Reaching the Cramer-Rao (CR) bound in this setting characterized dependent observations, the MLE $\hat{p}_n$ remains an asymptotically optimal estimator, as in usual iid cases where the CR bound is the inverse of the asymptotic variance. Nonetheless, while it dramatically beats the MC estimator (Figure 9) and other ones, it is affected by non-asymptotic bias (Figure 10). A bootstrap algorithm, based on a sequential surrogate of the surface Γ was shown to diminish significantly this bias (a monotonic network in [316], later modified using the SVM approach developed in [332]). An improvement of this M-estimator could be based on defining unbiased estimators of p that have a chance to reach the asymptotic variance. This topic is considered in next section.

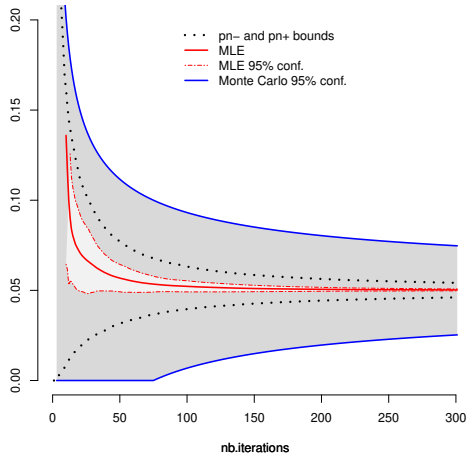


Figure 9 – Bounds (p_n^-, p_n^+) and MLE, with Monte Carlo and MLE 95%-confidence areas, in dimension $d = 2$, for $p = 5\%$. Empirical estimations are made over 300 parallel trajectories.

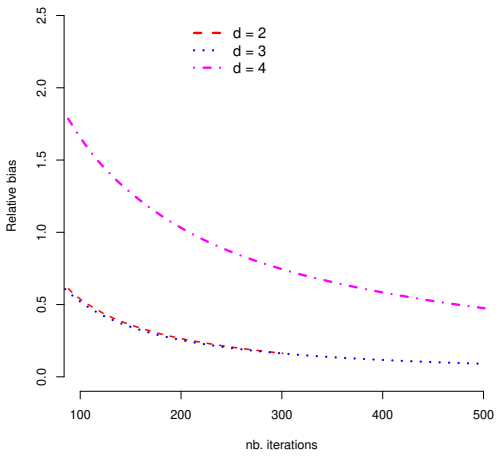


Figure 10 – Relative bias of the MLE $\hat{p}_n$ for the dimensions $d \in \{2, 3, 4\}$.

5.2.1.5 Towards sequentially nested importance sampling Beyond the simple case of nested uniform sampling, a generalization of Expression (51) allows to define a general class of unbiased L-estimators of p produced by sequentially nested importance sampling. Assume in the following that an initialisation step provides $\mathbb{U}_0 \subsetneq [0,1]^d$ and nontrivial initial bounds $\{p_0^-, p_0^+\}$ in $]0,1[^2$, and assume that $\forall n \geqslant 1$,

$$x_n \quad \sim \quad f_{n-1}(x).$$

Definition (9) and Proposition (8) allows to define a family of unbiased estimators of p, from which it is possible to derive optimal choices for the sequential importance densities $(f_k)_k$, such that these estimators alternatively reach a null variance, a classic result in importance sampling [285]. A key feature is the clear intractability of these densities. But it is often exhibited to guide the elaboration of a tractable importance density, which ideally converges in some sense towards this idealistic target [98]. Final estimation of p is produced by averaging *local estimators*, an usual technique to accelerate stochastic algorithms, popularized by Poliak and Rupert [256].

Definition 9. *Consider the family of estimators defined as a Cesaro weighted mean of local estimators $\bar{p}_k$, namely by a choice of weights ω_k (summing to 1) and*

$$\widehat{p}_n = \sum_{k=1}^{n} \omega_k \bar{p}_k \tag{53}$$

where

$$\bar{p}_k = p_{k-1}^+ - \frac{1}{f_{k-1}(X_k)} \mathbb{1}_{\{X_k \in \mathbb{U}^+\}}. \tag{54}$$

If f_{k-1} is the uniform distribution on $\mathbb{U}_{k-1}$, then estimators $\widetilde{p}_n$ and $\widehat{p}_n$ coincide.

Proposition 8. *If for all $k \geqslant 1$ and for all $x \in \mathbb{U}_{k-1} \cap \mathbb{U}^+$, $f_{k-1}(x) > 0$ then $\mathbb{E}[\widehat{p}_n] = p$. Moreover, if for all $k \geqslant 1$,*

$$f_{k-1}(x) = \frac{\mathbb{1}_{\{x \in \mathbb{U}^+ \cap \mathbb{U}_{k-1}\}}}{p_{k-1}^+ - p}, \tag{55}$$

then $Var[\widehat{p}_n] = 0$.

An approximation of this sequence of untractable optimal densities can be built from a sequence of estimates $\widehat{\mathbb{U}}_{k-1}^+$, based on response surface, of each unknown set $\mathbb{U}_{k-1} \cap \mathbb{U}^+$. To check the unbiasedness condition stated in Proposition 8, the set $\widehat{\mathbb{U}}_{k-1}^+$ must be such that for all $k \geqslant 1, \widehat{\mathbb{U}}_{k-1}^+ \subset \mathbb{U}^+ \cap \mathbb{U}_{k-1}$. To overcome this obstacle, it can be proposed at step $k \geqslant 1$ to simulate X_k uniformly in $\widehat{\mathbb{U}}_{k-1}^+$ with probability $1 - \varepsilon_{k-1}$ and uniformly in $\mathbb{U}_{k-1} \backslash \widehat{\mathbb{U}}_{k-1}^+$ with probability ε_{k-1}. So that, conditionally to $\mathcal{F}_{k-1}$ and ε_{k-1}, the density

of X_{k-1} is

$$f_{k-1}(x) = \frac{\varepsilon_{k-1}}{\mu(\mathbb{U}_{k-1}\backslash\widehat{\mathbb{U}}^+_{k-1})}\mathbb{1}_{\{x\in\mathbb{U}_{k-1}\backslash\widehat{\mathbb{U}}^+_{k-1}\}} + \frac{1-\varepsilon_{k-1}}{\mu(\widehat{\mathbb{U}}^+_{k-1})}\mathbb{1}_{\{x\in\widehat{\mathbb{U}}^+_{k-1}\}}, \tag{56}$$

which leads to

$$Var(\bar{p}_k|\mathcal{F}_{k-1}) = \frac{a_{k-1}}{\varepsilon_{k-1}} + \frac{b_{k-1}}{1-\varepsilon_{k-1}} - \left(p^+_{k-1} - p\right)^2,$$

where

$$a_{k-1} = \mu(\mathbb{U}_{k-1}\backslash\widehat{\mathbb{U}}^+_{k-1})\mu\left(\mathbb{U}^+ \cap \mathbb{U}_{k-1}\backslash\widehat{\mathbb{U}}^+_{k-1}\right),$$
$$b_{k-1} = \mu(\widehat{\mathbb{U}}^+_{k-1})\mu\left(\mathbb{U}^+ \cap \widehat{\mathbb{U}}^+_{k-1}\right).$$

Besides, the convergence properties of estimator $\widehat{p}_n$ can be explored using similar arguments to those developed in [316], using the property that

$$M_k = \sum_{j=1}^{k} \tilde{\omega}_j(\bar{p}_j - p)$$

is a zero-mean square-integrable martingale with

$$\mathbb{V}(M_n) = \sum_{k=1}^{n} \tilde{\omega}_k^2 Var\left(\bar{p}_k\right).$$

Azuma-Hoeffding and Cauchy-Schwarz inequalities (the latter being used to establish a Lindeberg condition in [316]) should appear to be the most relevant tools to obtain a martingale central limit theorem and, beyond, deriving a similar convergence theorem for $\widehat{p}_n$. However, technical difficulties related to the control of the divergence of $\sum_{k=1}^{n} \tilde{\omega}_k^4$ let think that the estimator $\widehat{p}_n$ most often converges too fast towards p to be controlled by an asymptotic normality theorem. This can be checked when the weights are made equal. It is likely that a decreasing sequence (e.g., exponentially decreasing) of deterministic weights might be chosen to reduce this speed. An open problem being currently considered with my coauthors, this choice is more guided by statistical considerations that operational constraints.

5.2.1.6 Towards accelerated quantile estimation Similar mechanisms can be applied to the dual problem of estimating a quantile associated with a probability-limit value. In addition to producing a statistical estimate faster than a classical Monte Carlo approach, it is also possible to produce deterministic bounds on this quantile. This work has been mainly carried out in the framework of the PhD thesis [226], but has not yet been published, as some remaining theoretical work is still ongoing.

5.2.2 Input penalization through quantile minimization

In the last contribution [28] achieved in the framework of N. Benoumechiara's thesis [27], my co-authors and I were interested in the problem of computing a *penalizing quantile* q_α^* (or so-called *worst case*) of the univariate output Y of a numerical model g, in connection with reliability and safety applications, given a probability level $\alpha \in (0,1)$. The penalizing (or *conservative*) characterization of q_α^* is related to the choice of the joint distribution of the inputs X, and in particular to the choice of the correlation structure between these inputs. Indeed, by positing $X = (X_1, \ldots, X_p) \in \mathbb{R}^d$, it is generally easier to obtain information on the marginal distributions of the X_i (data, expert opinion, etc.) than on the dependences between the X_i: correlated data can simply lack in numerous applications. More formally:

$$q_\alpha^* = \min_{P \in \mathcal{P}} \left\{ \inf \left\{ y \in \mathbb{R} : F_Y(y) \geqslant \alpha \ , Y = g(X) \ \text{ with } X \sim P \right\} \right\}$$

where F_Y is the cumulative distribution function (cdf) of Y and $\mathcal{P}$ denotes the set of all possible distributions. Assuming to know the marginal distributions (cdf) F_i of each X_i, a *vine copula-based* approach [175] was proposed in [28] to define a parametric subclass $\tilde{\mathcal{P}}(\theta)$ of $\mathcal{P}$, with $\theta \in \Theta \subset \mathbb{R}^d$, and producing:

- **(a)** an algorithm aiming at finding penalizing input correlation structures in $\tilde{\mathcal{P}}(\theta)$, accounting for possible other constraints (for instance due to correlation implausibility in structural reliability problems);
- **(b)** a consistent empirical estimator of q_α^*.

This work provides a more theoretical basis and a more operational practical framework for closely-related proposals made by [174, 339] in the recent years. The tools developed in this work have been implemented within the Python package `dep-impact`.

In more details, each member P_θ of $\tilde{\mathcal{P}}(\theta)$ was defined by the cdf $(F_1, \ldots, F_d)$ and a given vine copula function $C_\theta : (0,1)^d \to (0,1)$, combining bivariate copulas to create a multivariate one. The joint cdf of X can be written

$$F_\theta(x) \ = \ P(X_1 < x_1, \ldots, X_d < x_d) \ = \ C_\theta \left(F_1(X_1), \ldots, F_d(X_d) \right).$$

and its cfd can be written, following the (non unique) decomposition proposed by [73]:

$$\begin{aligned} f_\theta(x) &= \left[\prod_{i=1}^{d} f_i(x_i) \right] c_\theta \left(F_1(x_1), \ldots, F_d(x_d) \right), \\ &= \left[\prod_{i=1}^{d} f_i(x_i) \right] \prod_{j=1}^{d-1} \prod_{i=1}^{d-j} c_{\theta_{i,i+j}} \left(F_i(x_i, F_{i+j}(x_{i+j} | i+1, \ldots, i+j-1) \right) \end{aligned}$$

where the f_i are the marginal pdf, c_θ is the $d-$dimensional copula density function (cppf), and $c_{\theta_{i,i+j}}(\cdot, \cdot | i+1, \ldots, i+j-1)$ is the bivariate cppf conditional to $X_{i+1}, \ldots, X_{i+j-1}$.

Denote now respectively $F_{Y|\theta}$ and $F^{-1}_{Y|\theta}$ the cdf and quantile function of the push-forward distribution of P_θ by g. Then, for a given family of copula $(C_\theta)_{\theta\in\Theta}$, the minimum (penalizing) output quantile is defined by

$$q^*_{C,\alpha} \quad = \quad \inf_{\theta\in\Theta} F^{-1}_{Y|\theta}(\alpha).$$

and if it exists, denote $\theta^*_C = \arg\min_{\theta\in\Theta} F^{-1}_{Y|\theta}(\alpha)$.

Under regularity conditions $\mathcal{C}$ on $P_\theta \in \tilde{\mathcal{P}}(\theta)$ and g, and using a naive optimization algorithm based on grid-search, **(b)** is partially solved by the following theorem, which makes use (among other tools) of the Dvoretzky-Kiefer-Wolfowitz (DKW) inequality. In corollary, using the *coarea formula* [115] it can be shown that conditions $\mathcal{C}$ are verified if g is a Lipschitz and differentiable function with no null derivative everywhere in the neighborhood of $F^{-1}_{Y|\theta}(\alpha)$, and if, for any $\theta \in \Theta$ and any u in the support of the density of Y,

$$\int_{g^{-1}(u)} f_\theta d\mathcal{H}^{d-1} \quad < \quad \infty$$

where $\mathcal{H}^{d-1}$ is the $d-1$ dimensional Haussdorff measure.

THEOREM 6. **Consistency of the penalizing quantile estimator [28].**
Denote $\hat{q}_{\alpha,n}(\theta) = \hat{F}^{-1}_{Y|\theta,n}(\alpha)$ the empirical quantile computed from the $n-$sample $y_n = g(X_n)$ with iid $X_n \sim P_\theta$. Given a sequence of integers $(N_n)_{n\geqslant 1}$, denote $(\Theta_{N_n})_n$ a sequence of discrete finite grids of Θ of cardinality N_n such that

$$\sup_{\theta\in\Theta,\, \theta'\in\Theta_{N_n}} \|\theta - \theta'\|_2 \to 0 \qquad \text{as } N_n \to \infty.$$

Denote $\hat{\theta}_{N_n} = \arg\min_{\theta\in\Theta_{N_n}} \hat{F}^{-1}_{Y|\theta,n}(\alpha)$ the minimum quantile over the grid Θ_{N_n}.
Then, under $\mathcal{C}$, for all $\varepsilon > 0$,

$$P\left(\left|\hat{q}_{\alpha,n}\left(\hat{\theta}_{N_n}\right) - q^*_{C,\alpha}\right| > \varepsilon\right) \xrightarrow{n\to\infty} 0.$$

Moreover, if there exists a unique $\theta^* \in \Theta$ minimizing $\theta \mapsto F^{-1}_{Y|\theta,n}(\alpha)$, then for all $h > 0$

$$\mathbb{P}[|\theta^* - \theta^*_C| > h] \xrightarrow{n\to\infty} 0$$

Obviously, such a strategy becomes of exponential cost in d when d increases (without accounting for the possible additional cost of running the numerical model g), which implies that a algorithmic scheme answering to **(a)** be more sophistically designed to explore such problems. In [28], a *parsimony assumption* was made to decrease the exploring cost of the space of all correlations, related to the nature of vine copulas. This resulted in the proposal of a heuristic greedy algorithm. Its general principle is briefly given below.

Start using the independent copula $c_\theta^0(u_1,\dots,u_d) = \prod_{i=1}^{d} u_i$ where the $u_i \sim \mathcal{U}[0,1]$.

At each iteration k:

1. Denote Ω_k a set of (considered) dependent pairs, influential for quantile minimization, with $\Omega_0 = \varnothing$.
2. Let $\Omega_{-k} = \Omega / \Omega_k$ denote the set of candidate pairs
3. Find a promising candidate pair in Ω_{-k}:
   (i) Create a vine structure $\mathcal{V}_{i,j}$.
   (ii) Explore the set $\mathbf{B}$ of copula families and produce with a grid-search algorithm the minimizer (in the quantile sens) in $\mathbf{B}$

$$\begin{aligned} B_{i,j} &= \arg\min_{B\in\mathbf{B}} \hat{F}^{-1}_{Y|\hat{\theta}_B}(\alpha) \\ \hat{\theta}_{i,j} &= \hat{\theta}_{B_{i,j}}. \end{aligned}$$

4. Select the minimizing pair in Ω_{-k}

$$(i,j)^k = \arg\min_{(i,j)\in\Omega_{-k}} \hat{F}^{-1}_{Y|\hat{\theta}_{i,j}}(\alpha)$$

5. Update Ω_k, the retained conditional bivariate distributions and the structure of the whole vine copula.

Specifically designed (sub)algorithms were produced for carrying out the macro-steps of this algorithm, which are conceptually simple but require too much space for being explained in this summary. Basically, the complete algorithm allowed to successfully explore use cases up to the dimension $d = 10$, without taking into account the computational time required for each simulation of g: a few number of iterations ($k \geqslant 5$), using several common copula families (e.g., Gaussian, Archimedian) were found to be sufficient to find a minimum quantile on these various use cases.

Remark 4. *An important link must be made with the concerns about the monotonicity of g explored at § 5.2.1. In many situations where g is monotonic, the worst quantile is reached for perfect dependencies (Fréchet-Hoeffding bounds). More generally, when the function has a monotonic behavior with respect to many variables, it seems likely that the minimum output quantile be reached at the boundary of* Θ. *While it is clear that learning features on the geometry of g can certainly help to design faster algorithms for Task* **(a)**, *this intuition was disproved in [28], non-trivial high risk situations (associated with non-trivial copulas) being detected by the algorithm, on simple examples.*

Of course, this work remains a first approach to copula selection in a context where conservatism is required, in view of bounding the impact of uncertainties, while no prior information is available on the correlation structure. Apart trying to relax the hypotheses of Theorem 6, the grid-search estimation is clearly a limiting feature of the method, which arises from a lack of information about the convexity and the gradient of the quantile function. If stochastic recursive algorithms like Robbins-Munro could be relevantly tested, accelerated Monte Carlo approaches based on the problem geometry have to be developed or adapted. These avenues and other research possibilities are explained more in detail in [28], but we can retain that the optimization problem of quantile minimization could be also addressed within the more general framework of optimal transport theory, which knows a growing number of contributions today. Indeed, this problem can be seen as a multi-marginal transport problem, which remains however not well understood outside the bimarginal case ($d = 2$).

6 Perspectives

The work I envisage involves improving the scope of the results of previous work, the use of studied tools (as martingale theory) to get new results on tenacious issues, and the continuity of the reflections I am currently conducting, in particular on Bayesian modeling and the deep link existing between uncertainty quantification (UQ) and the problems of interpretability and robustness in machine learning.

1. **Specifically in risk analysis.** In collaboration with the Grenoble Alpes University, I have started a work program on the **establishment of convergence properties of classical statistical estimators** (maximum likelihood) **of ARA survival processes** (virtual age models) **defined from unique trajectories**. In this framework, the trajectory consists in the temporal sequence of deterioration measures of a single system of high reliability, subjected to maintenance. Virtual age is a modeling hypothesis that facilitates system diagnosis, which can increase or decrease depending on the quality of maintenance. Inferred, this type of Markov process thus provides both a quantitative and qualitative view of the impact of maintenance, and this modeling principle can easily be transposed to situations where a single renewable resource is exploited. The renewal is then allowed by a restrictive action on the extraction of the resource (ex: moratorium on fishing) and assimilated, within the framework of reliability, to an action of preventive maintenance. The theoretical difficulty posed by the existence of a single trajectory, typically characterizing a production component such as a nuclear reactor, or an endemic exploited species, is that the asymptotism required to establish the convergence of estimators is temporal, and not induced by the availability of a sample of similar systems. The study of estimators is then based on discrete time martingality and control properties by Azuma-Hoeffding type concentration inequalities, while the search for maintenance (or renewal) improvement strategies is based on the estimation of downtime and finite time stochastic control. We have already demonstrated that, in many cases, compliance with the L^2 and Lyapunov conditions necessary to reach convergence theorems induces overly restrictive assumptions about the system aging [327] (or, by analogy, the accelerated decline of the resource), and we are currently pursuing this work by seeking corrections to conventional estimators, possibly by adopting an objective Bayesian approach.
2. **Specifically for structural reliability analyses**, I also wish to extend my research on the **use of the properties of geometry and regularity of numerical models**, whether these are derived from mechanistic theories (ex : multiphysics problems) or constructed through correlation analyses (ex : deep neural networks). These properties are seen not only as variance reduction factors in Monte Carlo-type experiments, but also as necessary meta-modeling constraints (e.g. by constrained Gaussian processes) when it comes to inverting these models or facilitating their interpretation. Further steps to [331] have already been taken towards the use of experimental designs to "capture" certain local properties (monotony, convexity), still in collaboration with the Institut de Mathématique de

Toulouse (IMT). We aim to complete work to maximize the acceleration rate (relative to a Monte Carlo strategy) of a family of adaptive estimators of exceedance probabilities, in a monotonic framework, by minimizing the hypotheses of regularity as well as a close work on the estimation and the almost sure bounding of quantiles, following some first steps conducted in [226].

In the longer term, I envisage to work on learning local and global geometric properties of models that can take advantage of recent advances made in model approximation theory [163], which should be followed by applications in machine learning. Other avenues will be explored, such as the use of large scale constrained copulas capturing dependency relationships between data. Relevant collaborations could be established with the Massachusetts Institute of Technology, Ecole Centrale de Nantes, INRIA, Safran Tech, in addition to those I have built over the years with the IMT. Possible collaborations with the INRIA team DataShape will also be to examine, in order to identify possible links between manifold learning and topological data analysis.

3. In parallel, I am also interested in the **justification of the probabilistic approach to modeling epistemic uncertainty** permeating phenomenological models. This uncertainty suffers from a profound lack of formal definition, and is generally characterized by only one of its fundamental properties: it is reduced by the addition of knowledge. The notion of knowledge is an epistemological problem that goes beyond the mathematical framework, and many alternative theories to probability theory (e.g., possibility theory) have been proposed to represent this knowledge – the expertise that one typically wishes to inject into a model. Probabilistic justification is an important problem for many decision-makers in the industrial, economic or environmental world, because it implies that the probabilistic criteria used in risk management as well as most methodologies as *Verification, Validation and Uncertainty Quantification* (VVUQ) are fully relevant from a scientific point of view. I adopt the following point of view: handy knowledge depends on the theoretical tools currently available to model (such as Turing or Gandy machines [114]) and their limitations (discretization, consensus on Church-Turing's thesis). These tools define a notion of random (in the sense of Martin-Löf), coherent with Kolmogorov's axiomatics, which makes it possible to characterize a modeling error whose nature is directly related to that of epistemic uncertainty. Through arguments from algorithmic complexity theory (Chaitin-Kolmogorov), I have already demonstrated that a computable algorithm achieving the best achievable approximation (in polynomial time) of this error has characteristics similar to those of a random generator, in a non-trivial real situation [320]. I wish to continue this work by completing it with a discussion between subjective knowledge and objective information, and the application of Cox-Jaynes' representation theorems to this information. In this sense, I have initiated a collaboration with the CMLA at ENS Paris Saclay and the Department of Epistemology of the University of Rouen.
4. In a correlated way, I also aim to continue my work on the **establishment of**

prior modeling methodologies, feeding the Bayesian workflow program, as initiated in [321]. Collaborations could be started with research centers dedicated to this type of modelling, such as Columbia University or Sheffield University, through an international project bringing together a relatively fragmented French community. Alternatively, I wish to pursue a collaboration on this type of analysis applied to extreme statistical models recently initiated with Fabrizio Ruggeri (CMR IMATI Milan) [286] and formalize a strong link with uncertainty quantification techniques [337]. In the longer term, I would like to launch a textbook project about summarizing the latest methodological advances in Bayesian parametric modeling, taking advantage of the many theoretical and applied works conducted in recent years. It would complement with mathematical approaches the leading works in the field, which focus essentially on procedures of praise (knowledge extraction). Several works remain however to be produced, which relate in particular to the prior calibration and validation Bayesian models by the choice of parametric functions replacing imaginary statistics, the use of criteria of proximity between prior information and data and of Wasserstein-type metrics between probability measurements, among others.

5. The whole of this work in progress finally leads me to reflect on the **transfer of classical tools of uncertainty analysis** (sensitivity analysis, meta-modeling, construction of sequential experimental designs) and those that I have started to develop to take into account the geometrical characteristics of problems **in the field of machine / deep learning**, in order to respond to a growing theme of intelligibility of algorithms products in these fields. Recent methods as *Local-Interpretable Model-Agnostic* (LIME [274]) can be considered as (crude) meta-modeling techniques of a sampled sub-manifold, using simple regression functions. The selection of these functions according to local geometrical constraints constitutes a research track which seems to me important. Geometric properties, such as global and local monotony, appear to be strong levers of interpretability (if a given covariate of the problem evolves in a given direction, the proposed decision evolves in another given direction). De facto, a strong link between sensitivity analysis techniques with correlated inputs developed in UQ, such as Mara indices [214], HSIC indices [74], Shapley indices [247, 167] and PLI indices [197], and recently defined deep learning intelligibility tools, such as SHAP indices [209] and DeepLIFT methods [295], should be formally clarified. This work programme should also highlight the procedures for estimating these indices. Finally, this program cannot ignore the use of optimal transport tools in sensitivity analysis and model explanation. This program has in fact already begun and is part of the PhD work of M. Il Idrissi, whom I am co-supervising with IMT and EDF. It has already produced contributions [161, 162] that I believe to be important in this field at the crossroads of several mathematical disciplines.

7 Academic Curriculum Vitae

Current position

Senior Researcher & Head of industrial AI SINCLAIR Laboratory, EDF Lab
Associate Professor, Institut de Statistique (ISUP), Sorbonne Université

Areas of specialization

Bayesian modeling and learning, Monte Carlo methods, treatment of uncertainties. Applications to risk analysis, engineering and environmental issues.

Appointments held

2006-2007 INRIA Saclay (post-doctoral researcher)
2007-2008 Laval University & DFO, Canada (post-doctoral researcher)
2008-2013 EDF Lab (research engineer)
2013-2016 ENSAI (invited professor)
2013-2017 EDF Lab (expert research engineer)
2013-2017 Institut de Mathématique de Toulouse (associate researcher)
2017-2020 Quantmetry (scientific director)
2017-today Sorbonne Université (associate professor)
2020-today EDF Lab (Head of SINCLAIR Laboratory)
2022-today Univ. Paris Saclay & EDF Lab: Head of LARTISSTE Scientific Group
2023-today EDF Lab (Senior researcher)

Education

2003 Engineering degree in Applied Mathematics & Computer Science, ENSIMAG, Grenoble, France
2003 MSc in Applied Mathematics, Université Joseph Fourier, Grenoble, France
2006 PhD in Mathematics, Université Paris Saclay, France
2023 Auditor, The War School (*École de Guerre*), Paris, France

Grants, honours & awards

2014 IDEX Grant, Metadeb Project, Institut de Mathématique de Toulouse
2014 Best conference paper, Lambda-Mu Conference in Risk and Reliability
2016 ENBIS Young European Statistician Award
2019 AMIES PEPS Grant, Lymphomath Project
2023 Expert, Collège France Numérique 2030

Published articles and book chapters

1. Bousquet, N. Towards new formal rules for informative prior elicitation? A discussion on "Specifying Priors in Reliability Applications". *Applied Stochastic Models*

in Business and Industry, **40**: 92â102, 2024.

2. Ouzoulias, F., Bousquet, N., Genu, M., Gille, A., Spitz, J., Authier, M. Development of a new control rule for managing anthropogenic removals of protected, endangered or threatened species in marine ecosystems. *PeerJ*, 12:e16688, 2023.
3. Il Idrissi, M., Bousquet, N., Gamboa, F., Iooss, B., Loubes, J.-M. On the coalitional decomposition of parameters of interest. *Comptes-Rendus de l'Académie des Sciences*, **361**: 1653-1662, 2023.
4. A. Simoulin, N. Thiebaut, K. Neuberger, I. Ibnouhsein, N. Brunel, R. Viné, N. Bousquet, J. Latapy, N. Reix, S. Molière, M. Lodi, C. Mathelin (2023). From free text electronic health records to structured cohorts: Onconum, an innovative methodology for real-world data mining in breast cancer. *Computer Methods and Programs in Biomedicine*, **240**: 107693, , 2023.
5. Béthencourt, L., Dabachine, W., Dejouy, V., Lalmiche, Z., Neuberger, K., Ibnouhsein, I., Chéreau, S., Mathelin, C., Savy, N., Saint Pierre, P., Bousquet, N. Guiding measurement protocols of connected medical devices using digital twins: A statistical methodology applied to detecting and monitoring lymphedema. *IEEE Access*, **9**: 39444-39465, 2021.
6. Adon, R., Arthur, F., Baquiast, G., Hochard, G., Kaid Gherbi, A., Nègre, A., Simoulin, A., Talaouit- Mockli, F., Bousquet, N. (2021). Deep Learning : des usages contrastés dans le monde socio-économique. Une contextualisation de l'ouvrage de Goodfellow, Bengio et Courville. *Statistique et Société*, **8**: 55-108, 2021.
7. Benoumechiara, N., Bousquet, N., Michel, B., Saint-Pierre, P. Detecting and modeling critical dependence structures between random inputs of computer models. *Dependence Modeling*, **8**: 263-297, 2020.
8. Bousquet, N., Klein, T., Moutoussamy, V. Approximation of limit state surfaces in monotonic Monte Carlo settings, with applications to classification. *SIAM : Journal on Uncertainty Quantification*, **6**: 1-33, 2018.
9. Bousquet, N. Théorie bayésienne des extrêmes (and other chapters). In *Événements extrêmes naturels : théorie statistique et mitigation du risque* (N. Bousquet, P. Bernarada eds). Lavoisier, 2018.
10. Sonigo, C., Jankoswki, S., Yoo, O., Trassard, O., Bousquet, N., Grynberg, M., Beau, I., Binart, N. High-throughput ovarian follicle counting by an innovative deep learning approach. *Scientific Reports*, **8**: 13499, 2018.
11. Sonigo, C., Jankoswki, S., Yoo, O., Trassard, O., Bousquet, N., Grynberg, M., Beau, I., Binart, N. Comptage des follicules primordiaux par deep learning : l'intelligence artificielle au service de l'étude de la reproduction. *Annales d'Endocrinologie*, **79**: 225, 2018.
12. Bousquet, N. Extreme events modeling in energy companies. In *Wiley StatsRef: Statistics Reference Online* (eds N. Balakrishnan, T. Colton, B. Everitt, W. Piegorsch, F. Ruggeri and J.L. Teugels), 2018.

13. Pérot, N., Bousquet, N. Functional Weibull models of steel fracture toughness for structural risk analysis: estimation and selection. *Reliability Engineering and System Safety*, **165**: 355-367, 2017.
14. Fu,S., Couplet, M., Bousquet, N. An adaptative kriging method for solving nonlinear inverse statistical problems. *Environmetrics*, **28**: e2439, 2017.
15. Wolfson, L. Bousquet, N. Elicitation. In *Wiley StatsRef: Statistics Reference Online* (eds N. Balakrishnan, T. Colton, B. Everitt, W. Piegorsch, F. Ruggeri and J.L. Teugels), 2016.
16. Pasanisi, A., Roero, C., Bousquet, N., Remy, E. On the practical interest of discrete Inverse Polya and Weibull-1 models in industrial reliability studies. *Quality and Reliability Engineering International*, **31**: 1161-1175, 2015.
17. Keller, M., Popelin, A.-L., Bousquet, N., Remy, E. Nonparametric estimation of the probability of detection of flaws in an industrial component from destructive and non- destructive testing data, using Approximate Bayesian Computation. *Risk Analysis*, **35**: 1595-1610, 2015.
18. Lemaire, P., Sergienko, E., Arnaud, A., Bousquet, N., Gamboa, F., Iooss, B. Density modification based reliability sensitivity analysis. *Journal of Statistical Computation and Simulation*, **85**: 1200-1223, 2015.
19. Bousquet, N., Chassot, E., Duplisea, D., Hammill, M.O. Forecasting the major influences of predation and environnement on cod recovery in the northern Gulf of St Lawrence. *PLOS One*, **9**: e82836, 2014.
20. Archambault, B., Lepape, O., Bousquet, N., Rivot, E. Density dependence can be revealed by modeling the variance in the stock-recruitment process–an application to flatfishes. *ICES Journal of Marine Science*, **71**: 2127-2140, 2014.
21. E. Dortel, E. ,Sardenne, F., Bousquet, N., Rivot, E., Million, J., Le Croizier, G., Chassot, E. An integrated Bayesian modeling approach for the growth of Indian Ocean yellowfin tuna. *Fisheries Research*, **163**: 69-84, 2014.
22. Bousquet, N., Paroissin, C., Fouladirad, M. Grall, A. Bayesian Gamma processes for optimizing condition-based maintenance under uncertainty. *Applied Stochastic Models in Business and Industry*, **31**: 360-379, 2014.
23. Fu, S., Celeux, G., Bousquet, N., Couplet, M. Bayesian inference for inverse problems occurring in uncertainty. *International Journal for Uncertainty Quantification*, **5**: 73-98, 2014.
24. Parent, E., Pasanisi, A., Bousquet, N., Keller, M., Bernier, J. Considérations décisionnelles pour la construction d'un ouvrage de protection contre les crues. In: *Approches statistiques du risque* (J.J. Droesbeke, G. Saporta, C. Thomas-Agnan eds). Technip, 2014.
25. Dortel, E., Massiot-Granier, F., Rivot, E., Million, J., Hallier, J.-P., Morize, E., Munaron, J.-M., Bousquet, N., Chassot, E. Accounting for age uncertainty in growth modelling. The case study of yellowfin tuna (*Thunnus albacares*) of the Indian Ocean. *PLOS One*, **8**: e6088, 2013.

26. Bousquet, N. Accelerated Monte Carlo estimation of exceedance probabilities under monotonicity constraints. *Annales de la Faculté des Sciences de Toulouse*, **21**: 557-591, 2012.
27. Pasanisi, A., Fu, S., Bousquet, N. Estimating discrete Markov models from various missing data schemes. *Computational Statistics and Data Analysis*, **56**: 2609-2625, 2012.
28. Bousquet, N., Cadigan, N., Duchesne, T., Rivest, L.-P. Detecting and correcting underreported catches in fish stock assessment : trial of a new method. *Canadian Journal of Fisheries and Aquatic Sciences*, **67**: 1247-1261, 2010.
29. Bousquet, N. Eliciting vague but proper maximum entropy priors in Bayesian experiments. *Statistical Papers*, **51**: 613-628, 2010.
30. Chassot, E., Duplisea, D., Hammill, M., Caskenette, A., Bousquet, N., Lambert, L., Stenson, G. The role of predation by harp seals (*Phoca groenlandica*) in the collapse and non-recovery of northen Gulf of St. Lawrence cod (*Gadus morhua*). *Marine Ecology Progress Series*, **379**: 279-297, 2009 .
31. Bousquet, N., Duchesne, T., Rivest L.-P. Redefining the maximum sustainable yield for the Schaefer population model including multiplicative environmental noise. *Journal of Theoretical Biology*, **254**: 65-75, 2008 .
32. Bousquet, N. Diagnostics of prior-data agreement in applied Bayesian analysis. *Journal of Applied Statistics*, **35**: 1011-1029, 2008.
33. Bousquet, N. Bertholon, H., Celeux, G. An alternative competing risk model to the Weibull distribution in lifetime data analysis. *Lifetime Data Analysis*, **12**: 481-504, 2006.
34. Introducing a new way of eliciting prior distributions for Weibull inference in an industrial context. *Communications in Dependability and Quality Management*, **8**: 23-29, 2005.

Scientific Books

1. *Extreme Value Theory with Applications to Natural Hazards. From Statistical Theory to Industrial Practice.* Springer, 2021 ((500p). Main editor, co-redactor of several chapters.
2. *Événements extrêmes naturels : théorie statistique et mitigation du risque.* Lavoisier, 2018 (561 p). Main editor, co-redactor of several chapters.
3. **Book translation**: *L'apprentissage profond*, a French translation of *Deep Learning*, by I. Goodfellow, Y. Bengio and A. Courville (MIT Press). Éditions Florent Massot, 2018. I supervised this translation, the first scientific textbook translated using an AI tool mixing regex and deep learning techniques, produced by a public-private consortium.

Articles submitted or in revision

1. Bousquet, N., Blazère, M., Cerbelaud, T. (2023). Covariance constraints for stochastic inverse problems of computer models. In revision for the *Electronic Journal of Statistics*: `https://arxiv.org/abs/1806.03440`
2. Il Idrissi, M., Bousquet, N., Gamboa, F., Iooss, B., Loubes, J.-M. (2024). Hoeffding's decomposition of black box models with dependent inputs. Submitted: `https://arxiv.org/abs/2310.06567`
3. Il Idrissi, M., Bousquet, N., Gamboa, F., Iooss, B., Loubes, J.-M. (2022). Quantile-constrained Wasserstein projections for robust interpretability analyses of numerical and machine learning models. In revision for the *Electronic Journal of Statistics*: `https://arxiv.org/abs/2209.11539`
4. Ben Jebria, N., de Oliveira, E., Carmigniani, R., Drouineau, H., Bousquet, N. (2023). Preliminary Bayesian assessment of outmigration velocity regime of river smolt from hybrid simulation modeling. Submitted to *Science of the Total Environment*.
5. Lodi, M., Bousquet, N., Valverde, P., De la Ferrière, M., Neuberger, K., Jankowski, S., Chenard, P.P., Reix, N. Heitz, D., Tomasetto, C.,Mathelin, C. (2024). Artificial Intelligence-based analysis of breast cancer in elderly women among a cohort of 7965 patients: surprising results and unforeseen implications. Submitted to *Annals of Oncology*.
6. Jankowski, S., Lodi, M., Simoulin, A., Thiebaut, N., Ibnouhsein, I., Bousquet, N., Reix, N. Molière, S. Neuberger, K., Mathelin C. (2023). Artificial intelligence in breast cancer analysis : the study of 15,000 patients in the Senometry trial. Submitted to *Annals of Oncology*.

Articles in preparation

1. Bousquet, N. (2024). Computing conservative probability of rare events with surrogates.
2. Guillaume, A., Franchi, G., Bousquet, N. (2024). TIME: Temporal Similarity Embedding for State Forecasting.
3. Belay Ketema, B., Sueur, R., Iooss, B., Bousquet, N., Gamboa, F., Costantino, F. (2024). On the Fisher distance between truncated distributions for robustness analyses in uncertainty quantification.
4. Ouzoulias, F., Autthier, M., Bousquet, N. (2024). Estimating anthropic pressure on non-commercial species through bycatches and stochastic surplus production models.
5. Ancelet, S., Bousquet, N., Parent, E. (2024). Bayesian calibration of physical designs of experiments, with application to the estimation of steel fracture toughness.

6. Bousquet, N. (2024). Justifying random modelling of epistemic uncertainties in the exploration of computer models.
7. Ruggeri, F., Bousquet, N. (2024). Robust Bayesian analysis of extreme value models. ———————————————————————

Selected conference proceedings

1. Ouzoulias, F., Bousquet, N., Genu, M., Gilles, A., Spitz, J. (2024). Développement d'un modèle stochastique de surplus-production en vue de gérer les prises accessoires d'espèces protégées. *Journées de Statistiques 2024*, Bordeaux.
2. Il Idrissi, M., Bousquet, N., Gamboa, F., Iooss, B., Loubes, J.-M. (2023). Cooperative game theory and importance quantification. *EYSM'23*, Ljubljana.
3. Ketema, B. and Costantino, F. and Gamboa, F. and Sueur, R. and Bousquet, N. and Iooss, B. (2023). Optimization on Riemannian Manifolds for Uncertainty Quantification. *CJC-MA Conference*.
4. Il Idrissi, M., Bousquet, N., Gamboa, F., Iooss, B., Loubes, J.-M. (2022). Projection de mesures de probabilité sous contraintes de quantile par distance de Wasserstein et approximation monotone polynomiale. *Actes des Journées de Statistique*, Lyon.
5. Roussel, J., Ancelet, S., Djazoubi, S., Bernier, M.-O., Ricard, D., Bousquet, N (2020). Inférence bayésienne de l'évolution de l'atropie cérébrale et de plages de leucopathie à partir de séquences IRM 3D non homogénéisées. *Actes des Journées de Statistique*, Nice.
6. Bousquet, N. (2019). Bayesian validation of basins of attraction for block maxima extreme value models. *Proceedings of MMR 2019 Conference*, Hong-Kong.
7. R. Sueur, N. Bousquet, B. Iooss and J. Bect. Perturbed-Law based sensitivity indices for sensitivity analysis in structural reliability (2016). *Proceedings of SAMO 2016 Conference*, Reunion Island.
8. Blazère, M., Bousquet, N. (2016). A new look on well-posedness conditions in stochastic inverse problems. *Proceedings of SAMO 2016 Conference*, Reunion Island.
9. Bousquet, N., Corset, F. (2015). Exploring the consistency of MLE of imperfect repair ARA1 models computed from a single trajectory. *Proceedings of MMR 2015 Conference*, Tokyo, Japan.
10. Bousquet, N., Douard, F. (2014) Analyse bayésienne d'intensités de défaillance pour les études de gestion d'actif. Lambda-Mu Reliability & Risk Conference. *Best article award.*
11. J. Bect, N. Bousquet, B. Iooss, S. Liu, A. Mabille, A-L. Popelin, T. Rivière, R. Stroh, R. Sueur, E. Vazquez (2014). Uncertainty quantification and reduction for the monotonicity properties of expensive-to-evaluate computer models. *Uncertainty in Computer Models 2014 Conference*, Sheffield.

12. Moutoussamy, V., Rochet, P., Bousquet, N., Gamboa, F., Iooss, B., Klein, T. (2013). Comparing conservative estimations of failure probabilities using sequential designs of experiments in monotone frameworks. *Proceedings of ICOSSAR 2013*, New York.
13. Fouladirad, M., Paroissin, C., Bousquet, N., Grall, A. (2013). Bayesian optimization of condition-based maintenance. *Proceedings of MMR 2013*, Stellenbosch,
14. Bousquet, N., Chassot. E., Dortel, E. et al. (2012). Estimating tuna mortality rates from a Bayesian Brownie-Petersen model. Indian Ocean Tagging Symposium, Mauritius.
15. Keller, M., Bousquet, N.(2012). Estimation of a flaw size distribution from a data mixture based on destructive tests and non-destructive in-service inspections. . *Proceedings of the European Safety and Reliability Conference (ESREL)*, Helsinki.
16. Pasanisi, A., Fu, S., Bousquet, N. (2010). Estimation de modèles markoviens discrets dans un cadre industriel fiabiliste à données manquantes. *Actes des Journées de Statistique*, Marseille.
17. Pasanisi, A., de Rocquigny, E., Parent5, E., Bousquet, N. (2009). Some useful features of the Bayesian setting while dealing with uncertainties in industrial practice. *Proceedings of the European Safety and Reliability Conference (ESREL)*, Prague.
18. Bousquet, N. (2009). Calibrating Weibull priors using virtual data. *Proceedings of the MMR Conference*, Moscow.

Selected invited talks

1. Bousquet, N. (2023). Modélisation bayésienne auditable pour les études de risque. *Institut des Actuaires*, Paris, April.
2. Bousquet, N. (2022). Auditable Bayesian modeling for quality measurements. *(Keynote speaker) MATHMET Conference*, Paris, November.
3. Bousquet, N. (2022). Risk analysis, uncertainty and robust decision-making: an introduction. *(Keynote speaker) MASCOT-NUM Workshop on Statistical methods for Safety and Decommissioning*, Avignon, November.
4. Bousquet, N. (2022). Copulas in uncertainty quantification *(Keynote speaker) MEXICO Network Conference*, Bordeaux, November.
5. Bousquet, N. (2022). Decision-helping for stopping machines before rare and fast natural hazards: statistical approaches can beat machine learning rules. *(Keynote speaker) FINS Worshop on "Recent Developments in Dependence Modelling with Applications in Finance and Insurance*, Agistri, September.
6. Bousquet, N. (2021). Powerful deep approaches to follicle counting in fertility studies through medical imaging. *GDR REPRO Days*, Université Versailles-Saint-Quentin, April.

7. Bousquet, N. (2020).Online alert systems againsts rare events affecting devices interacting with the environment: machine learning vs vine-copulas-based approaches. *TIES Virtual Conference*, December.
8. Bousquet, N. (2020). Connections between sensivity analysis and well-posed problems in stochastic inversion. *Saclay Seminar on Uncertainty Quantification* (UQSay), Paris-Saclay University (October).
9. Bousquet, N. (2019). Prior constraints for well-posed nonlinear inversion problems. *Toulouse III Mathematical Seminar* (January)
10. Bousquet, N. (2019). Detecting and monitoring an asymmetric industrial risk using R-Vine copula-based classification rules. *IAPau*, France (October).
11. Bousquet, N. (2019). Statistical calibration of the digital twin of a connected health object. SIS 2019 Conference, Milano.
12. Bousquet, N. (2019). About the geometry of black-box models. *Invited Lecture at Academia Sinica*, Taipei (Taiwan), February 13.
13. Bousquet, N. (2018). How treating uncertainty in black-box complex models? *LIP6-IPSL Seminar on Artificial Intelligence and Climate*, November
14. Bousquet, N. (2017). Expert elicitation methods. Course at *ETICS-17 Summer School*, Porquerolles, October.
15. Bousquet, N. (2017). High-dimensional copulas for solving unbalanced classification problems in industrial risk mitigation. *5th Symposium on Games and Decisions in Reliability and Risk*, Madrid.
16. Bousquet, N. (2016). Convexity and monotonicity in accelerated Monte Carlo methods for structural reliability. Keynote speaker at *ALT Conference 2016* (Troyes).
17. Bousquet, N. (2016). How geometric properties of black box computer models can help to provide conservative assessments in time-consuming situations. *ENBIS 2016 Conference* (Sheffield).
18. Bousquet, N. (2014). Quelques principes de modélisation bayésienne utiles en traitement des incertitudes. *Mascot-Num Workshop*, CEA/DAM, Bruyères-le-Châtel, France.
19. Bousquet, N. (2014). Some methodological aspects of Bayesian prior modelling in industrial and natural resources management. *Elicitation Working Group*. University of Pavia, Italy.
20. Bousquet, N. (2010). Bounding and estimating probabilities of extreme events in natural ressource management. *NCIDS Workshop on Statistics using Spatial and Geographical Data*. Montréal.

Patents

1. N. Pérot, N. Bousquet, M. Marques (2015). Method for determining the strength distribution and the ductile-brittle transition temperature of a steel product subjected to thermal variations. *European Patent Database WO-2015-165962*.

2. D. Vautrin, N. Bousquet, E. de Oliveira, K. Zoubert-Ousseni, N. Paul (2016). Managing water-supply pumping for an electric production plant circuit. *European Patent Deposit WO-2017-86603*.

Ph.D. students

1. Marouane Il Idrissi (Univ. Toulouse III - Co-advisors: JM Loubes / F. Gamboa / B. Iooss): *Unification of sensitivity analysis and post-hoc interpretability methods in machine learning* (2021-2024).
2. Nazih Benoumechiara (Sorbonne Université - Co-advisors: G. Biau/P. Saint-Pierre/B. Michel/R. Sueur) : *Sensitivity analysis of reliability indicators involving the quantification of stochastic dependencies* (2015-2018).
3. Vincent Moutoussamy (Univ. Toulouse III - Academic co-advisor: T. Klein): *Accelerated Monte Carlo computation of failure probabilities and quantiles in monotonic frameworks* (2012-2015).
4. Paul Lemaître (Univ. Bordeaux - Co-advisors: P. Del Moral / B. Iooss): *Global sensitivity analysis of failure probabilities in structural reliability analysis (computer experiments)* (2010-2013).
5. Shuai Fu (University Paris XI - Co-advisors : G. Celeux/M. Couplet) : *Nonlinear stochastic inversion for time-consuming computer codes* (2009-2012)

Other students

1. **Msc students:**
   (a) Baptiste Ferrère (ENSAE), jointly with Bertrand Iooss : Estimating generalized Hoeffding-based interpretability indices for machine learning problems (2024).
   (b) Ibrahim Seydi (Sorbonne Université), jointly with Julien Pelamatti : Classification of credible flood hydrograms (2024).
   (c) Antoine Lhomme (Mines Nancy), jointly with Charlotte Géry: *A pipeline for converting LateX contributions to a HTML wiki website* (2023).
   (d) Fanny Ouzoulias (La Rochelle Université), jointly with Matthieu Authier: *Stochastic Surplus Production Models for Bycatches Management* (2022).
   (e) Bastien Bergère (Sorbonne Université) : *Improvements on accelerated Monte Carlo techniques in monotonic frameworks* (2021).
   (f) Loïc Béthencourt (Ecole Centrale Lyon) : *Calibration of a digital twin of a connected object for medical test assessement* (2019)
   (g) Tangi Salaün (Ecole Centrale Lille, ENSAE) and Nicolas Peltre (CentraleSupélec): *Links between global sensitivity analysis tools and Shapley-based interpretation methods in machine learning* (2018).
   (h) Samy Djazoubi (ENSAI) : *Unsupervised segmentation of brain tumors from clinical routine data* (2017).

(i) Fariath Soule (Univ. Lyon 1) : *Estimating noisy extreme value models for natural hazards* (2014).

(j) Kersane Zoubert-Ousseni (ENSAI) : *Detecting massive plugs in nuclear plants using machine learning* (2014).

(k) Vincent Moutoussamy (Univ. Paris V): *Adaptive accelerated Monte Carlo methods* (2012).

(l) Shuai Fu (Univ. Paris VI), jointly with Alberto Pasanasi: *Discrete Markovian models and adaptive MCMC* (2009).

(m) Amine Bennabi (CNAM) : *Bayesian prior elicitation in structural reliability problems* (2008).

2. **Post-doctoral junior researchers**

   (a) Sophie Ancelet (ENGREF - Academic advisor : E. Parent) : *Bayesian optimized designs of physical experiments* (2011-2012).

   (b) Paul Rochet (Univ. Toulouse III - Academic advisor : T. Klein) : *Bounding quantiles in monotonic Monte Carlo settings* (2012-2013).

   (c) Mélanie Blazère (Univ. Toulouse III - Academic advisor: J.-M. Loubes) : *Well-posed stochastic inversion problems* (2015-2016).

   (d) Antoine Guillaume (ENSTA - Advisor : G. Franchi) : *Alert systems trained from noisy and unbalanced multivariate series* (2022-2024)

3. **Organizations:**

   (a) Scientic Advisory Board, Statistics and Uncertainty Department, *Laboratoire National de Métrologie et d'Essais* (LNE) , from 2020.

   (b) The International Environmetrics Society (TIES), 2021-2022

   (c) French Statistical Days (JDS), 2018 & 2022.

Research project management

1. From 2022 (EDF): Creation of the LARTISSTE (Saclay) scientific interest grouping, dedicated to the management of uncertainties in numerical models used for design and safety studies. Stakeholders: Paris-Saclay University, SigmaClermont, EDF, Framatome, CEA, Safran, ONERA, Airbus, IFPEN, LNE, IRT SystemX, CERFACS ($\sim$ 80 researchers).
2. From 2020 (EDF): leading the research activities of the SINCLAIR AI industrial laboratory (EDF, TotalEnergies, Thales), with specializations on trustwhorty AI, reinforcement Learning and AI for boosting industrial simulation ($\sim$ 25 researchers).
3. 2017-2020 (Quantmetry) : Creation and supervision of 15 major projects of Applied Machine / Deep Learning mainly focused in Medical Imaging, Natural Language Processing for Medicine (Electronic Health Records) and Industry, and Outlier detection in nonstationary time series (involving >34 data architects, engineers and scientists). Collaborations with Berkeley Lab NERSC (US),

OURANOS, Montreal Institute of Learning Algorithms (Québec), IRSN, Hôpital Kremlin-Bicêtre, INSERM, University Hospitals of Strasbourg (HUS), INRIA Lille, ENSAI, Université de Caen, Telecom Paris Tech (France).

4. 2019-2020 (Sorbonne Université): PEPS-AMIES project LYMPHOMATH (Sorbonne Université, HUS, Quantmetry, IMT Toulouse) - Developing a digital twin to monitor a clinical study on a connected heath device detecting lymphoedema (7 researchers).
5. 2016-2017 (EDF): Coordinator of the MATIS EDF Project: "Integrated Management of Cooling Systems" (21 research engineers): Simulation of plants and robustness studies in projected environments (climate change, regulatory modifications, socio-economic patterns, computer code chaining) .
6. 2014-2016 (Univ Toulouse III) : Coordinator of IDEX project METADeb : ?Metamodels of metabolism for Indian Ocean tunas?, including resources from IRD (5 researchers).

Involvement in other research projects with public funding

1. 2016-2019 : H2020 MATCHING European Project : "Materials Technologies for Performance Improvement of Cooling Systems in Power Plants": Task leader (building socio-economic patterns based on climate change): `http://matching-project.eu/`
2. 2012-2015: ANR project AMMSI : "Ageing and Maintenance in reliability: Modelling and Statistical Inference", coordinated by Laboratoire Jean Kuntzmann : `http://www-ljk.imag.fr/AMMSI/`
3. 2013-2012 : FP7 ERA-NET BioDiverSa2 European project CoForTips project (Europe / Africa) "Congo basin forests: tipping points for biodiversity conservation and resilience of forested social and ecological systems", coordinated by CIRAD (C. Garcia). Scientific advisor: `http://www.fordev.ethz.ch/research/active/CoForTips`
4. 2012-2015: ANR project EMOTION : "Estimation of Maternal effects On the sustainability of large pelagic population", coordinated by IRD : `http://www.poleaquimer.com/en/projects/63-emotion.html`

Software development

1. From 2022: supervising the development in Python of the MACADAM AI tool for industrial alert systems.
2. From 2022: supervising the development of a SINCLAIR Guide on Explainable AI including Python and R routines.
3. 2013-2016: co-supervising the development of the R package `mistral` for accelerated Monte Carlo methods in structural reliability frameworks (EDF).
4. 2010-2016: co-supervising the development of the MATLAB software `WOLF3` for steel toughness quantification (joint work with CEA).

5. 2009-2011: development of the the R/C/Excel software SIMCAB (*Seal Impact on Cod ABundance*) for Fisheries & Oceans Canada (DFO).
6. 2007-2010: developing the software tool QuANFRE (XML/R/C) for EDF: estimation of lifetime models from censored data in both frequentist and Bayesian frameworks.

Teaching

1. From 2017: MSc Course: Bayesian Modeling and Computational Statistics (with applications in Statistical Learning). Sciences & Technologies. Sorbonne Université.
2. 2019: Invited MsC Course: "Quantifying uncertainties and managing inference problems". Bordeaux University.
3. 2017: Bayesian modelling for the treatment of uncertainties. Professional course at ETICS Research Summer School, Porquerolles.
4. 2014-2015: Junior Graduate Bachelor - "Stochastic uncertainties in engineering". Baranquilla University, Colombia. PEERS-AIRD program.
5. 2012: MSc Course - Special Course on Bayesian Decisional Statistics in Uncertainty Frameworks - University City of Caracas, Venezuela.
6. 2013-2016: MSc Course - Bayesian Modeling and Computational Statistics, ENSAI.
7. 2010: Professional course “Accounting for Uncertainties", coorganized with the French Statistical Society (SFdS) (Multivariate statistics, probabilistic inversion, accelerated Monte Carlo methods)
8. 2010-2012: Junior Graduate Bachelor - Bayesian Statistics, University Rennes II.
9. 2008-2011: Level-2 Exploratory Statistics, University Paris Dauphine.
10. 2003-2006: Level-1 Algebra and Numerical Analysis, University Paris XI (Saclay).

Service to the profession

Reviewing

1. **PhD defense**
   (a) Max Halford (Univ. Toulouse III). *Machine Learning Applied to Large Scale Database Optimisation*, September, 2020
   (b) Marion Belloni (IRSN ; Paris-Saclay University). *Hierarchical Bayesian approach for the estimation of radio-induced risks*, March 2021.
   (c) Sosthène Akia (IRD - Université Montpellier II). *Tuna species interactions in Atlantic-African coasts*, September, 2021.
   (d) Jérémie Donà (Sorbonne Université). *Statistical Learning of Physical Dynamics*, May, 2022.

(e) Thomas Cerbelaud (Mines Paris Tech). *Automatic building energy model calibration for building performance monitoring during operation*, November, 2023.

(f) Salim Amoukou (Université Paris Saclay). *Trustworthy Machine Learning: Explainability and Distribution-Free Uncertainty Quantification*, December, 2023.

(g) Madina Traoré (École Centrale Nantes). *Machine learning methods to improve nuclear power plants manoeuvrability*, April, 2024.

(h) Théo Defontaine (CERFACS). *Machine learning for flood forecasting : extension of the time frame and little heterogeneous data*, May, 2024.

(i) Bérénice-Alexia Jocteur (Université Lyon I). *Random forest methods for estimating CATE*, July, 2024.

(j) Bilel Bensaid (CEA/ University of Bordeaux). *Neural network optimization from a PDE perspective*, September, 2024.

2. **Scientific journals:**

(a) Computational Statistics and Data Analysis (CSDA), Electronic Journal of Statistics (EJS), Computational Statistics (CS), Journal of Statistical Computation and Simulation (JSCS), ESAIM, Brazilian Journal of Statistics, Bayesian Analysis (BA), IEEE Transactions on Reliability, TEST, Entropy, Stat (*International Statistical Institute*), Ecological Modelling, Statistical Modelling, ICES Journal of Marine Science, Journal of Theoretical Biology, Statistics in Medicine, ASCE-ASME Journal of Risk and Uncertainty in Engineering, Systems, Part A: Civil Engineering, ASME Journal of Mechanical Design, Applied Stochastic Models in Business and Industry.

3. **Conference proceedings:**

(a) NeurIPS, ESREL Proceedings, Statistical Days (JDS, France)....

4. **National and international research grants / projects:**

(a) Spring 2024: Member of the Scientific Committee of the 1st CapGemini Invent hydrological hackathon

(b) From 2023: France 2030

(c) INRIA Team DataShape (*topological data analysis*) (2023-2024)

(d) INRIA Team Statify (*statistical modeling of systems involving data with a complex structure*) (2022-2023)

(e) The French-American FullBright Commission (2019-2020)

(f) The French National Association for Research and Technology (2018-2020)

(g) Various FP7 and H2020 projects (European Commission ; 2013-2019).

(h) French Agency for Food, Environmental and Occupational Health and Safety (ANSES ; 2019).

(i) Deutsches Zentrum für Luft- und Raumfahrt e. V. (DLR) German Aerospace Center Project Management Agency (2016).

5. **Recruitment jury member:**

(a) 2018: INRIA Saclay Recruitment Commission (Full-time junior researchers).

(b) 2014 : IFREMER Recruitment Commission.

6. **Scientific tour:**

   (a) Under the authority of the French Ministry of Foreign Affairs and International Development and other institutions, I co-organized and led a mission on artificial intelligence in Russia, in November 2019, including visits of public and private AI laboratories in Moscow, Saint-Petersburg and Novosibirsk, with a team of 8 French academic and industrial specialists (ENIB-CNRS, 3IA ANITI, INRIA, Airbus, Naver Labs, Renault Group, Quantmetry). The aim was to determine the level of technological readiness, training and maturity of the ecosystem in order to favor future French-Russian cooperation in the field of AI. A confidential report was produced for the French government and the main French industrial companies and laboratories.

7. **International organization:**

   (a) I was invited by the Food & Agriculture Organization (FAO) during some months in 2012 for reviewing the mathematical aspects of the estimation of the natural mortality of Indian Ocean tunas, and as a keynote speaker invited at the Indian Ocean Tagging Symposium, Mauritius.

Editing

1. 2021-2022: Guest Editor for the Special Issue of peer-reviewed journal *Entropy* on "Information Theory-Based Statistical Relational Artificial Intelligence".
2. 2020-2022: Associate Editor for the peer-reviewed journal *Statistique & Société*.

Organizing

1. From 2023 : Director, GIS LARTISSTE (`https://uq-at-paris-saclay.github.io`)
2. April, 2023 : Chairman and co-organizer, (*MASCOT-NUM Conference*), Le Croisic.
3. From September 2021 : Co-organizer of the UQSay (*Uncertainty Quantification in Saclay*) Seminar, with CentraleSupélec and ENS Paris-Saclay.
4. 2020-2021: Co-organizer of the annual EDF Lab Data Science and AI Seminar.
5. March 2020: Co-organizer of the Colloquium on Understanding, predicting and mastering environmental risks using statistics of the French Statistical Society (SFdS): `https://www.sfds.asso.fr/fr/environnement_et_statistique/658-journee_risque/`
6. 2019-2022: Head of the Environment Group at the French Statistical Society (SFdS)

7. June 2019 Organizer of the first open access Quantmetry R&D Half-day on the Future of Artificial Intelligence.
8. September 2018: Organizer of the Special Session on Machine Learning for automatic diagnosis and decision-making at the ENBIS-18 Conference (Nancy).
9. May 2018 : Co-organizer of the SFdS-DMA Day on Interpretability and Fairness of Machine Learning and Artificial Intelligence : `https://www.sfds.asso.fr/fr/data_mining_et_apprentissage/484-actualites/`
10. June 2018: Co-organizer with EDF RD and the Young Statisticians Group of the first Data Challenge at the Annual SFdS Congress (Saclay)
11. May 2018: Co-organizer of the Seminar on Transparency and Intelligibility of Artificial Intelligence, on behalf of the French Statistical Society (Institut Henri Poincaré)
12. 2016 - 2018: Co-Head of Working Group “Risk and Uncertainty" within the GdR MASCOT-NUM
13. Jun 2016: Special Workshop on the Calibration of Computer Models for Industry and Environment (September, Paris). Organizer on behalf of the SFdS and the GdR MASCOT-NUM
14. Nov 2015: Organizer of the GDR Mascot-Num Seminar on Complex Model Exploration, Institut Henri Poincaré, Paris: `http://www.gdr-mascotnum.fr/nov15.html`
15. 2012-2013: Co-organizer of the First and Second French career fairs specialized in Mathematics: `http://smai.emath.fr/forum-emploi`
16. 2013-2019: Elected Member and Treasurer of the SFdS Group “Environment".
17. 2012-2014: Member of the Group “Industry and Environment" within the International Society for Bayesian Analysis (ISBA), session co-organizer at the annual congress
18. 2011-2012: Co-founder and Deputy Head of the SFdS Young Statisticians Group
19. 2011: Organizer of the special workshop on the Use of Expert Judgment in Risk Analysis (SFdS - LNE), Paris.

Vulgarizing

1. Blanchard, J.-B., Chocat, R., Damblin, G., Baudin, M., Bousquet, N., Chabridon, V., Iooss, B., Keller, M., Pelamatti, J., Sueur, R. (2023). Fiches pédagogiques sur le traitement des incertitudes dans les codes de calcul. Rapport de l'Institut Tripartite EDF-CEA-FRAMATOME / BIAS : HAL-042055632.
2. Chabridon, V., Ajenjo, A., Baudin, M., Bousquet, N., Fekhari, E., Iooss, B, Keller, M., Muré, J., Pelamatti, J., Remy, E., Sueur, R. (2023). UQ and ML in Industry: Current Practices and Challenges in Industrial Applications for Low-Carbon Electricity Production. AISSAI Workshop on Artificial Intelligence and the Uncertainty Challenge in Fundamental Physics, November 27th, SCAI, Paris.

3. Bousquet, N., Guggiola, A. (2019). Preface to the White Paper "AI, explain yourself!", Quantmetry: `https://www.quantmetry.com/quant-lab/`
4. Bousquet, N. (2018). Poissons, équations. In: *Manifeste pour l'océan*, with F. Agostini, F. Gabart, N. Hulot, P.-Y. Cousteau *et al.* Éditions Autrement.
5. Bousquet, N. (2018). What is practical artificial intelligence? Interviews with three leaders in business innovation. Blog `www.quantmetry.com/blog`.
6. Bousquet, N. (2014). For a maximized but sustainable fishing. In: *Brève de maths: Mathématiques pour la planète Terre*. Nouveau Monde Éditions.
7. Bousquet, N. (2014). How sizing the dike? In: *Brève de maths: Mathématiques pour la planète Terre*. Nouveau Monde Éditions.
8. Bousquet, N. (2009). Advantages and challenges of Bayesian statistical analysis in industrial lifetime. Lettres Techniques de l'Ingénieur: Risques Industriels, **23**: 5-8.

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